Convective-Reactive Nucleosynthesis in Oxygen-Carbon Shell Mergers

by

Joshua Issa
B.Sc., University of Waterloo, 2023

A Thesis Submitted in Partial Fulfillment of the
Requirements for the Degree of

MASTER OF SCIENCE

in the Department of Physics and Astronomy

Convective-Reactive Nucleosynthesis in Oxygen-Carbon Shell Mergers

by

Joshua Issa
B.Sc., University of Waterloo, 2023

Supervisory Committee

Dr. F. Herwig, Supervisor
(Department of Physics and Astronomy University of Victoria)

Dr. I. Dillmann, Supervisor
(Department of Physics and Astronomy University of Victoria, Research Scientist at TRIUMF)

Abstract

Massive stars models at the end of their evolution can have a merger between the convective O-burning shell and C-burning shells. This is known as an O-C shell merger and occurs in a large number of 1D models. The O-shell at this time in the evolution is convective-reactive, meaning that the timescales for advection and nuclear reactions are equal. This type of environment allows for nucleosynthetic pathways to occur that would not normally happen because material that would otherwise react can mix deeper to hotter temperatures. During the merger, the material in the C-shell which includes C-burning ashes and stable isotopes heavier than Fe are ingested into the convective-reactive O-burning shell where temperatures are much hotter than C-burning conditions. Stellar models that have O-C shell mergers uniquely produce the stable isotopes of phosphorus, potassium, chlorine, and scandium which are otherwise not produced by stellar models to match the solar observations (Ritter et al., 2018b; Roberti et al., 2025). O-C shell mergers are also a site for the production of the p nuclei (Rauscher et al., 2002; Ritter et al., 2018b; Roberti et al., 2023). The p nuclei are 35 stable isotopes that are classically understood to not be produced by the neutron capture processes, but in the temperatures of the O-shell they can be produced by photodisintegration reactions of pre-existing stable isotopes (Burbidge et al., 1957; Woosley and Howard, 1978). Stellar models calculate the stellar evolution and nucleosynthesis of the merger in 1D (Ritter et al., 2018b), but 3D hydrodynamic simulations show that there are significant disagreements with the structure of the flow in both the O-burning shell and during the merger. Convective velocities in 3D are much higher than 1D, the radial component of the velocities have a gradual downturn at the boundaries not found in 1D, the flow has large scale non-radial asymmetries, and the rate of entraining C-shell material can be significantly different (Meakin and Arnett, 2006; Jones et al., 2017; Andrassy et al., 2020; Yadav et al., 2020). Because of the O-shell is a convective-reactive environment, the macrophysical uncertainties in the 3D hydrodynamic simulations can have a significant impact on the nucleosynthesis of the p nuclei as the mixing timescale could be different. In this thesis, insights from 3D hydrodynamic simulations are implemented in 1D simulations to determine how macrophysical uncertainties in the O-C shell merger impact the nucleosynthesis in a convective-reactive environment, particularly for the p nuclei.

Table of Contents

Supervisory Committee ii

Abstract iii

Table of Contents iv

List of Tables vi

List of Figures vii

Acknowledgements xii

Dedication xiii

1 Introduction 1

1.1 Motivation . . . 1
1.2 Stars and the Origin of the Elements . . . 2
1.3 Stellar Nucleosynthesis . . . 2
1.3.1 Nuclear Burning and Convective Mixing Interaction . . . 3
1.3.2 Neutron Captures . . . 4
1.4 p-Nuclei Nucleosynthesis . . . 5
1.5 Massive Star Evolution . . . 6
1.5.1 O-C Shell Mergers . . . 8
1.6 1D Stellar Modelling . . . 9
1.6.1 Mixing Length Theory Approximation . . . 14
1.6.2 Stellar Evolution with `MESA` . . . 15
1.6.3 Post-Processing with `mppnp` . . . 15
1.6.4 Convective-Reactive Nucleosynthesis . . . 16
1.7 3D Macrophysics . . . 20
1.7.1 Insufficiencies of MLT . . . 20

1.7.2 3D Simulations of Convective O-Shell Burning and C-Shell Entrainment 22
1.7.3 1D Implementation of 3D Hydrodynamics 25
1.8 Thesis Outline 26

2 Impact of 3D-macrophysics and nuclear physics on p-nuclei nucleosynthesis in O–C shell mergers 27
2.1 Abstract 27
2.2 Introduction 28
2.3 Methodology 31
2.3.1 Initial Model and Post-Processing Setup 31
2.3.2 1D implementation of 3D macrophysics 35
2.3.3 Determining impact of varying nuclear reactions 38
2.4 Results 40
2.4.1 Convective-reactive production of the *p* nuclei 40
2.4.2 Impact from a downturn and boosting mixing speeds 45
2.4.3 Impact from varying the ingestion rate 47
2.4.4 Impact from dips from convective quenching 48
2.4.5 Nuclear physics impact and mixing dependencies 49
2.5 Discussions and Conclusion 54
2.6 Appendix 57
2.6.1 Correlations of nuclear reaction rates 57
2.6.2 Results of varying ingestion rate 59
2.6.3 Results of varying the input nuclear physics 61

3 Investigations Beyond the *p*-Nuclei 74
3.1 Light Odd-Z Isotope Production 74
3.2 Advective-Reactive *r*-process 76

4 Summary and Conclusion 78
4.1 Mixing, Nuclear Physics, and the *p*-Nuclei 78
4.2 Light Odd-Z Isotope Production and Macrophysics 78

Bibliography 80

List of Tables

Table 1.1 Details about hydrostatic burning phases in massive stars taken from the 13-25$M_\odot$ stars in Table 1 of Woosley et al. (2002) 3

Table 2.1 $\langle \mathrm{OP} \rangle$ for each mixing scenario and the average spread ($\mathrm{OP}_{\mathrm{max}} - \mathrm{OP}_{\mathrm{min}}$) for the Monte Carlo simulations for the p nuclei. All Monte Carlo simulations are calculated with an ingestion rate of 4×10^{-3} $M_\odot s^{-1}$. 40

Table 2.2 Reactions correlated with the production/destruction of an isotope unique to an individual mixing scenario. 52

Table 2.3 Reactions correlated with the production/destruction of an isotope shared across all mixing scenarios. 53

Table 2.4 Correlations and ζ slopes between mass fraction and reaction rates for the MLT mixing scenario. 61

Table 2.5 Correlations and ζ slopes between mass fraction and reaction rates for the 3D-inspired mixing scenario. 63

Table 2.6 Correlations and ζ slopes between mass fraction and reaction rates for the 3×3D-inspired mixing scenario. 65

Table 2.7 Correlations and ζ slopes between mass fraction and reaction rates for the 10×3D-inspired mixing scenario. 67

Table 2.8 Correlations and ζ slopes between mass fraction and reaction rates for the 50×3D-inspired mixing scenario. 69

List of Figures

Figure 1.1 Taken from https://blog.sdss.org/2017/01/09/origin-of-the-elements-in-the-sol Periodic table shaded according to the origin of the elements in the solar system. 2

Figure 1.2 Taken from Prialnik (2009) Figure 9.19: Schematic structure of a supernova progenitor star. 7

Figure 1.3 Kippenhahn diagram showing the merger of the convective O and C-burning shells for the 15 $M_\odot$ $Z = 0.02$ model from Ritter et al. (2018b). The O-burning shell extends from 1.55−1.95 $M_\odot$. The first convective C-burning shell sits directly on top from 1.96−2.11 $M_\odot$ and additional convective C-burning shells that are ingested are above. The merger onsets at $\log_{10}(t - t_{\rm end})/{\rm yr} \approx -3.85$ and reaches full extent at ≈ -4. . 8

Figure 1.4 Evolution of the mass fraction of ^{12}C, ^{16}O, ^{20}Ne, and ^{35}Cl in a convective-reactive environment at the same simulation time with different timesteps. The solid line corresponds to a simulation with timestep of $\Delta t = 0.5$s, the dashed line corresponds to $\Delta t = 0.1$s, and the dotted line corresponds to $\Delta t = 0.01$s. 17

Figure 1.5 $\tau_{\rm MLT}$, $\tau_{\rm react}$, and $\tau_{\rm scale}$ for the 15 $M_\odot$ $Z = 0.02$ stellar model from Ritter et al. (2018b) at $\log_{10}(t - t_{\rm end}) \approx -3.85$ (model number 9200). The black line is $\tau_{\rm MLT}$. The solid red line is $\tau_{\rm react}$ for ^{12}C, and the jagged red line is $\tau_{\rm scale}$ for ^{12}C. Likewise, the green lines are $\tau_{\rm react}$ and $\tau_{\rm scale}$ for ^{16}O, and the purple lines are $\tau_{\rm react}$ and $\tau_{\rm scale}$ for ^{20}Ne. Black circles are provided where $D_\alpha = 1$ for ^{12}C and ^{20}Ne. 19

Figure 1.6 O- and C-burning shells in the 15$M_\odot$ $Z = 0.02$ model from Ritter et al. (2018b) at model number 9200 right before the merger with the C-shell is fully realized. The left y-axis shows the diffusion coefficient in blue, and the right y-axis shows the mixing length in black. The O-shell is from $2.79{-}7.49 \times 10^6$ m, the merger region is from $7.49{-}8.56 \times 10^6$ m, and the C-burning shell is from $8.56{-}18 \times 10^6$ m. 21

Figure 1.7 Taken from Jones et al. (2017): Time-averaged radial diffusion coefficient profile calculated from the spherically averaged abundance profiles by the method described in Section 3.5 (brown solid line; black solid line is a fit to the noisy region). The convective velocities computing using MLT agree with the spherically averaged 3D velocities to within about a factor of 2 inside the convection zone but are too large in the vicinity of the convective boundary, resulting in an overestimation of the diffusion coefficient there. Limiting the mixing length to the distance from the convective boundary reproduces the fall-off of the diffusion coefficient inside the convection zone approaching the boundary which is seen in the spherically averaged 3D simulation results. 23

Figure 2.1 Kippenhahn diagram showing the merger of the convective O and C-burning shells. The O-burning shell extends from $1.55M_\odot$ to $1.95M_\odot$. The first convective C-burning shell ingested sits directly on top from $1.96M_\odot$ to $2.11M_\odot$ and the other convective C-burning shells are above. A red guideline has been provided to mark the thin radiative layer separating the O- and C-shells. The merger onsets at $\log_{10}(t - t_{\rm end})/{\rm yr} \approx -3.85$ and reaches full extent at ≈ -4. A black triangle marks the location where the initial p nuclei are taken from and a white star marks the location where the ingested C-shell material is taken from. 32

Figure 2.2 Mass fractions of the p nuclei from $\log_{10}(t-t_{\rm end})/{\rm yr} = -3.856$. used for initial O-shell composition and ingested C-shell material. Markers are the same as those used in Figure 2.1. 34

Figure 2.3 The diffusion coefficient profile and mixing length at model number 9200 for the $M_{\rm ZAMS} = 15\ M_\odot$, $Z = 0.02$ model. The light blue line is D from MESA, the orange line the smoothed D used for the MLT mixing scenario in this paper, and the grey line is the mixing length. Black dashed lines mark the shell boundaries for this paper. 35

Figure 2.4 The diffusion coefficient profiles for the MLT and 3D-inspired gradual downturn scenarios. The dashed orange line is the $D_{\rm MLT}$ profile, and the dashed light grey line, dotted light blue line, solid grey line, and dashed dark blue line are the downturn profiles with boost factors of 1, 3, 10, and 50 respectively. 37

Figure 2.5 The quenched mixing scenario convective profiles. The dashed light orange line and dotted light grey line are the GOSH-like profiles with a dip centred at $r = 4.95$ Mm. The solid red line and dashed dark grey line are the partial merger profiles with a dip centred $r = 7.5$ Mm. The MLT profile is omitted for clarity. 39
Figure 2.6 Mass fractions without radiogenic contributions at $t = 110$ sec for the MLT mixing scenario without C-shell ingestion. 41
Figure 2.7 Reaction fluxes f_{ij} for ^{156}Dy and mass fractions X_i for $^{156-158}$Dy without radiogenic contributions for the MLT scenario with no ingestion after $t = 110$ s. The direction of reactions $i \rightarrow j$ are written as left to right in the legend. 42
Figure 2.8 Reaction fluxes f_{ij} for ^{156}Dy and mass fractions X_i for $^{156-158}$Dy without radiogenic contributions for the MLT scenario with an ingestion rate of 4×10^{-3} $\mathrm{M_\odot s^{-1}}$ after $t = 110$ s. The direction of reactions $i \rightarrow j$ are written as left to right in the legend. 43
Figure 2.9 Chart of reactions between isotopes at $m = 1.64$ $\mathrm{M_\odot}$ [$T = 2.28$ GK] for the same conditions as Figure 2.8. Both arrow colour and size indicate f_{ij}, and arrows point in the direction of the reaction. 44
Figure 2.10 The overproduction compared to initial of the p nuclei for the MLT and 3D-inspired mixing scenarios. The average spread in production $\mathrm{OP_{max}} - \mathrm{OP_{min}} = 0.96$ dex. $\mathrm{OP} = 0$ is the initial amount. 45
Figure 2.11 The overproduction compared to initial of the p nuclei for the MLT mixing scenario for no ingestion, 4×10^{-5} $\mathrm{M_\odot s^{-1}}$, 4×10^{-4} $\mathrm{M_\odot s^{-1}}$, and 4×10^{-3} $\mathrm{M_\odot s^{-1}}$. The average spread in production $\mathrm{OP_{max}} - \mathrm{OP_{min}} = 1.22$ dex excluding the no ingestion case. $\mathrm{OP} = 0$ is the initial amount. 47
Figure 2.12 The overproduction compared to initial of the p nuclei for the MLT, GOSH-like, and partial merger scenarios with ingestion rate 4×10^{-3} $\mathrm{M_\odot s^{-1}}$. The average spread in production $\mathrm{OP_{max}} - \mathrm{OP_{min}} = 0.51$ dex. $\mathrm{OP} = 0$ is the initial amount. 48
Figure 2.13 Histogram showing the spread due to varying (γ, p), (γ, n), (γ, α) and corresponding capture rates for unstable n-deficient isotopes from Se–Po for the MLT mixing scenario. Colour and size both correspond to the logarthimic binning of Monte Carlo runs. The average spread $\mathrm{OP_{max}} - \mathrm{OP_{min}} = 0.56$ dex. $\mathrm{OP} = 0$ is the initial amount. 50

Figure 2.14 Bars representing the maximum and minimum OP across all mixing scenarios, excluding those without C-shell ingestion. The average spread $\mathrm{OP}_{\mathrm{max}} - \mathrm{OP}_{\mathrm{min}} = 2.45$ dex. OP $= 0$ is the initial amount. 55
Figure 2.15 Examples of strongly correlated mass fractions with reaction rates for four species for the MLT mixing scenario. The orange dots are the values and the black line is the best linear fit to $\mathrm{OP}_{\mathrm{var}}$ and $\log_{10}$(variation factor). The top left plot and bottom right plot show strong correlations for ^{74}Se and ^{196}Hg that directly impact the mass fraction. The top right plot shows a correlation for ^{98}Ru where there is considerable spread in the mass fraction. The bottom left plot shows a correlation for ^{184}Os where the mass fraction is strongly affected only for multiplication factors less than 1. 58
Figure 2.16 The overproduction compared to initial of the p nuclei for the 3D-inspired mixing scenario for no ingestion, $4{\times}10^{-5}$ $\mathrm{M}_{\odot}\mathrm{s}^{-1}$, $4{\times}10^{-4}$ $\mathrm{M}_{\odot}\mathrm{s}^{-1}$, and $4{\times}10^{-3}$ $\mathrm{M}_{\odot}\mathrm{s}^{-1}$. The average spread in production $\mathrm{OP}_{\mathrm{max}} - \mathrm{OP}_{\mathrm{min}} =$ 1.58 dex. OP $= 0$ is the initial amount. 59
Figure 2.17 The overproduction compared to initial of the p nuclei for the 3×3D-inspired mixing scenario for no ingestion, $4{\times}10^{-5}$ $\mathrm{M}_{\odot}\mathrm{s}^{-1}$, $4{\times}10^{-4}$ $\mathrm{M}_{\odot}\mathrm{s}^{-1}$, and $4{\times}10^{-3}$ $\mathrm{M}_{\odot}\mathrm{s}^{-1}$. The average spread in production $\mathrm{OP}_{\mathrm{max}} - \mathrm{OP}_{\mathrm{min}} =$ 1.64 dex excluding the no ingestion case. OP $= 0$ is the initial amount. 59
Figure 2.18 The overproduction compared to initial of the p nuclei for the 10×3D-inspired mixing scenario for no ingestion, $4{\times}10^{-5}$ $\mathrm{M}_{\odot}\mathrm{s}^{-1}$, $4{\times}10^{-4}$ $\mathrm{M}_{\odot}\mathrm{s}^{-1}$, and 4×10^{-3} $\mathrm{M}_{\odot}\mathrm{s}^{-1}$. Arrows denote OP out of bounds and the true OP is written above. The average spread in production $\mathrm{OP}_{\mathrm{max}} - \mathrm{OP}_{\mathrm{min}} =$ 1.78 dex excluding the no ingestion case. OP $= 0$ is the initial amount. 60
Figure 2.19 The overproduction compared to initial of the p nuclei for the 50×3D-inspired mixing scenario for no ingestion, $4{\times}10^{-5}$ $\mathrm{M}_{\odot}\mathrm{s}^{-1}$, $4{\times}10^{-4}$ $\mathrm{M}_{\odot}\mathrm{s}^{-1}$, and 4×10^{-3} $\mathrm{M}_{\odot}\mathrm{s}^{-1}$. Arrows denote OP out of bounds and the true OP is written above. The average spread in production $\mathrm{OP}_{\mathrm{max}} - \mathrm{OP}_{\mathrm{min}} =$ 1.84 dex excluding the no ingestion case. OP $= 0$ is the initial amount. 60
Figure 2.20 Histogram showing the spread due to varying (γ, p), (γ, n), (γ, α) and corresponding capture rates for unstable n-deficient isotopes from Se–Po for the 3D-inspired mixing scenario. Colour and size both correspond to the logarthimic binning of Monte Carlo runs. The average spread $\mathrm{OP}_{\mathrm{max}} - \mathrm{OP}_{\mathrm{min}} = 0.59$ dex. OP $= 0$ is the initial amount. 61

Figure 2.21 Histogram showing the spread due to varying (γ, p), (γ, n), (γ, α) and corresponding capture rates for unstable n-deficient isotopes from Se–Po for the 3×3D-inspired mixing scenario. Colour and size both correspond to the logarthimic binning of Monte Carlo runs. The average spread $\mathrm{OP}_{\mathrm{max}} - \mathrm{OP}_{\mathrm{min}} = 0.69$ dex. $\mathrm{OP} = 0$ is the initial amount. 72

Figure 2.22 Histogram showing the spread due to varying (γ, p), (γ, n), (γ, α) and corresponding capture rates for unstable n-deficient isotopes from Se–Po for the 10×3D-inspired mixing scenario. Colour and size both correspond to the logarthimic binning of Monte Carlo runs. The average spread $\mathrm{OP}_{\mathrm{max}} - \mathrm{OP}_{\mathrm{min}} = 0.76$ dex. $\mathrm{OP} = 0$ is the initial amount. . . 72

Figure 2.23 Histogram showing the spread due to varying (γ, p), (γ, n), (γ, α) and corresponding capture rates for unstable n-deficient isotopes from Se–Po for the 50×3D-inspired mixing scenario. Colour and size both correspond to the logarthimic binning of Monte Carlo runs. The average spread $\mathrm{OP}_{\mathrm{max}} - \mathrm{OP}_{\mathrm{min}} = 0.79$ dex. $\mathrm{OP} = 0$ is the initial amount. . . 73

Figure 3.1 Impact on the production of the light odd-Z stable isotopes of P, Cl, K, and Sc comparing MLT and convective dowturn models. 75

Figure 3.2 Fraction of trajectories with distance for enough time to exchange mass. Each coloured line represents a size for L_{mix}. 77

Acknowledgements

This thesis did not happen by accident, nor by some unique gift I alone have. I would like to thank of primary importance Dr. Falk Herwig and Dr. Iris Dillmann. Without your patience, teaching and supporting me through all my doubts and mistakes, this work would not have happened. Thank you for all the doors you have opened and the opportunities you have given me. I would also like to thank Dr. Pavel Denissenkov for his constant availability to answer all my programming questions and showing me how to use all the tools I needed to do this work. I am deeply grateful to have learned from you all.

To my parents, who have always been supportive of me. Thank you to my mother who taught me how to critically engage in the world and refuse to take things at face value. Thank you to my father who taught me what a life long dedication to learning looks like and insisting that I become a doctor of stars.

To Sneha, my companion. Words can't begin to express how grateful I am for your support and love, always being there and excited for every step that I have taken in this journey. Thank you for the FaceTime calls to wake me up in the morning, and the late night scrolls through Instagram together. Thank you for the motivation every day to work hard and shoot for the stars.

To Mr. Hordyk who told me I would do great things, Mr. Koiter who taught me physics and sparked my love for it, and Mr. Blyleven who who took me from a student who couldn't program Hello World to helping me land my first programming job, thank you for shaping me to be the scientist I am today. I still hear your advice to this day, and I am the better for it.

To my friends in Victoria and Vancouver who have been there through the frustrations and the joys. Thank you Praneet, Maeve, Mallory, Daniel, Bryn, Diego, Dhwani, Layla, Aviv, Ruxin, and Jono. Grad school without you all would have been a sad and lonely place.

I acknowledge and respect the Ləḱʷəŋəŋ (Songhees and Xʷsepsəm/Esquimalt) Peoples on whose territory I live and work on, and the Ləḱʷəŋəŋ and W̱SÁNEĆ Peoples whose historical relationships with the land continue to this day.

Well, I am just a humble theologian, and as far as I am concerned the whole of astronomy can be summed up by saying 'Twinkle, twinkle little star, how I wonder what you are.'

- Karl Barth

Dedication

To my parents who taught me how to see the world, my high school teachers who gave me my passion for computational physics, and to my fiancée who has been my biggest supporter in grad school and in life. Thank you.

Chapter 1

Introduction

The origin of the elements in the universe is a fundamental question in astrophysics. Most elements are produced in stars through various nuclear reactions during their lifetimes and deaths. Nuclear astrophysics studies the processes responsible for the synthesis of these elements. The seminal work by Burbidge et al. (1957) laid the foundation for stellar nucleosynthesis, describing the formation of all isotopes in stars via a series of burning stages and nuclear processes, and proposing likely stellar conditions where they happen. Subsequent observational and computational advances have reinforced the importance of this work in the field. Shortly after, Henyey et al. (1964) introduced a numerical method to model helium formation in stars via the CNO cycle. Since then, computational nuclear astrophysics has progressed significantly, enabling detailed modeling of stellar evolution across a wide range of masses and metallicities (Pignatari et al., 2016b; Ritter et al., 2018b; Battino et al., 2019).

1.1 Motivation

As far as computational nuclear astrophysics has come, there are still many open questions regarding the origin of the elements. The stellar models that calculate the nucleosynthesis of elements rely on one dimensional models that assume stars are spherically symmetric. While these assumptions are adequate for early in a star's life, they no longer hold as the star reaches the end of its life, particularly for massive stars. The goal of this thesis is to investigate how different the formation of elements in massive stars can be when accounting for the three dimensional nature of stars at the end of their lives.

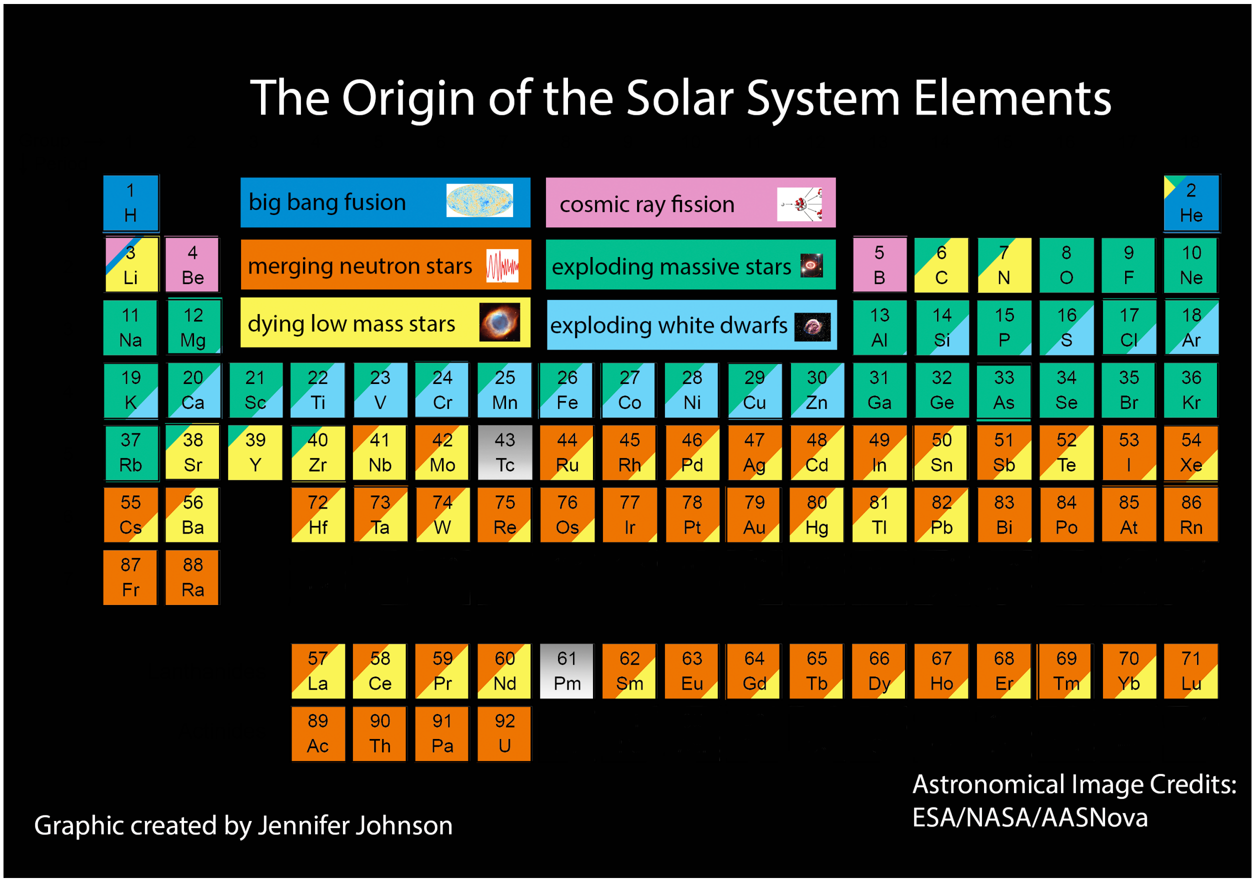

Figure 1.1: Taken from https://blog.sdss.org/2017/01/09/origin-of-the-elements-in-the-solar-system/: Periodic table shaded according to the origin of the elements in the solar system.

1.2 Stars and the Origin of the Elements

The origin of the elements in our solar system. Give a big picture of stars smaller than 8 $M_\odot$.

[*JI: Include a Figure of s/r/p percentages*]

1.3 Stellar Nucleosynthesis

Stellar nucleosynthesis describes how elements are formed within stars through nuclear reactions. Beginning with the fusion of hydrogen into helium in stellar cores, progressively heavier elements are synthesized as stars evolve and undergo successive stages of nuclear burning.

1.3.1 Nuclear Burning and Convective Mixing Interaction

Nuclear burning in convective regions of stars occurs largely in hydrostatic equilibrium, where the energy produced by nuclear reactions balance the gravitational forces and allowing the star to maintain hydrostatic equilibrium. Each of these burning phases is characterized by a specific temperature, density, and timescale where the nucleosynthesis takes place.

Table 1.1: Details about hydrostatic burning phases in massive stars taken from the 13-25$M_{\odot}$ stars in Table 1 of Woosley et al. (2002)

Burning Phase	**Temperature (MK)**	**Density (g cm^{-3})**	**Timescale (yrs)**
H-burning	34.4 - 38.1	6.66 - 3.81	13.5 - 6.70 $\times 10^7$
He-burning	172 - 196	1.73 - 0.762 $\times 10^3$	2.67 - 0.839 $\times 10^7$
C-burning	815 - 841	3.13 - 1.29 $\times 10^5$	2.82 - 0.522 $\times 10^3$
Ne-burning	1690 - 1570	10.8 - 3.95 $\times 10^6$	0.341 - 0.891
O-burning	1890 - 2090	8.19 - 3.60 $\times 10^6$	4.77 - 0.402
Si-burning	3280 - 3650	4.83 - 3.01 $\times 10^7$	48.8 - 2.0 $\times 10^{-3}$

The following section describes the burning stages following Woosley et al. (2002) and Iliadis (2015). H-burning is the first stage of hydrostatic burning which occurs in massive stars primarily by the CNO cycle.

$$^{12}\mathrm{C}(p,\gamma)^{13}\mathrm{N}(\beta^+)^{13}\mathrm{C}(p,\gamma)^{14}\mathrm{N}(p,\gamma)^{15}\mathrm{O}(\beta^+)^{15}\mathrm{N}(p,\alpha)^{12}\mathrm{C}$$

In the CNO cycle, carbon, nitrogen, and oxygen act as a catalyst for the fusion of four ^{1}H isotopes into ^{4}He.

Next is He-burning which has two main processes

$$^{4}\mathrm{He}(\alpha,\gamma)^{8}\mathrm{Be}(\alpha,\gamma)^{12}\mathrm{C}, \quad ^{12}\mathrm{C}(\alpha,\gamma)^{16}\mathrm{O}$$

Therefore, the main products of He-burning are ^{12}C and ^{16}O.

Following this is C-burning which has the following main processes

$$^{12}\mathrm{C} +^{12}\mathrm{C} \rightarrow^{24}\mathrm{Mg} + n$$

$$^{12}\mathrm{C} +^{12}\mathrm{C} \rightarrow^{20}\mathrm{Ne} + \alpha$$

$$^{12}\mathrm{C} +^{12}\mathrm{C} \rightarrow^{23}\mathrm{Na} + p$$

$$^{12}\mathrm{C} + ^{16}\mathrm{O} \rightarrow ^{24}\mathrm{Mg} + \alpha$$

After is Ne-burning which undergoes

$$^{20}\mathrm{Ne}(\gamma,\alpha)^{16}\mathrm{O}$$

$$^{20}\mathrm{Ne}(\alpha,\gamma)^{24}\mathrm{Mg}$$

$$^{20}\mathrm{Ne} + ^{20}\mathrm{Ne} \rightarrow ^{16}\mathrm{O} + ^{24}\mathrm{Mg}$$

Next is O-burning which undergoes

$$^{16}\mathrm{O}(^{16}\mathrm{O},p)^{31}\mathrm{P}$$

$$^{16}\mathrm{O}(^{16}\mathrm{O},2p)^{30}\mathrm{Si}$$

$$^{16}\mathrm{O}(^{16}\mathrm{O},\alpha)^{28}\mathrm{Si}$$

$$^{16}\mathrm{O}(^{16}\mathrm{O},2\alpha)^{24}\mathrm{Mg}$$

$$^{16}\mathrm{O}(^{16}\mathrm{O},d)^{30}\mathrm{P}$$

$$^{16}\mathrm{O}(^{16}\mathrm{O},n)^{31}\mathrm{S}$$

In the final stage of hydrostatic burning, Si-burning occurs by a series of (α,γ) reactions:

$$^{28}\mathrm{Si}(\alpha,\gamma)^{32}\mathrm{S}(\alpha,\gamma)^{36}\mathrm{Ar}(\alpha,\gamma)^{40}\mathrm{Ca}(\alpha,\gamma)^{44}\mathrm{Ti}(\alpha,\gamma)^{48}\mathrm{Cr}(\alpha,\gamma)^{52}\mathrm{Fe}(\alpha,\gamma)^{56}\mathrm{Ni}$$

1.3.2 Neutron Captures

The neutron capture processes are the primary mechanism for the production of elements heavier than iron in stars (Burbidge et al., 1957). There are three main regimes to the neutron capture processes: the slow, intermediate, and rapid neutron capture processes, commonly referred to as the s-, i-, and r-processes. Generically, in the neutron capture processes, a nucleus in a neutron rich environment captures neutrons and becomes increasingly neutron-rich and unstable. The unstable nucleus then decays via β^- decay until it reaches a stable state. This process is controlled by the timescale of neutron capture compared to the timescale of β^- decay (Burbidge et al., 1957; Iliadis, 2015).

The s-process in massive stars occurs during the He- and C-burning phases where ^{22}Ne from $^{14}\mathrm{N}(\alpha,\gamma)^{18}\mathrm{F}(\beta^+)^{18}\mathrm{O}(\gamma,\alpha)^{22}\mathrm{Ne}$ undergoes the reaction $^{22}\mathrm{Ne}(\alpha,n)^{25}\mathrm{Mg}$ and produces neutron densities of 10^6–$10^8\,\mathrm{cm}^{-3}$ (Pignatari et al., 2010; Käppeler et al., 2011).

The i- and r-processes are not well-determined in massive stars as the neutron densities needed (10^{13}–10^{15} and $\geq 10^{20}\,\mathrm{cm}^{-3}$) are not reached in these conditions. Denissenkov et al. (2018); Battino et al. (2020) find these conditions can be met in rapidly accreting white dwarfs where a white dwarf accretes H from a companion main sequence star leading to H-flashes where $^{12}\mathrm{C}(p,\gamma)^{13}N(\beta^{+})^{13}\mathrm{C}(\alpha,n)^{16}\mathrm{O}$ is a source for neutrons. The r-process is thought to occur in neutron star mergers (Mumpower et al., 2018) where the neutrons densities are high enough to produce the heaviest elements in the universe.

1.4 p-Nuclei Nucleosynthesis

The p nuclei are a set of 35 n-deficient stable isotopes heavier than iron that were all originally considered to be produced in a process other than neutron captures. Burbidge et al. (1957) proposed that the p nuclei were produced in a series of (p,γ) reactions in equilibrium with (γ,p) and (γ,n) reactions in the H-rich envelope of a Type II supernova in a process they labelled the "p process". Woosley and Howard (1978) later proposed that the temperature conditions were not sufficient in the H-envelope of a supernova to produce the p nuclei, and instead proposed that in explosive O-burning during the supernova explosion, the p nucleicould be produced in a process they labelled the "γ process". The γ process is a secondary process that requires pre-existing seed nuclei up to Pb from the s process that series of (γ,n) and (γ,α) reactions along with β^{+} decays that occur during explosive O-burning, or potentially C-burning, at $2-3\times10^{9}$K. However, they also showed that a single temperature is not sufficient to produce all of the p-nuclei as at cooler temperatures the heavier p-nuclei are more easily produced while at hotter temperatures the lighter p-nuclei are more easily produced.

Since then, many astrophysical sites have been proposed for the production of the p-nuclei. Woosley and Hoffman (1992) proposed that the lighter p-nuclei could be produced in the α-process in Type II supernovae. Fröhlich et al. (2006); Arcones and Montes (2011) found that ν induced (n,p) reactions in the νp-process could produce the lighter p-nuclei in the winds of a supernova. Schatz et al. (2001) likewise found that they could produce the lighter p-nuclei in the rp-process in a H-rich accretion disk around a neutron star where they are made by rapid (p,γ) and β^{+} decay analogous to the neutron capture process. Goriely et al. (2002) found taht they could produce both the light and heavy p-nuclei in a He-detonation of a C-O white dwarf in a cousin of the rp-process called the pn-process where (α,p) and (α,n) reactions create the needed sources of neutrons and protons. Rauscher et al. (2002) found that the γ-process could take place in both the pre-explosive and explosive O-burning phases of massive stars and could produce all the p-nuclei, but underproduced ^{92}Mo,^{94} Mo,^{96} Ru and ^{98}Ru. Travaglio et al. (2011, 2015) found that they could produce γ-process conditions that could produce all of the p-nuclei in Type Ia supernovae including the underproduced Mo and Ru isotopes. Battino et al. (2020) found that the p-nuclei could be produced in the He-shell flahes of a rapidly accreting white dwarf by the γ-process during the i-process.

Not all the p-nuclei are produced in these ways, however. Despite the claims of Burbidge et al. (1957), ^{152}Gd, ^{164}Er, ^{180m}Ta have a majority of their production in the s-process (Bisterzo et al., 2011), and ^{113}In and ^{115}In can be made during the r-process (Dillmann et al., 2008). ν-capture reactions have been found to play a role in the production of ^{113}In, ^{138}La, and ^{180m}Ta (Goriely et al., 2001; Arnould and Goriely, 2003; Sieverding et al., 2018).

The γ-process is the mode in which the p-nuclei are produced in massive stars. In the O-burning conditions between the temperatures of $1.5{-}3 \times 10^9$K, the γ-process occurs via a series of (γ, n), (γ, α), and (γ, p) reactions (Rauscher et al., 2002; Rapp et al., 2006; Rauscher et al., 2013). Pre-explosive O-C shell mergers have been found to be a critical site for the production of the p-nuclei regardless of the energy of the supernova explosion (Ritter et al., 2018a; Roberti et al., 2023, 2024). The explosive γ-process has been found to underproduce the p-nuclei with $A = 90 - 130$, especially the Mo and Ru isotopes (Arnould and Goriely, 2003; Woosley and Heger, 2007). Further, it has been argued that the γ-process is too weak to produce ^{113}In, ^{115}Sn, ^{152}Gd, ^{164}Er, and ^{180m}Ta and these isotopes should be considered completely separate from the other p-nuclei (Rayet et al., 1990; Dillmann et al., 2008).

1.5 Massive Star Evolution

Stars with an inital mass $> 8\mathrm{M}_\odot$ undergo the series of nuclear burning phases described above.

$$\mathrm{H} \rightarrow \mathrm{He} \rightarrow \mathrm{C} \rightarrow \mathrm{Ne} \rightarrow \mathrm{O} \rightarrow \mathrm{Si} \rightarrow \mathrm{Fe} \rightarrow \text{core collapse.}$$

These burning phases lead to the structure of the star being separated convective shells with primary burning structures, as seen in Figure 1.2.

After the formation of the Fe-core the star begins to rapidly contract as nuclear burning no longer can balance the gravitational force. This leads to a core collapse supernova explosion where further shockwave induced nucleosynthesis can occur (Prialnik, 2009). The core of the star leaves behind a neutron star or black hole, and all other material beyond the mass cut is ejected into the interstellar medium. The exact border of where mass is ejected from changes the yields of massive star nucleosynthesis. Ritter et al. (2018b) place this mass cut for massive stars around the interface of the Si/O-shell.

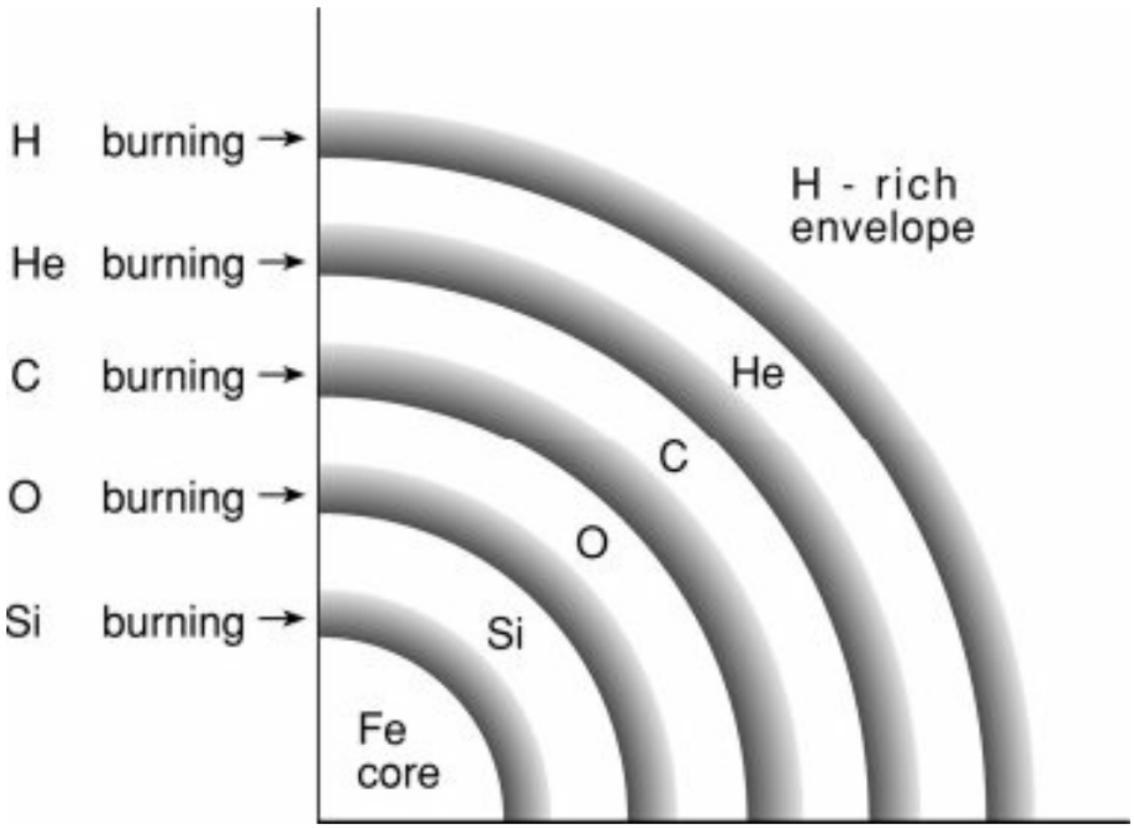

Figure 1.2: Taken from Prialnik (2009) Figure 9.19: Schematic structure of a supernova progenitor star.

1.5.1 O-C Shell Mergers

In the final stages of advanced O-shell stellar burning, hours before the core collapse of the star, anywhere from 20% to 40% of 1D massive star models have been found to have a merger of the O-, Ne-, and C-shells (Rauscher et al., 2002; Ritter et al., 2018a; Collins et al., 2018; Roberti et al., 2023, 2025). This phenomenon is known as an "O-C shell merger". Figure 1.3 shows a Kippenhahn diagram of the merger of the O- and C-burning shells for a 15 $M_\odot$ $Z = 0.02$ model from Ritter et al. (2018b).

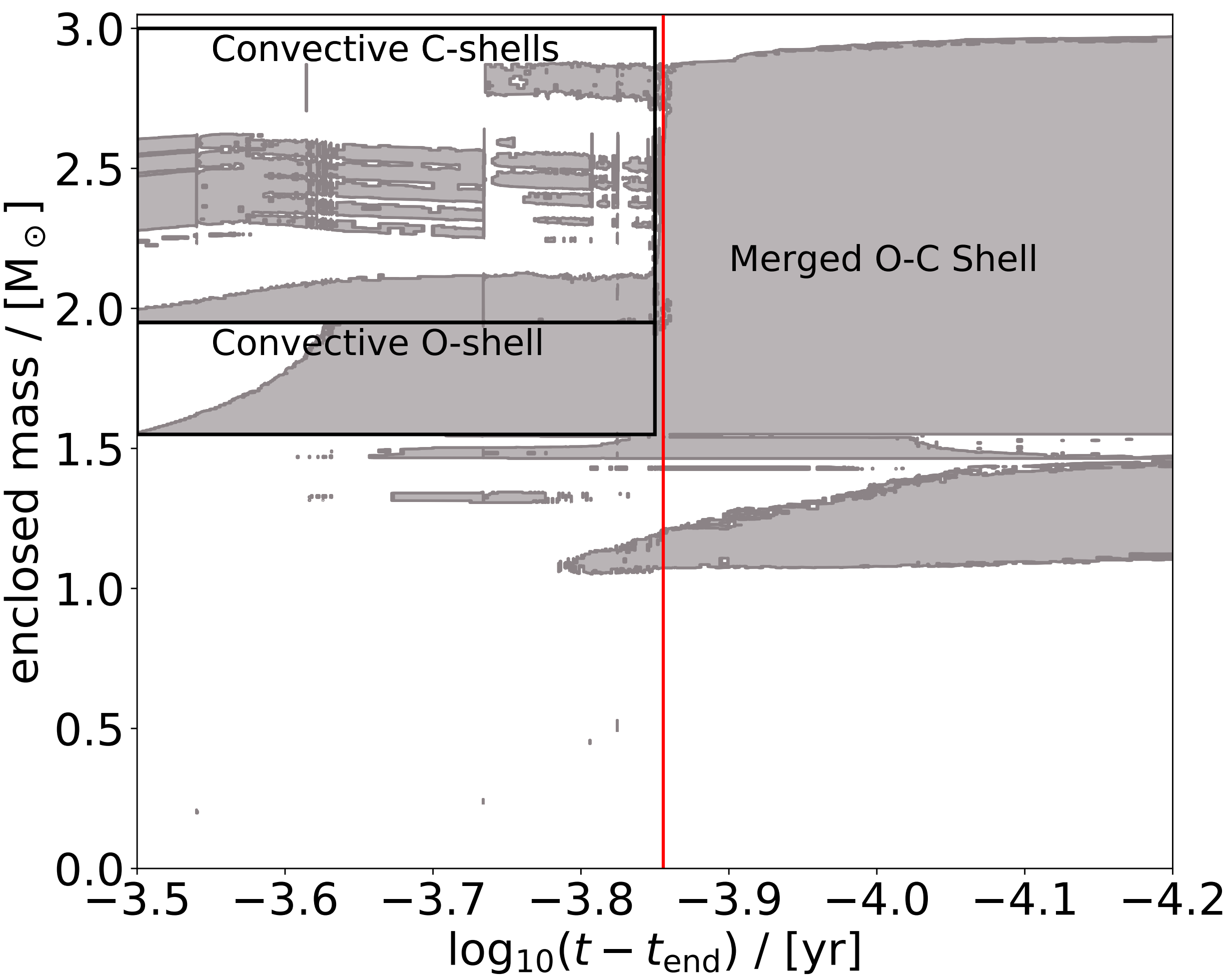

Figure 1.3: Kippenhahn diagram showing the merger of the convective O and C-burning shells for the 15 $M_\odot$ $Z = 0.02$ model from Ritter et al. (2018b). The O-burning shell extends from 1.55−1.95 $M_\odot$. The first convective C-burning shell sits directly on top from 1.96−2.11 $M_\odot$ and additional convective C-burning shells that are ingested are above. The merger onsets at $\log_{10}(t - t_{\mathrm{end}})/\mathrm{yr} \approx -3.85$ and reaches full extent at ≈ -4.

These mergers occur when the convective O- and C-shells grow large enough to overlap, forming a single extended convective region. These mergers are known to have peculiar markers of nucleosynthesis because the convective-reactive environment of the O-shell, such as the production of the odd-Z elements P, Cl, K, and Sc which are underproduced in galactic chemical evolution calculations (Ritter et al., 2018a; Roberti et al., 2025). In addition to these isotopes, the O-C shell merger has been a location for the production of the p-nuclei (Rauscher et al., 2002; Ritter et al., 2018a; Roberti et al., 2023). As C-shell material containing stable s- and r-process seeds is ingested into the much hotter O-shell temperatures, they undergo γ-process. The O-C shell merger nucleosynthesis has been found to dominate this production independent of the peak energy of the supernova explosion (Roberti et al., 2024).

O-C shell mergers have also been understood to be important for seeding asymmetries important for the supernova (Müller, 2016; Collins et al., 2018; Müller, 2020; Yadav et al., 2020; Andrassy et al., 2020). The shell mergers are highly asymmetric and provide perturbations necessary for the supernova explosion in the O-shell (Collins et al., 2018).

1.6 1D Stellar Modelling

Stars are 3D fluids that are governed by the Navier-Stokes equations where the fluid is undergoing a series of nuclear reactions. The hydrodynamical equations are given in Kippenhahn et al. (2013).

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0 \tag{1.1}$$

$$\rho \left(\frac{\partial \vec{v}}{\partial t} + (\vec{v} \cdot \nabla)\vec{v} \right) = -\nabla P - \rho \nabla \Phi \tag{1.2}$$

$$\frac{dE}{dt} = -\frac{P}{\rho}\nabla \cdot \vec{v} + \epsilon - \frac{1}{\rho}\nabla \cdot \vec{F} \tag{1.3}$$

$$\nabla^2 \Phi = 4\pi G \rho \tag{1.4}$$

where ρ is the density, $\mathbf{v}$ is the velocity, P is the pressure, Φ is the gravitational potential, E is the specific internal energy, ϵ is the energy generation rate, and $\vec{F}$ is the energy flux.

These equations are solvable in 3D for simulations, to couple them with a nuclear network becomes too computationally expensive to solve. Some studies are able to calculate both the hydrodynamical and nuclear netowrk equations together (Rizzuti et al., 2024b), but even involve limited nuclear networks. Additionally, 3D simulations are prohibited by the computational cost of evolving a star through its entire evolution for the amount of time necessary. Because of this, it is necessary to have a 1D approximation to be able to study the whole of stellar evolution.

1D stellar modelling requires the assumption of spherical symmetry. Under this assumption, all variable become functions of the radial coordinate r and time t. However, it is typical in stellar evolution to consider a mass coordinate m rather than the radial coorinate r. This means that all of our stellar evolution variables can be written as a function of mass and time.

$$\rho = \rho(m, t) \quad P = P(m, t) \quad T = T(m, t) \quad L = L(m, t) \quad r = r(m, t)$$

We can derive each of these quantities relevant for stellar structure. First, radius is simplest as it can be determined by writing mass in terms of a spherical density $m = \frac{4}{3}\pi r^3 \rho$, differentiating, and rearranging.

$$\frac{\partial r}{\partial m} = \left(4\pi r^2 \rho(m, t)\right)^{-1} \tag{1.5}$$

Next, let us impose a fundamental assumption in 1D stellar modelling: hydrostatic equilibrium. Under this assumption, the mass shells of the star are static so that the star does not grow or shrink in time because pressure and gravitational force are in equilbrium which would mean $\vec{v} = 0$. Taking both spherical symmetry and hydrostatic equilibrium, Equations 1.2 and 1.4 reduce to

$$\frac{\partial P}{\partial r} + \rho \frac{\partial \Phi}{\partial r} = 0$$

$$\frac{1}{r^2} \frac{d}{dr}\left(r^2 \frac{d\Phi}{dr}\right) = 4\pi G \rho$$

We can use the final equation here to get an explicit form of $\frac{\partial \Phi}{\partial r}$ by integrating and using the relationship between r and m to get that

$$\frac{\partial \Phi}{\partial r} = \frac{Gm}{r^2}$$

Using the form of $\frac{\partial \Phi}{\partial r}$ with Equation 1.5 we can rewrite the pressure equation to get

$$\frac{\partial P}{\partial m} = -\frac{Gm}{4\pi r^4} \tag{1.6}$$

Let's now considering how to get the luminosity in a similar form. Applying spherical symmetry, hydrostatic equilibrium, the relationship $L = 4\pi r^2 F$, and Equation 1.5 to Equation 1.3 we find with some rearranging that

$$\frac{dL}{dm} = \varepsilon - \frac{dE}{dt}$$

ϵ in stars comes from nuclear reactions that generate energy and lose energy from neutrinos and so can be written as $\epsilon = \varepsilon_{\text{nuc}} - \varepsilon_\nu$. The energy E can be derived from therodynamics and when fully written out the evolution of luminosity following Equation 10.3 in Kippenhahn et al. (2013) is

$$\frac{\partial L}{\partial m} = \varepsilon_{\text{nuc}} - \varepsilon_\nu - c_P \frac{\partial T}{\partial t} + \frac{\delta}{\rho}\frac{\partial P}{\partial t} \tag{1.7}$$

Consider now the temperature of the star. Let us assert that there is a generic temperature gradient to the star

$$\nabla \equiv \frac{T}{P}\frac{dP}{dT} = \frac{T}{P}\frac{\partial P}{\partial m} \cdot \frac{\partial m}{\partial T}$$

Using Equation 1.6, we can rearrange and find that

$$\frac{\partial T}{\partial m} = -\frac{GmT}{4\pi r^4 P}\nabla \tag{1.8}$$

The temperature gradient that we assert describes the transport of energy by either radiation or adiabatic movement of fluid elements and can be described by $\nabla = \nabla_{\text{rad}} + \nabla_{\text{ad}}$. The steeper the gradient, the more difficult it is for energy to be transported from r_1 to r_2.

Let us consider the scenario where energy is being more efficiently transported by photons ($\nabla_{\text{rad}} < \nabla_{\text{ad}}$). We can derive an explicit form for the temperature gradient from first principles following the derivation in Chapter 5 of Kippenhahn et al. (2013). First, suppose that we have photons travelling in a medium that have some mean free path that they travel before interacting. It is clear that the only interaction inside of the star would be with the plasma. The more dense the star is, the shorter the mean free path will be, hence

$$\ell_{\text{mfp}} = (\kappa\rho)^{-1}$$

where ρ is the density of the star at r and κ is the opacity which by dimensional analysis has units of cm^2 g^{-1}.

Assuming that we are in local thermal equilibrium and that the mean free path is much smaller than the size of the star, we can assume that we are able to describe the motions of the photons by a diffusive process. Generically, in diffusive processes the flux can be described by

$$\mathbf{j} = -D\ \nabla n$$

where n is the particle density and $D = \frac{1}{3}v\ell$ is the diffusion coefficient where v is the mean velocity and ℓ is the mean free path of those particles.

Since we are describing photons and not particles, let us consider the energy density $U = aT^4$, where a is the radiation constant, rather than n and its flux F rather than $\mathbf{j}$. This also means that the velocity $v = c$ and that the mean free path $\ell = (\kappa\rho)^{-1}$. Putting these together we find that

$$F = -\frac{1}{3}\frac{c}{\kappa\rho}\ \nabla(aT^4) = -\frac{4ac}{3}\frac{T^3}{\kappa\rho}\frac{\partial T}{\partial r}$$

Knowing that the flux can be described by $F = 4\pi r^2 L$, we can rearrange our formula to find that the radiative temperature gradient is given by

$$\nabla_{\rm rad} = \left.\frac{\partial T}{\partial r}\right|_{\rm rad} = -\frac{3}{16\pi ac}\frac{\kappa\rho L}{r^2T^3} \tag{1.9}$$

Let us now consider the scenario where energy is transported mostly efficiently by the adibatic motions of fluid elements ($\nabla_{\rm rad} > \nabla_{\rm ad}$). This would describe the scenario in a convective region where warmer fluid elements are rising to the top and cooler elements are sinking to the bottom. Assuming that the fluid in stars can be described by the ideal gas law, we can find that

$$P = \frac{\mathscr{R}}{\mu}\rho T \to \ln P = \ln\rho + \ln T + \ln\left(\frac{\mathscr{R}}{\mu}\right) \to \frac{d\ln P}{dr} = \frac{d\ln\rho}{dr} + \frac{d\ln T}{dr}$$

Assuming that the gas behaves adiabatically, following Chapter 19 of Kippenhahn et al. (2013) we know that

$$P \propto \rho^\gamma \to \frac{d\ln P}{dr} = \gamma\frac{d\ln\rho}{dr}$$

where γ is the polytopic index. We can insert the relationship for $d\ln\rho$ into the ideal gas law to find that

$$\frac{d\ln P}{d\ln r} = \frac{1}{\gamma}\frac{d\ln P}{d\ln r} + \frac{d\ln T}{dr} \to \frac{d\ln T}{dr} = \left(1 - \frac{1}{\gamma}\right)\frac{d\ln P}{d\ln r}$$

The generic form $\frac{d\ln A}{dx} = \frac{1}{A}\frac{dA}{dx}$ and so we can rewrite this as

$$\nabla_{\mathrm{ad}} = \left.\frac{\partial T}{\partial r}\right|_{\mathrm{ad}} = \left(1 - \frac{1}{\gamma}\right)\frac{T}{P}\frac{dP}{dr} \tag{1.10}$$

In the radiative transport case, the only quantity that evolves is the temperature as photons are moving around, but in the adiabatic transport case fluid elements that have their own unique chemical composition μ in addition to their own temperature are moving around. Therefore, in the radiative case only the temperature is equilibriated, but in the adiabatic case temperature and chemical composition are equilibriated. Additionally, for the motions of a fluid to be adiabatic, it is assumed that it has a constant entropy gradient. If a region was radiative but slowly becomes adiabatic, it must be the case that the entropy gradient also equilibriates as the region forms.

Stellar modelling is also concerned with the evolution of the chemical composition of the star. The chemical composition in a star can be described by the both the nuclear reactions that occur to change the composition and the motions to mix the composition. The following follows the lecture content of Herwig (2023) and description by Langer et al. (1985).

The mass fraction of an individual isotope is given as X_i where the sum of all isotopes $\sum_i X_i = 1$, and is related to the mole fraction $Y = X/A$ where A is the atomic mass of the isotope. The energy generation of a star comes from the nuclear reactions that occur in the star and can be described by

$$\left.\frac{d\mathbf{X}_j}{dt}\right|_{\mathrm{burn}} = \hat{F}_j \cdot \mathbf{X}_j \tag{1.11}$$

where $\mathbf{X}_j$ is the vector of mass fractions at some position j and $\hat{F}_j$ are the nuclear reactions that the isotopes undergo. These reactions are typically written in number density as opposed to mass fraction which is given by $N_i = Y_i \rho N_A$ where N_A is the Avogadro number and $Y_i = X_i/A_i$ is the mole fraction and A_i is the mass number.

$$\frac{dN_i}{dt} = N_l N_k \langle\sigma v\rangle_{lk} - N_i N_j \langle\sigma v\rangle_{ij} + \ldots + N_n \lambda_n - N_p \lambda_p \tag{1.12}$$

where $\langle\sigma v\rangle_{ij}$ is the average of the product of the cross section and relative velocities in the centre of mass frame and λ is rate of β decays.

The mixing of isotopes in the star is described by the diffusion equation:

$$\left.\frac{dX_i}{dt}\right|_{\text{mix}} = \frac{\partial}{\partial m}\left[(4\pi r^2 \rho)^2 D \frac{dX_i}{dm}\right] \tag{1.13}$$

where X_i is the mass fraction of a particular isotope.

1.6.1 Mixing Length Theory Approximation

Prandtl (1925) was first to describe the turbulent flow of water through a channel by considering a characteristic length scale the eddies would travel before mixing with their surroundings. Following his insight, Böhm-Vitense (1958) developed the first stellar model using the idea of a characteristic length scale to describe the convective motions of the Sun.

Turbulent motions are complex 3D flows involving eddies of different shapes, velocities, and distances they travel before mixing with their environment. Rather than describing the flow of every individual fluid element, it models the average distance each fluid element travels, ℓ, before mixing with its surroundings. It also assumes that the average size of the fluid elements is equal to the travel distance ℓ. This is the "mixing length theory" (MLT) approximation. Fluid elements also have other average properties such as the velocity $\bar{v}$ they travel at and therefore an average timescale they travel for before becoming mixed with their surroundings $\tau = \ell/\bar{v}$ (Kippenhahn et al., 2013; Cox and Giuli, 1968; Joyce and Tayar, 2023). In stars, the average distance traveled ℓ is the mixing length ℓ_{MLT} and is given by:

$$\ell_{\text{MLT}} = \alpha_{\text{MLT}} \cdot H_P \tag{1.14}$$

where H_P is the pressure scale height and α_{MLT} is a dimensionless free parameter and can be chosen to match stellar evolution tracks to empirical data (Joyce and Tayar, 2023).

Prandtl (1925) used this assumption to describe the turbulent flows by a diffusion process, and this description of the motion is still the understood physical picture that MLT relies on (Joyce and Tayar, 2023). In the MLT approximation, the motion of fluid elements is characterized by a diffusion coefficient:

$$D_{\text{MLT}} = \frac{1}{3} v_{\text{conv}} \times \ell_{\text{MLT}} \tag{1.15}$$

where v_{conv} are the average convective velocities of the fluid elements and ℓ_{MLT} is the mixing length. The timescale can also be re-written in terms of this diffusion coefficient and the mixing length (Herwig et al., 2011):

$$\tau_{\mathrm{MLT}} = \ell_{\mathrm{MLT}}^2 / D_{\mathrm{MLT}} \tag{1.16}$$

1.6.2 Stellar Evolution with MESA

The Modules for Experiments in Stellar Astrophysics (MESA) code is an open-source 1D stellar evolution framework used in stellar astrophysics that allows the computation of the structure and evolution of stars from pre-main sequence to advanced stages of burning up to the core collapse (Paxton et al., 2010). One of MESA's distinguishing features is its implicit, fully coupled solution of the structure, nuclear burning, and mixing equations. The code employs a Newton-Raphson method to simultaneously solve the coupled set of equations of 1D stellar structure, diffusion of species, and nuclear reactions. The coupled equations allow MESA to resolve multi-timescale problems. In convective regions, MESA adopts a time-dependent diffusive mixing approximation, solving the mixing (subscript "mix") and nuclear burning (subscript "nuc") processes at the same time with operator coupling:

$$\left(\frac{dX_i}{dt}\right)\mathrm{total} = \left(\frac{dX_i}{dt}\right)_{\mathrm{nuc}} + \left(\frac{dX_i}{dt}\right)_{\mathrm{mix}}, \tag{1.17}$$

where X_i is the mass fraction of isotope i, and both terms are evaluated and advanced implicitly in a coupled fashion. Species mixing in MESA is taken to be diffusion (Paxton et al., 2010). MESA can include nuclear networks of hundreds of isotopes. However, solving large coupled equations takes a lot of computational power, and so stellar evolution calculations typically use reduced networks to capture the reactions that are relevant to energy generation without tracking full nucleosynthetic yields.

1.6.3 Post-Processing with mppnp

To obtain detailed nucleosynthesis predictions, it is necessary to post-process stellar models. One such code is multi-zone post-processing network parallel (mppnp) created by the NuGrid collaboration[1]. mppnp is a parallelized multi-zone code that takes as input an already evolved stellar structure and species from MESA and calculates the full nucleosynthesis of thousands of isotopes and tens of thousands of reactions.

Unlike the coupled equations of MESA, mppnp adopts an operator-split approach where it solves nuclear reactions and mixing separately in alternating steps:

[1] Details about the NuGrid collaboration and their codes can be found at https://nugrid.github.io/.

1. The abundances X_i are updated via an implicit solver for nuclear reactions, using the supplied temperature and density histories.

2. The resulting abundances are then mixed by diffusion.

3. Optionally, an ingestion step can be included where a species is ingested at the top of region and mixed in.

By decoupling the nuclear physics from the stellar structure, `mppnp` is able to calculate large nuclear networks, but it also poses a problem for convective-reactive nucleosynthesis. In regular convective burning, D_α is much smaller than 1 as mixing occurs much faster than nuclear burning (and vice versa for radiative burning). In convective-reactive environments D_α is equal to 1, and since these timescales are near equal, the evolution of the mass fraction is coupled to both operators. Therefore, it is important when calculating the results of a convective-reactive environment, that the timestep is small enough that there is no significant change in the final mass fraction. This would mean that we have resolved both the mixing and burning timescales.

As shown in Figure 1.4, there is a noticable change in the final mass fraction of the isotopic abundances. While these lighter isotopes ^{12}C, ^{16}O, and ^{20}Ne are not drastically changed, the final production of heavier isotopes such as the p-nuclei can be orders of magnitude.

In addition to the importance for timescale resolution, there is also spatial resolution. Changing the number of mass zones in the simulation can change the final result as the mixing of the isotopes depends on the number of fluid elements that can be resolved. By ensuring that the number of mass zones is high enough, the mixing can be resolved to a point where the final result is not changed by the number of mass zones.

1.6.4 Convective-Reactive Nucleosynthesis

Typically in the stellar environment, the timescales for nuclear burning and convective mixing are vastly different. In a convective zone, the timescale for convective mixing is much shorter than the timescale for nuclear burning and in the radiative zone vice versa. To describe the relationship between the timescales of nuclear burning and convective mixing, we can use the Dämkohler number (Dimotakis, 2005):

$$D_\alpha = \frac{\tau_{\rm mix}}{\tau_{\rm burn}}. \tag{1.18}$$

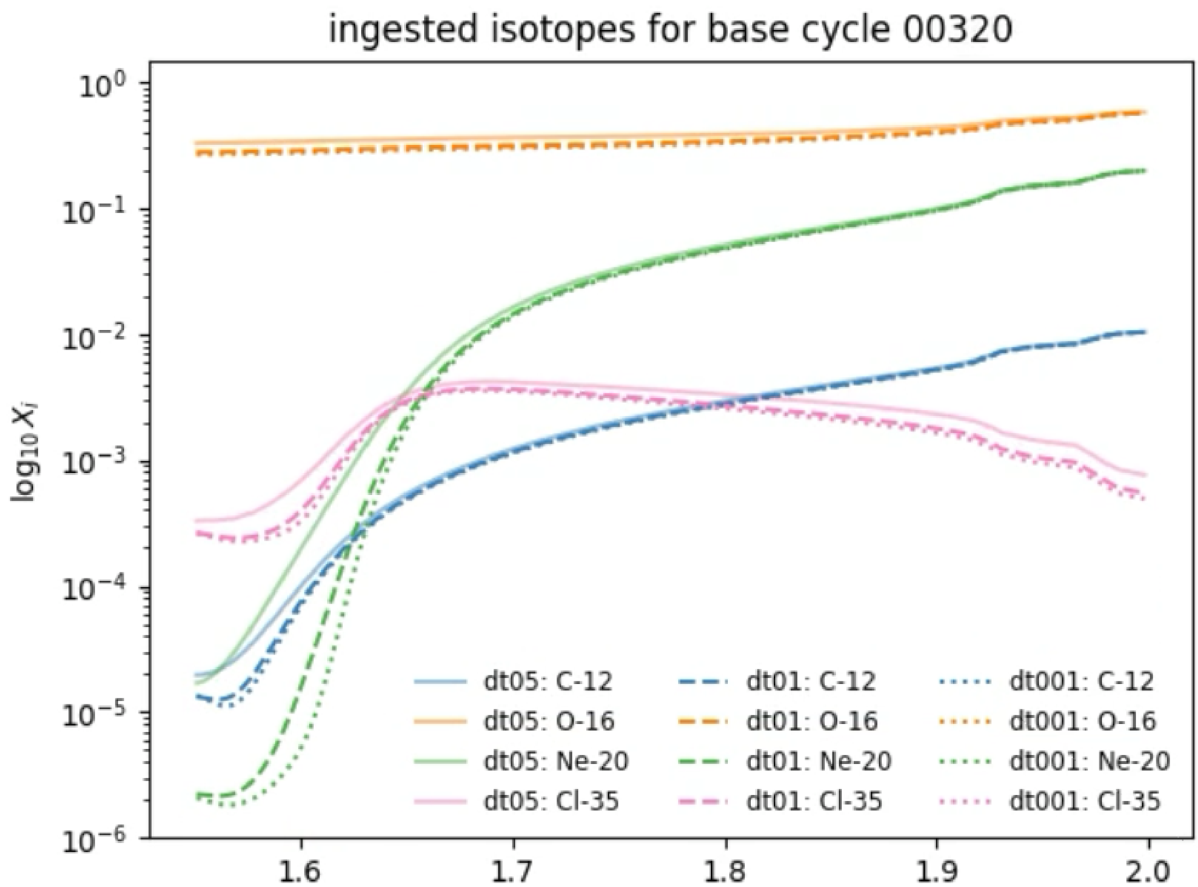

Figure 1.4: Evolution of the mass fraction of ^{12}C, ^{16}O, ^{20}Ne, and ^{35}Cl in a convective-reactive environment at the same simulation time with different timesteps. The solid line corresponds to a simulation with timestep of $\Delta t = 0.5$s, the dashed line corresponds to $\Delta t = 0.1$s, and the dotted line corresponds to $\Delta t = 0.01$s.

For the MLT scenario, $\tau_{\rm mix}$ is given by Equation 1.16 and $\tau_{\rm burn}$ is given by the timescale for nuclear burning, which can be approximated as the inverse of the energy generation rate:

$$\tau_{\rm react} = \sum_j \frac{1}{\rho N_A \langle \sigma v \rangle X_j / A_j} \tag{1.19}$$

where ρ is the density, N_A is Avogadro's number, $\langle \sigma v \rangle_j$ the reaction rates contributing to some species, X_j is the mass fraction of the reacting species, and A_j is the mass number of the reacting species.

Hydrostatic burning in convective regions is characterized by $D_\alpha \ll 1$. In this case, the mixing timescale is much shorter than the nuclear burning timescale, and the convective mixing is fast enough that the isotopes are mixed before they can undergo significant nuclear burning. Convective-reactive nucleosynthesis occurs when nuclear burning and convective mixing operate on comparable timescales, $D_\alpha = 1$ (Dimotakis, 2005; Herwig et al., 2011; Ritter et al., 2018b). In these conditions, the mass fractions of species are not well-mixed, and can undergo significant nuclear burning as they are being mixed. Reactions are not balanced across the region, and species are able to have different nucleosynthetic pathways depending on the temperature where $D_\alpha = 1$.

Herwig et al. (2011) found that in a convective-reactive environment, the MLT mixing timescale is not always appropriate for describing the mixing timescale. They propose a reaction scale height timescale using the generalized principles of scale heights (Chapman, 1961):

$$H_\phi = \left(\frac{d ln \phi}{dr} \right)^{-1} \tag{1.20}$$

where ϕ is the quantity of interest, in this case $\phi = \rho N_a \langle \sigma v \rangle$. The timescale associated with this scale height is given by (Herwig et al., 2011):

$$\tau_{\rm scale} = \frac{H_\phi^2}{D_{\rm MLT}}. \tag{1.21}$$

and there would likewise be a Dämkohler number for the scale height where $\tau_{\rm mix}$ is $\tau_{\rm scale}$.

Figure 1.5 shows that both ^{12}C and ^{20}Ne experience convective-reactive nucleosynthesis in the O-shell, but that ^{16}O does not. This is because $D_\alpha = \tau_{\rm MLT}/\tau_{\rm react} = 1$ and $D_\alpha = \tau_{\rm MLT}/\tau_{\rm scale} = 1$ for ^{12}C and ^{20}Ne but $D_\alpha < 1$ for ^{16}O. The principal species ingested into the shell will experience convective-reactive nucleosynthesis.

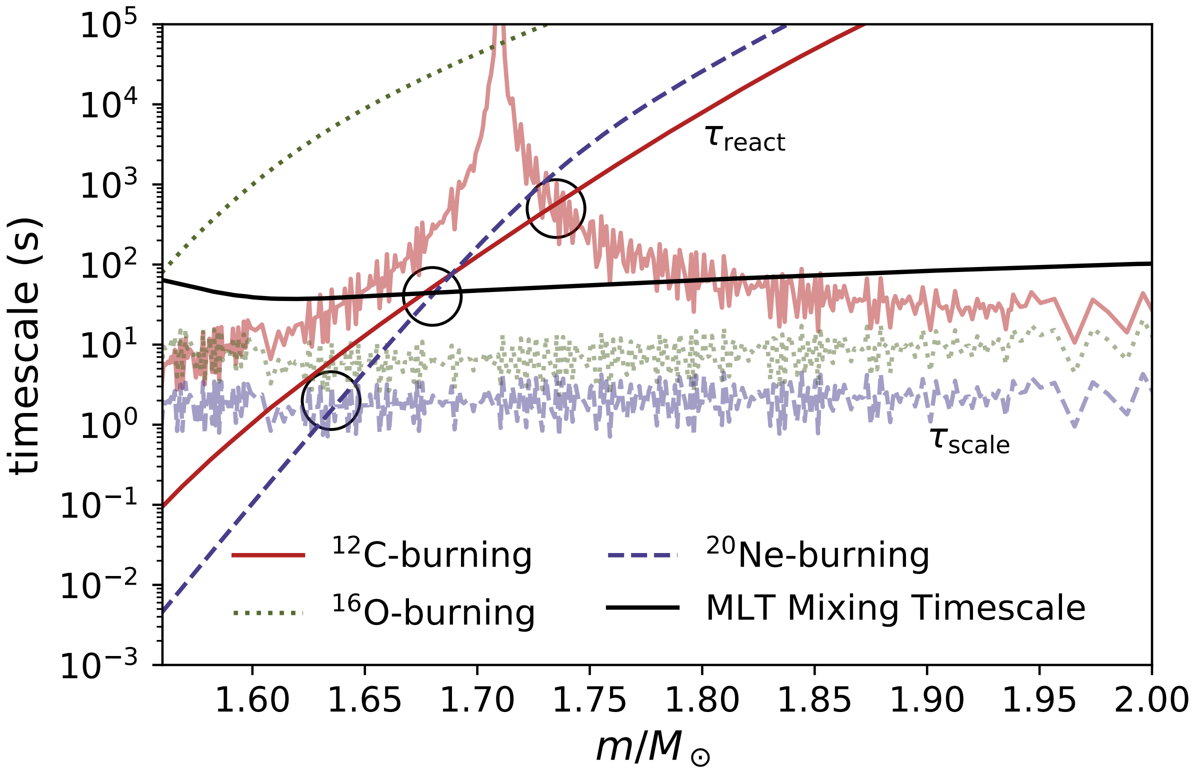

Figure 1.5: $\tau_{\rm MLT}$, $\tau_{\rm react}$, and $\tau_{\rm scale}$ for the 15 $\rm M_\odot$ $Z = 0.02$ stellar model from Ritter et al. (2018b) at $\log_{10}(t - t_{\rm end}) \approx -3.85$ (model number 9200). The black line is $\tau_{\rm MLT}$. The solid red line is $\tau_{\rm react}$ for ^{12}C, and the jagged red line is $\tau_{\rm scale}$ for ^{12}C. Likewise, the green lines are $\tau_{\rm react}$ and $\tau_{\rm scale}$ for ^{16}O, and the purple lines are $\tau_{\rm react}$ and $\tau_{\rm scale}$ for ^{20}Ne. Black circles are provided where $D_\alpha = 1$ for ^{12}C and ^{20}Ne.

1.7 3D Macrophysics

Despite the value of 1D stellar evolution codes, stars are 3D objects that exhibit non-spherical features. The simplifications made by spherical symmetry and the MLT approximation do not sufficiently capture the 3D physics of convection. Because of this, there are modelling inconsistencies between 1D MLT predictions and 3D hydrodynamic simulations.

1.7.1 Insufficiencies of MLT

As explained before, MLT assumes that there is some average length ℓ_{MLT} that all fluid elements travel in a convective environment before mixing with their surroundings.

One of the problems with MLT is how it treats the the boundary between a convective region and a radiative region. As explained in Section 1.6 as defined by Equations 1.9 and 1.10, convective regions are described by $\nabla_{\mathrm{ad}} < \nabla_{\mathrm{rad}}$ and radiative regions where $\nabla_{\mathrm{ad}} > \nabla_{\mathrm{rad}}$. The boundary between these two regions is defined by the Schwarzschild criterion where $\nabla_{\mathrm{ad}} = \nabla_{\mathrm{rad}}$. According to Equation 1.15, the diffusive coefficient for mixing is equal to the convective velocities. Outside the boundary of a convective region, suddenly $v_{\mathrm{conv}} = 0$ which would imply an unphysical infinite acceleration to stop the fluid at the edges. To more accurately treat the boundaries, stellar evolution codes like `MESA` implement a convective overshooting parameterization where

$$D_{\mathrm{OV}} = D_{\mathrm{conv},0} \times \exp\left(-\frac{2z}{f\lambda_{P,0}}\right) \tag{1.22}$$

where $D_{\mathrm{conv},0}$ is the diffusion coefficient at the boundary, $\lambda_{P,0}$ is the pressure scale height at that location, z is the position in the radiative zone, and f is a free parameter so that the diffusion drops off exponentially rather than immediately (Paxton et al., 2010).

As described in Renzini (1987), there are some fundamental issues in using MLT to describe a convective region. One big issue is that the mixing length is assumed to be both the average distance a fluid element travels and its size. If the convective region is smaller than the mixing length, then it cannot resolve the mixing since it is smaller than a single fluid element. This applies to both convective regions and the overshooting region, which is a fraction of the pressure scale height. Therefore, for MLT to be a valid description, ℓ_{MLT} must be sufficiently small.

As recognized by Bazan and Arnett (1994), the O-burning shell cannot adequately be described by MLT. For example, Figure 1.6 shows the $15\mathrm{M}_{\odot}$ $Z = 0.02$ model from Ritter

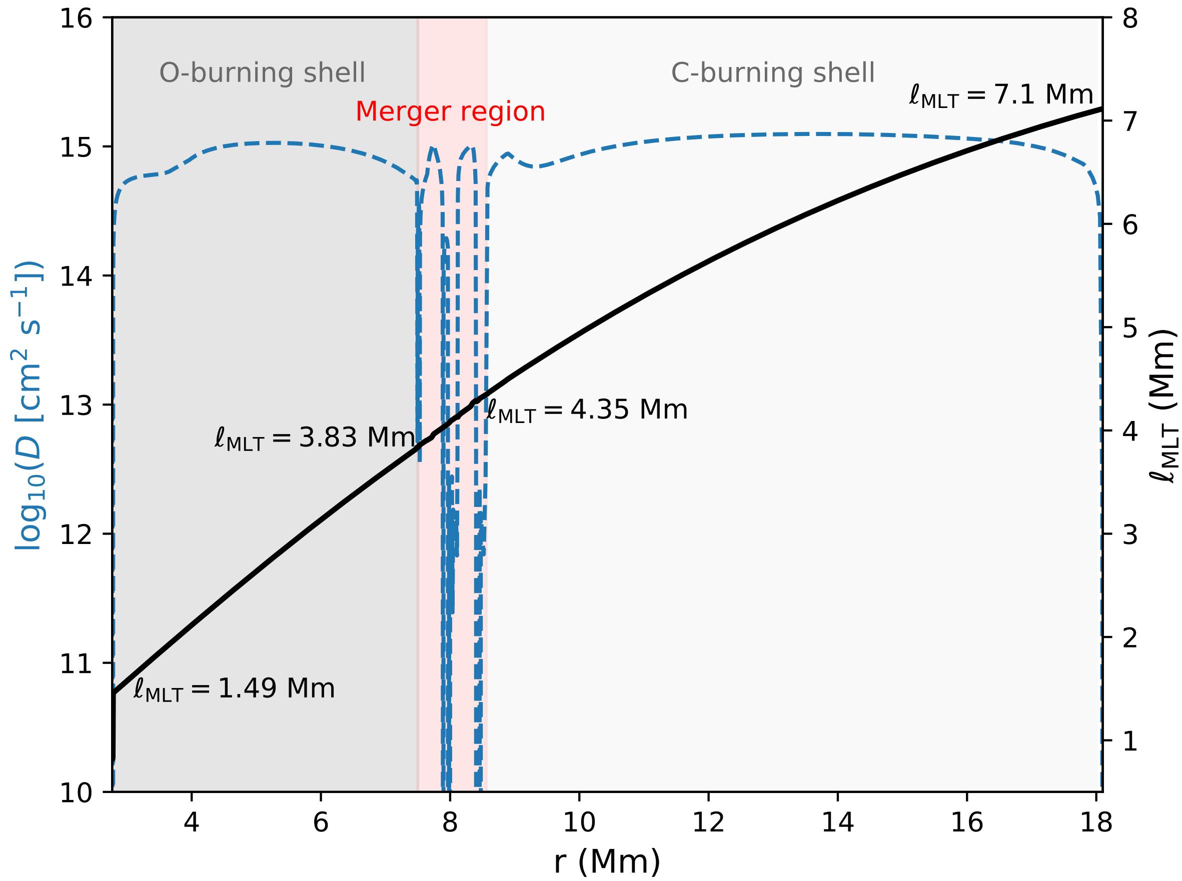

Figure 1.6: O- and C-burning shells in the $15\mathrm{M}_{\odot}$ $Z = 0.02$ model from Ritter et al. (2018b) at model number 9200 right before the merger with the C-shell is fully realized. The left y-axis shows the diffusion coefficient in blue, and the right y-axis shows the mixing length in black. The O-shell is from $2.79{-}7.49 \times 10^6$ m, the merger region is from $7.49{-}8.56 \times 10^6$ m, and the C-burning shell is from $8.56{-}18 \times 10^6$ m.

et al. (2018b) as it is merging. The O-burning shell has a size of 4.7×10^6 m, merging region 1.07×10^6 m, and C-burning shell 9.44×10^6 m. Since the fluid element in MLT is assumed to be the same size as the mixing length, it can be estimated that each region can resolve 3.2, 0.3, and 2.2 fluid elements respectively by dividing the lowest mixing length by the size of the region in Figure 1.6. Clearly, these regions are not well resolved by MLT, especially the region where the O- and C-shells are merging.

1.7.2 3D Simulations of Convective O-Shell Burning and C-Shell Entrainment

Multidimensional hydrodynamic simulations have been done of O-shell burning and C-shell entrainment in massive stars. Bazan and Arnett (1994) performed a 2D simulation of O-burning where they show that the convective motions and temperature are non-radially, spherically asymmetric and mix through radiative layers by a penetrative convection rather than a simple overshoot.

Meakin and Arnett (2006) performed a 2D simulation of both the O- and C-burning shells with a limited nuclear network and compared to a 3D simulation of just the O-shell. They found that the non-convective region between the two shells had comparable velocities to the convective shells and that the convective shells have large, round vortices and asymmetric density perturbations. Additionally, they found that the convective motions excited waves in the radiative layer that allowed for C-shell entrainment into the O-shell at $\sim 10^{-4} \mathrm{M}_{\odot} s^{-1}$. Because of this entrainment, they found there would be sudden O-flames burning in the O-shell that would turn on and off. However, they note that their 3D O-shell simulation show plumes rather than vortices to describe the flow and caution against using 2D simulations to describe the convective motions in this scenario.

Meakin and Arnett (2007) also find this mixing difference between 2D and 3D simulation of O-shells, and they also find that the 3D radial velocity decreases gradually at the top of the shell and sharply at the bottom, which cannot be explained by a constant α_{MLT}. This is explained by the distance to the convective boundary acting as an upper limit to the mixing length, as fluid elements cannot travel a full mixing length if the remaining space they have is less than it. They also find turbulent entrainment of material from the stable layer above the O-shell.

Jones et al. (2017) in their 3D simulation of O-shell burning find that the 3D radial velocity profile exhibits the same gradual decrease at the upper boundary and find an entrainment from the stable layer of $1.33 \times 10^{-6} \mathrm{M}_{\odot} s^{-1}$. As they show in Figure 1.7, the diffusion coefficient from MLT does not match the shape of what is estimated in 3D and underestimate

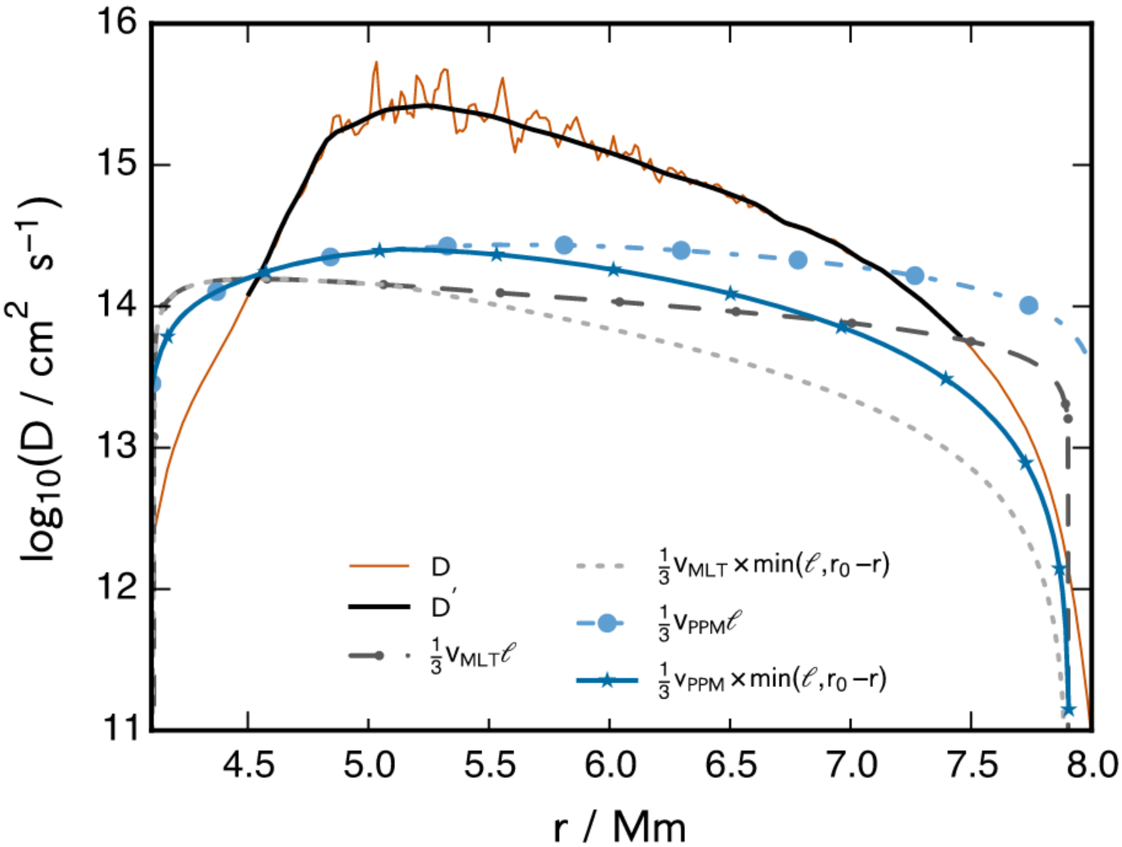

Figure 1.7: Taken from Jones et al. (2017): Time-averaged radial diffusion coefficient profile calculated from the spherically averaged abundance profiles by the method described in Section 3.5 (brown solid line; black solid line is a fit to the noisy region). The convective velocities computing using MLT agree with the spherically averaged 3D velocities to within about a factor of 2 inside the convection zone but are too large in the vicinity of the convective boundary, resulting in an overestimation of the diffusion coefficient there. Limiting the mixing length to the distance from the convective boundary reproduces the fall-off of the diffusion coefficient inside the convection zone approaching the boundary which is seen in the spherically averaged 3D simulation results.

velocities. However, they provide a recommendation for 1D modelling to better match the 3D diffusion profile shape:

$$D = v_{\rm MLT} \times \min(\alpha H_{\rm P}, |r - r_{\rm SC}|) \tag{1.23}$$

where $\alpha H_{\rm P} = \ell$ is the mixing length, r is the radius in the convective region, $r_{\rm SC}$ is the distance to the convective boundary, and $v_{\rm MLT}$ are the velocities predicted by MLT. This is rooted in the same insight by Meakin and Arnett (2007) that fluid elements may be too close to the convective boundary to move a full mixing length. In addition to this, they apply the same insight of distance to convective boundary to convective overshooting and find good agreement with the spherically average 3D diffusion coefficient.

Arnett et al. (2019) follows the critiques of Renzini (1987) by listing the many issues with the assumptions made by MLT and comparing the results of 3D hydrodynamic simulations to MLT. Among other things, they find that 3D simulations have a gradual decrease in the radial velocity profile as it reaches the bottom of their shell, and that there is entrainment of material from the stable layers. These are behaviours which cannot be captured by MLT.

Yadav et al. (2020) in their 3D simulation of an O-Ne shell merger find that 3D simulations have much faster convective velocities than predicted in 1D, and as the Ne-shell merges strong plumes carry the material to the bottom of the O-shell where burn in a convective-reactive environment. The asymmetric plume distribution leads to local hotspots of burning causing as difference in the Si, O, and Ne mass fractions when compared to 1D.

Andrassy et al. (2020) performed 3D simulation of C-ingestion into a convective O-shell and found a range of possible ingestion rates from $9.5 \times 10^{-8} \mathrm{M}_{\odot} s^{-1}$ to $1.18 \times 10^{-8} \mathrm{M}_{\odot} s^{-4}$ and convective velocities of 38 km s^{-1} to 181 km s^{-1} depending on the luminosity of the $^{12}\mathrm{C} +^{12} \mathrm{C}$, $^{16}\mathrm{O} +^{16} \mathrm{O}$, and $^{12}\mathrm{C} +^{16} \mathrm{O}$ reactions. In their simulations they find large scale asymmetries in the luminosity and density during entrainment. The radial velocity profile they find exhibits a gradual decline at the top of the O-shell, but also a less steep decline at the bottom of the shell. In addition to this, they boost the energy of C-burning and find large scale asymmetric oscillations like the Global Oscillation of Shell H-ingestion (GOSH) as found in Herwig et al. (2014). GOSHs can lead to increased entrainment of material, but also quench convective motions.

Rizzuti et al. (2024a) performed a 3D hydrodynamic simulation of an O-C shell merger with a limited nuclear network and find significant differences from their 1D model. In the 1D model, the top convective boundary of the O-shell grows until it merges with the Ne/C-shell above it, but in the 3D simulation the bottom of the C-shell grows until it merges with the Ne and O-shells under it. Like Jones et al. (2017), they find that 1D convective velocities are an order of magnitude lower than the 3D simulation. Because of the differences in 1D and 3D they find that there are significant differences in the mass fraction of species like ^{20}Ne, ^{24}Mg, and ^{28}Si.

1.7.3 1D Implementation of 3D Hydrodynamics

Despite the insufficiencies of 1D stellar evolution, 3D hydrodynamic codes are unable to calculate the nucleosynthesis for complex processes such as the γ-process where thousands of isotopes can be involved. Nucleosynthetic yields have to be calculated then in 1D, but the insufficiencies of 1D can be mitigated by adopting the results of 3D hydrodynamics and testing a variety of mixing scenarios. Specific to O-shell burning in O-C shell mergers, the following steps can be taken to investigate in 1D how macrophysical uncertainties impact nucleosynthesis:

- The shape of the radially averaged 3D diffusion coefficient profile exhibits a downturn at the boundaries of the O-shell that is not seen in 1D (Meakin and Arnett, 2007; Jones et al., 2017).
- Convective velocities are higher in 3D hydrodynamic simulations than predicted by mixing length theory (Jones et al., 2017; Rizzuti et al., 2024a).
- The entrainment rate of C-shell material into the O-shell could be anywhere from 10^{-8} M$_\odot$s^{-1} to 10^{-4} M$_\odot$s^{-1} (Andrassy et al., 2020).
- Asymmetric non-radial instabilities are present due to energy feedback which could lead to convective quenching (Andrassy et al., 2020).

To address the shape a gradual downturn to D_{conv} Equation 1.15 from Jones et al. (2017) is adopted, but at the bottom of the O-shell instead of the top. This is done because we are simulating the nucleosynthesis during an O-C shell merger, and so the upper convective boundary is already connected to the C-shell above. To address lower convective velocities in 1D, a series of boost factors can be applied to the diffusion coefficient. To address the variability of C-shell entrainment, a variety of entrainment rates can be adopted in a 1D post-processed model. Finally, to address energy feedback such as the GOSH-like events seen in Andrassy et al. (2020), a simple dip in the convective profile can be adopted:

$$D_{\mathrm{dip}} = D_{\mathrm{MLT}} - (D_{\mathrm{MLT}} - c) \times \exp\left[-\frac{(r-a)^2}{w^2}\right] \tag{1.24}$$

where c is the depth of the dip, a is the location of the dip, w is width of the dip, and r is the radius in the region. All variables are in 10^6 m. Using the results of 3D hydrodynamic simulations, we can explore the possible impact on nucleosynthesis from macrophysics.

1.8 Thesis Outline

Chapter 2: Impact of 3D-macrophysics and nuclear physics on p-nuclei nucleosynthesis in O-C shell mergers

In Chapter 2, I describe the impact of a 1D implementation of 3D hydrodynamic concerns on the production of the p nuclei. I explore how the nature of convective-reactive affects the nucleosynthesis of the p nuclei, how the mixing concerns mentioned in Section 1.7.3. In addition to this, I present the results of a nuclear reaction correlation study and investigate how the different mixing scenarios have different reaction rates correlated with the production of p nuclei. I also provide a discussion where I explain the potential impacts and limitations of this work.

Chapter 3: Additional Work

In Chapter 3, I describe how this work can continue to be applied to the production of the light odd-Z isotopes. In particular, I highlight how the isotopes ^{39}K, ^{40}K, and ^{41}K are impacted by these macrophysical uncertainties. This has relevance to whether exoplanets could form habitable atmospheres, as the decay of the radionuclide ^{40}K is a significant heating source in early planet formation (Frank et al., 2014; O'Neill et al., 2020). I also provide some preliminary work investigating whether advective-reactive nucleosynthesis matters for the r-process in the post-processing of trajectories that have escaped a black hole.

Chapter 2

Impact of 3D-macrophysics and nuclear physics on p-nuclei nucleosynthesis in O–C shell mergers

The following work has been written and prepared for submission to peer-review. This paper was written in collaboration with my supervisor Dr. Falk Herwig, Dr. Pavel Denissenkov, and Dr. Marco Pignatari. Drs. Herwig and Pignatari provided comments on the manuscript and editing feedback. Dr. Denissenkov supplied the initial code used to create the post-processed models, analysis tools, and training necessary to understand how to use the code. I have modified this code and created new tools to perform the analysis, but his contributions were essential to the inital stages. Dr. Herwig has also provided guidance and support throughout the project in his role as my supervisor.

2.1 Abstract

O-C shell mergers in massive stars are a site for producing the p nuclei by the γ process, but 1D stellar models rely on mixing length theory, which does not reproduce the velocity profiles found in 3D hydrodynamic simulations. We investigate how 3D macro physics informed mixing impacts the nucleosynthesis of p nuclei. We post-process the O-shell of the M_{ZAMS} =15 $\mathrm{M}_{\odot}$, Z = 0.02 model from the NuGrid stellar data set (Ritter et al., 2018b). Applying a downturn to velocities at the boundary and increasing velocities across the shell (Jones et al., 2017), we find non-linear, non-monotonic increase in p-nuclei production with a spread of 0.96 dex, and find that isotopic ratios can change. Reducing C-shell ingestion rates (Andrassy et al., 2020) suppresses production, with spreads of 1.22−1.84 dex across MLT and downturn scenarios. Applying dips to the diffusion profile to mimic quenching events (Andrassy et al., 2020) also suppresses production, with a 0.51 dex spread. We also analyze the impact of varying all photo-disintegration rates of unstable n-deficient isotopes from Se–Po by a factor of 10 up and down. The nuclear physics variations for the MLT and

downturn cases have a spread of 0.56−0.78 dex. We also provide which reaction rates are correlated with the p nuclei, and find few correlations shared between mixing scenarios. Our results demonstrate that uncertainties in mixing arising from uncertain 3D macro physics are as significant as nuclear physics and are critical for understanding p-nuclei production during O-C shell mergers quantitatively.

2.2 Introduction

The origin of the solar pattern of the 35 stable p nuclei has been a long-standing problem in nucleosynthesis. Burbidge et al. (1957) first identified these isotopes as a distinct group that were not produced by the s or r process, but instead the p process by (p, γ), (n, γ), and (γ, n) reactions during a Type II supernova. Since then, a variety of processes and astrophysical sites have been discussed, but no single mechanism has been identified for the production of all p nuclei. Woosley and Hoffman (1992) found that ^{74}Se–^{92}Mo could be produced during the α-rich freezeout of a supernova. Fröhlich et al. (2006) and Arcones and Montes (2011) found high neutrino fluxes during a supernova can create ^{74}Se–^{108}Cd by (n, p), (n, γ), and (p, γ) reactions in the νp process. Schatz et al. (1998) suggest that a hydrogen-rich accretion disk around a neutron star could undergo a series of rapid proton captures in the rp process to produce ^{74}Se–^{98}Ru. Xiong et al. (2024) proposed that neutrino induced reactions of r-process material in a νr process could produce ^{78}Kr–^{138}La in the winds of a proto-neutron star. Goriely et al. (2002) proposed that a proton-poor and neutron boosted region could undergo proton-captures could produce all p nuclei by the pn process during He-detonation of a C-O white dwarf's ejected envelope. Woosley and Howard (1978) argued that all of the p nuclei could be produced by a series of (γ, n) and (γ, α) reactions on the s- and r process seeds in the γ processduring explosive C- or O-shell burning, and Arnould (1976) argued that hydrostatic O-burning could also be a site. Rauscher et al. (2002) found that the γ process could produce the p nuclei in massive stars during the core collapse supernova (CCSN), but also prior if the O-shell merges the C-shell, but that it underproduced ^{92,94}Mo and ^{96,98}Ru. Ritter et al. (2018a) found that the p nuclei could be produced during an O-C shell merger. The γ process describes the flow of (γ, n), (γ, p), and (γ, α) reactions on the stable isotopic seeds that are already present in the shell at temperatures of $1.5 \leq T \leq 3 \times 10^9$ K (Rauscher et al., 2013). However, the γ process in massive stars underproduces not only the Mo and Ru, but all p nuclei with $A = 90 - 130$ (Arnould and Goriely, 2003; Woosley and Heger, 2007) and Woosley and Heger (2007). Travaglio et al. (2011) found that the γ-process could produce all p nuclei of s-process material in a Type Ia supernova, without the underproduction of

^{92,94}Mo and ^{96,98}Ru. Travaglio et al. (2015) found that modifying the distribution of s-process material significantly influenced the production of the p nuclei, especially the heaviest ones, and that the lightest three were metallicity dependent. Battino et al. (2020) found that the H-flashes of rapidly accreting white dwarfs undergo the i process which modifies the seed distribution to produce p nuclei with $96 < A < 196$ by the γ process during the subsequent SNIa. Pignatari et al. (2016a) argue that in massive stars the weak s process can modify the seed distribution to boost production of the p nuclei by the γ process during the CCSN. The initial seed distribution is important for the γ process as it sets the initial conditions for photo-disintegration and subsequent nucleosynthesis pathways.

Since Burbidge et al. (1957), it has been found that not all p nuclei in the solar pattern are produced by a single process. Bisterzo et al. (2011) found that ^{152}Gd, ^{164}Er, and ^{180}Ta have significant contributions of 70.5%, 75.5%, and 74.5% from the s process. Dillmann et al. (2008) argue that ^{113}In and ^{115}Sn are made by β-decays after the r process through isomeric states. Goriely et al. (2001) argue that (γ, n) was too weak to produce ^{138}La, and instead that it is made by ν_e-capture on ^{138}Ba during the CCSN, and Arnould and Goriely (2003) similarly argue $^{180\mathrm{m}}$Ta could also have ν-induced contributions. Sieverding et al. (2018) found that the ν-process is important for the nucleosynthesis of ^{113}In, ^{138}La, and ^{180}Ta.

Existing massive star models that include p-nuclei production are 1D models that rely on mixing length theory (MLT) to describe the mixing of material in convective shells. Multi-dimensional hydrodynamic simulations of convective O-shell burning predict higher convective velocities than MLT and a gradual downturn to the velocities at the shell boundaries instead of a sharp cutoff like MLT predicts (Bazan and Arnett, 1994; Meakin and Arnett, 2007; Jones et al., 2017). In addition to this, 3D O/Ne/C shell mergers have large scale non-radially asymmetries and could have C-shell ingestion rates significantly lower than those predicted by 1D models (Andrassy et al., 2020; Yadav et al., 2020). The 1D models in Ritter et al. (2018b) have O-C shell mergers when the upper boundary of the O-shell rapidly changes it entrains ^{12}C and ^{20}Ne, which burn in the O-shell and flatten the entropy gradient. In contrast, Rizzuti et al. (2024a) find in their 3D simulations that the bottom boundary of the C-shell extends down and subsumes the O-shell. Yadav et al. (2020) find similarly that 3D mergers occur differently than 1D because entropy is created due to nuclear burning hotspots lowering entropy in the Ne-shell and that material being carried in by convective downdrafts into the O-shell raising its entropy. The shell merger occurs late in the stellar evolution, and the dynamic behaviour of these supernova progenitors are not well-described in 1D models (Arnett and Meakin, 2011; Müller, 2016; Yadav et al., 2020).

The O-shell is a convective-reactive environment where mixing and nuclear burning timescales are equal and species are not well-mixed (Ritter et al., 2018a). If these timescales are equal, species could either react or advect to another location and react there. If these reactions release little energy, the flow would not be significantly affected, such as ^{7}Li production by the Cameron-Fowler mechanism (Cameron and Fowler, 1971; Sackmann and Boothroyd, 1992). However, if there is significant energy released the flow would be modified (Dimotakis, 2005), which occurs during H-ingestion into He layer (Herwig et al., 2011, 2014) or O-C shell mergers with violent mixing (Andrassy et al., 2020; Yadav et al., 2020).

The Damköhler number (Dimotakis, 2005) can be used to quantify whether a region is convective-reactive or not.

$$D_{\alpha} \equiv \frac{\tau_{\mathrm{mix}}}{\tau_{\mathrm{react}}} \tag{2.1}$$

where τ_{mix} is the mixing timescale and τ_{react} is the nuclear reaction timescale. Convective regions are regions where $D_{\alpha} \ll 1$, so the mixing is much faster than the nuclear burning and species are well-mixed across the region. Convective-reactive regions have $D_{\alpha} \sim 1$ and the behaviour described above occurs. Mixing in 1D models is treated as a diffusive process with a coefficient $D_{\mathrm{MLT}} = \frac{1}{3} v_{\mathrm{MLT}} \ell$, where v_{MLT} is the convective velocity and ℓ is the mixing length given by $\ell = \alpha H_p$, with H_p the pressure scale height and α a free parameter (Joyce and Tayar, 2023). The mixing timescale is then given by

$$\tau_{\mathrm{mix}} = \frac{\ell^2}{D_{\mathrm{MLT}}} \tag{2.2}$$

as in (Herwig et al., 2011). The reaction timescale is defined as

$$\tau_{\mathrm{react}} = \frac{1}{\rho N_A \langle \sigma v \rangle Y_j} \tag{2.3}$$

where ρ is the local density, N_A is Avogadro's number, $\langle \sigma v \rangle$ is the thermally averaged reaction rate, and Y_j is the molar abundance of the interacting species.

Although 3D hydrodynamic simulations may not be solving all relevant equations, such as a robust nuclear network, the 1D predictions lack the ability to describe the mixing during an O-C shell merger. This is because in 3D simulations, the mixing of ingested C-shell material occurs by convective plumes which are non-radial and result in spherically asymmetric instabilities (Meakin and Arnett, 2006; Andrassy et al., 2020; Yadav et al., 2020). Further, 1D simulations do not capture how a merger happens between the O- and C-shells. This is a problem for p-nuclei production as Roberti et al. (2023) found that p nuclei with $A > 110$ were largely produced during the merger rather than during explosive burning.

This result was found to be independent of the peak energy of the CCSN (Roberti et al., 2024). To explore the impact of the uncertainties from MLT to describe the mixing in convective shells, we adopt in this paper a 3D hydrodynamic inspired set of modifed radial mixing profiles to determine the impact of mixing on the γ process in the O-shell as C-shell material is entrained. In addition to this, the impact of nuclear physics in the O-shell will also be explored.

In Section 2.3 we present the methods of how the Ritter et al. (2018b) model is post-processed, the way 3D hydrodynamic results are implemented in 1D, and how the nuclear physics impact is determined. In Section 2.4.1 we explore the nucleosynthesis pathways of the γ process in the convective-reactive environment. Next, in Sections 2.4.2–2.4.4 we explore how different mixing details can impact the production of the p-nuclei: Section 2.4.2 explores the impact of gradual downturn to velocities at the lower boundary and boosting velocities in the O-shell, Section 2.4.3 considers how lowering the C-shell ingestion rate, and Section 2.4.4 looks at the impact of dips in the diffusion profile. Section 2.4.5 explores the impact from the nuclear physics and its relationship to the mixing details. Finally, in Section 2.5 we summarize the results and discuss the implications of this work.

2.3 Methodology

2.3.1 Initial Model and Post-Processing Setup

In this paper, we post-process the massive stellar model $M_{\mathrm{ZAMS}} = 15\mathrm{M}_{\odot}$, $Z = 0.02$ in the NuGrid data set (Ritter et al., 2018b). This 1D stellar model was computed with `MESA` (Paxton et al., 2010) without rotation. Convective boundary mixing is treated using an exponential-diffusive prescription (Freytag et al., 1996; Herwig, 2000) with $f = 0.022$ at all boundaries except at the base of convective shells where $f = 0.005$ until the end of core He burning, after which $f = 0$. This model has a merger of its convective O and C-burning shells late in its evolution as shown in Figure 2.1. During this merger, the C-burning ashes and stable isotopic material are ingested into the much hotter O-burning shell.

The detailed nucleosynthesis is calculated with the 1D multi-zone post-processing code `mppnp` (Pignatari et al., 2016b). `mppnp` is a multi-zone post-processing code that uses stellar structure calculated by stellar evolution codes to calculate the full nucleosynthesis of a stellar model. `mppnp` treats mixing, nuclear burning, and ingestion separately rather than the coupled treatment by a code like `MESA` Paxton et al. (2010). A convergence test was performed by decreasing the timesteps and increasing the number of mass zones. It found that a timestep of $\Delta t = 0.01$ s and 400 equidistant mass zones with 4 additional zones at the bottom

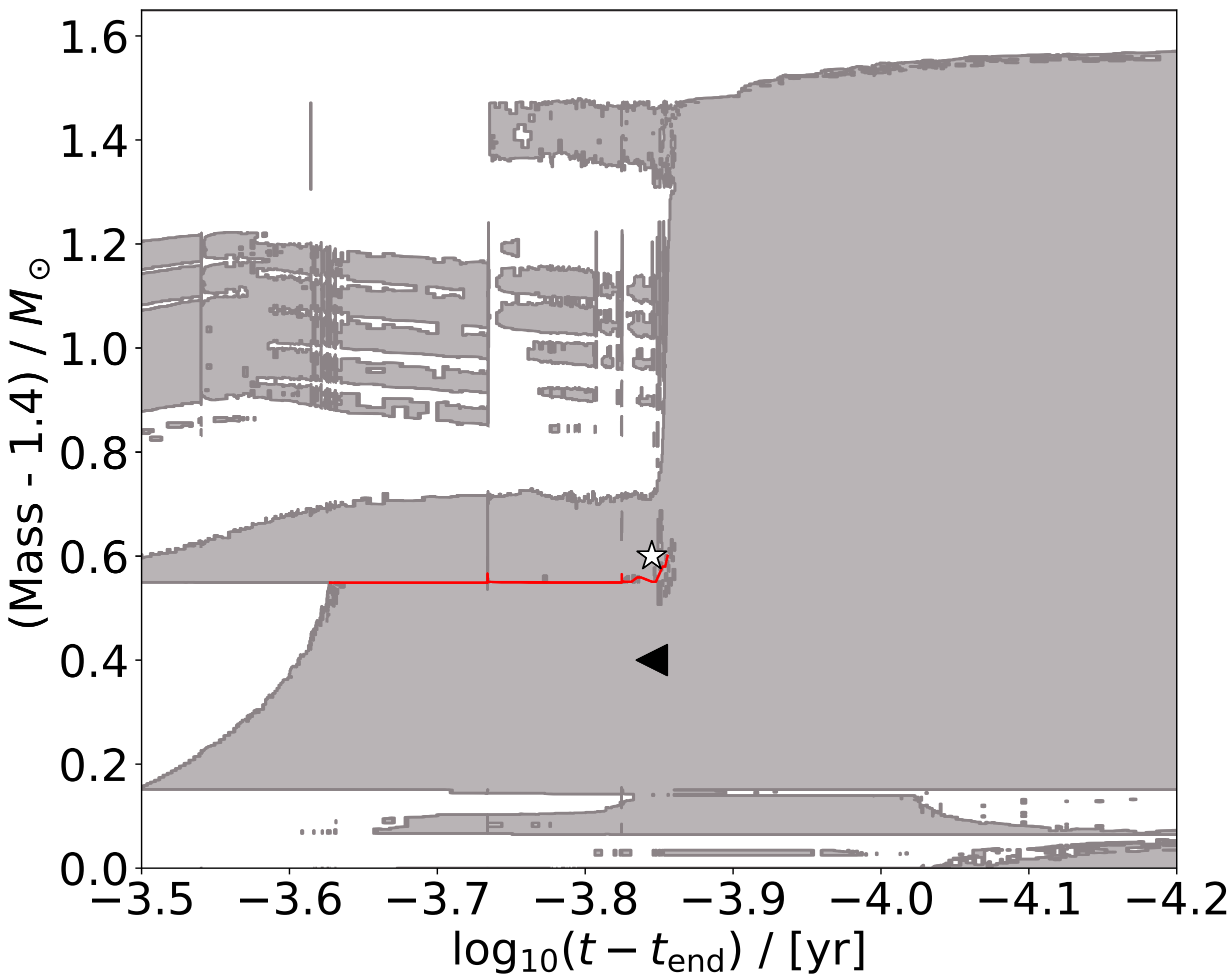

Figure 2.1: Kippenhahn diagram showing the merger of the convective O and C-burning shells. The O-burning shell extends from $1.55M_\odot$ to $1.95M_\odot$. The first convective C-burning shell ingested sits directly on top from $1.96M_\odot$ to $2.11M_\odot$ and the other convective C-burning shells are above. A red guideline has been provided to mark the thin radiative layer separating the O- and C-shells. The merger onsets at $\log_{10}(t - t_{\rm end})/\rm{yr} \approx -3.85$ and reaches full extent at ≈ -4. A black triangle marks the location where the initial p nuclei are taken from and a white star marks the location where the ingested C-shell material is taken from.

of the O-shell were sufficient to resolve both the burning and mixing timescales. Calculations were initially done with a 5234 isotope network, but many of these were unnecessary as they were n-rich and not needed for the calculations. All necessary isotopes were included, which amounted to the network of 1470 species, focusing on the p-rich side allowing for faster calculations. The network adopted did not include isomeric states, so the nucleosynthesis of $^{180\mathrm{m}}$Ta is not calculated although we calculate the ground state ^{180}Ta. A single simulation costed approximately 8 hours on 40 cores, for a total of 180 core years for this paper.

The `mppnp` code calculates both the undecayed mass fraction as a function of mass, and mass-averaged radiogenic decay contribution to the isotopes at a temperature of T = 100 MK which does not include explosive contributions. The mass fraction X_i of species i is defined as the ratio of the mass of that species to the total mass of the stellar material, such that $\sum_i X_i = 1$ for all isotopes. To analyze the macrophysical impact during the merger on nucleosynthesis, the isotopic mass fractions are taken right before onset of merger at $\log_{10}(t - t_{\mathrm{end}})/\mathrm{yr} = -3.845$ from $m = 1.8\ \mathrm{M}_{\odot}$. The ingested C-shell material is taken from the same time from $m = 2.0\ \mathrm{M}_{\odot}$. The results are presented in terms of an overproduction OP compared to the initial amount given by the equation

$$\mathrm{OP} = \log_{10}\left(\frac{X_f}{X_i}\right) \tag{2.4}$$

where X_f is the final mass-averaged decayed mass fraction of a species in the O-shell and X_i is the inital mass fraction. The average overproduction factor is calculated as the arithmetic mean of individual OP values:

$$\langle \mathrm{OP} \rangle = \frac{1}{N} \sum_i^N \mathrm{OP}_i \tag{2.5}$$

where N is the number of p nuclei considered.

Earlier in the evolution of this stellar model, it produces the p nuclei by the γ process in the first convective O-shell which extends from $1.55-1.95\ \mathrm{M}_{\odot}$ during model numbers 6200–7200. The initial mass fractions of the p nuclei are set here and are not modified during subsequent radiative C- or O-burning. The mass fractions used in this study are shown in Figure 2.2.

The stellar structure from the onset of the merger at $\log_{10}(t - t_{\mathrm{end}})/\mathrm{yr} = -3.856$. To be able to clearly analyze the impact of mixing alone, we have kept the stellar structure constant. The stellar structure is not static, but the change to the temperature, density, and entropy between these timesteps is less than 5%. The merger at $\log_{10}(t - t_{\mathrm{end}})/\mathrm{yr} = -3.856$

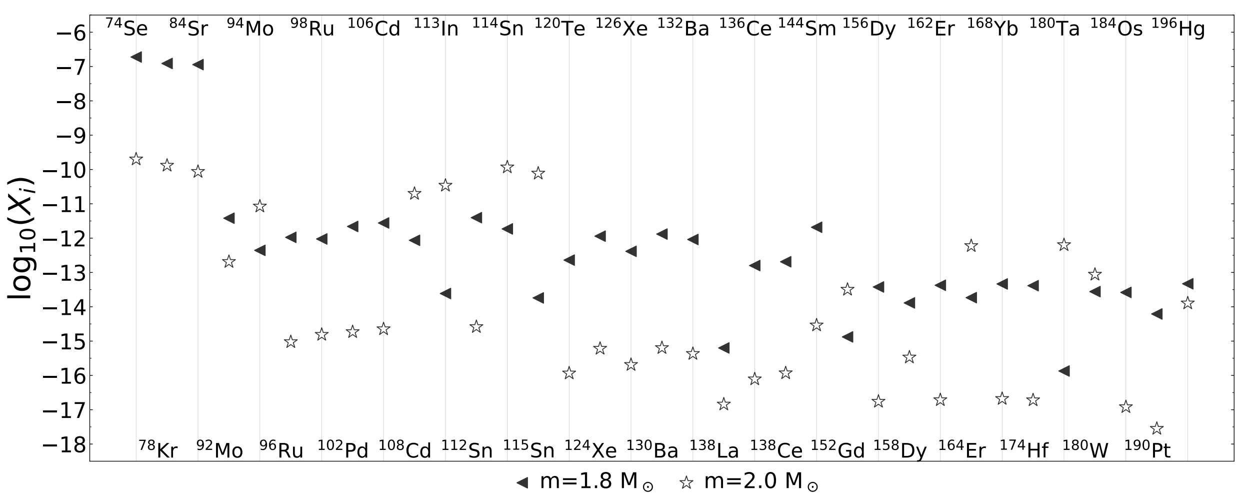

Figure 2.2: Mass fractions of the p nuclei from $\log_{10}(t - t_{\rm end})/{\rm yr} = -3.856$. used for initial O-shell composition and ingested C-shell material. Markers are the same as those used in Figure 2.1.

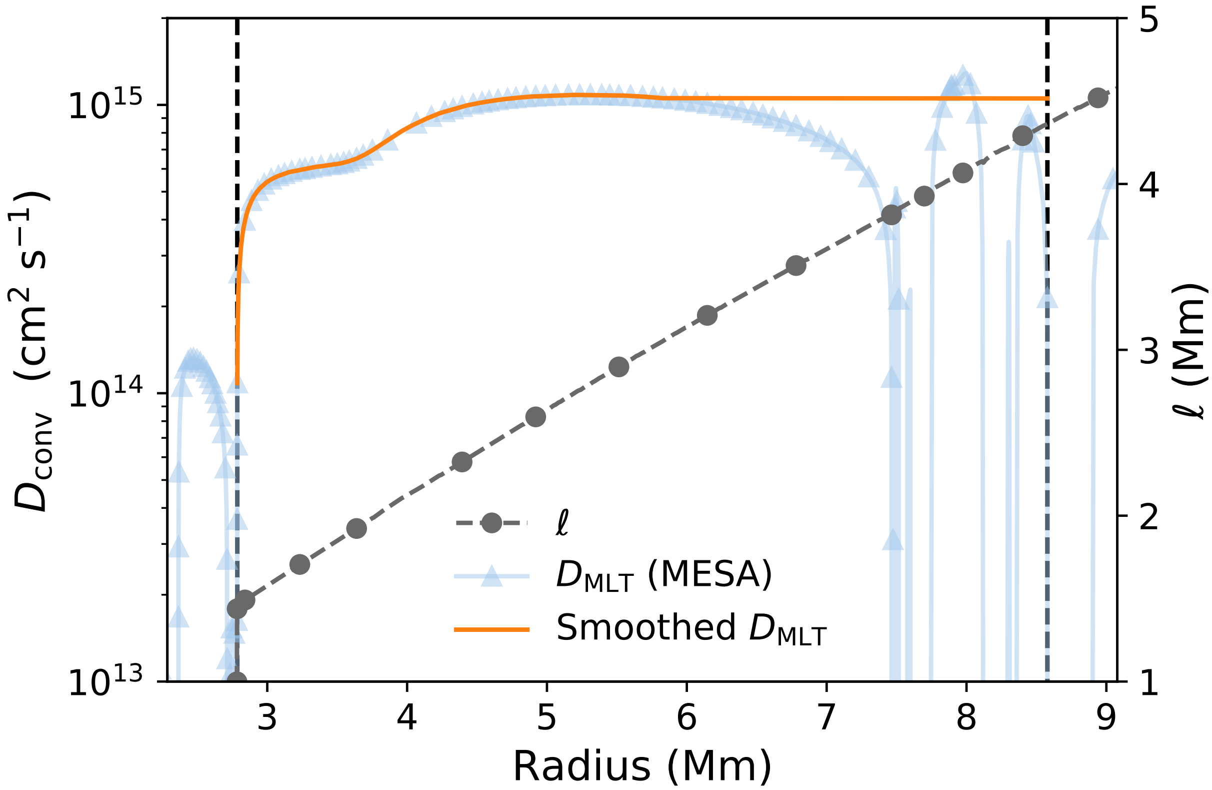

Figure 2.3: The diffusion coefficient profile and mixing length at model number 9200 for the $M_{\rm ZAMS} = 15\ M_\odot$, $Z = 0.02$ model. The light blue line is D from MESA, the orange line the smoothed D used for the MLT mixing scenario in this paper, and the grey line is the mixing length. Black dashed lines mark the shell boundaries for this paper.

is not fully developed, but `MESA` cannot accurately model the merger in 1D. Therefore, the convective velocity profile is smoothed at the top as if the C-shell is merged with the O-shell as seen in Figure 2.3.

2.3.2 1D implementation of 3D macrophysics

MLT predictions of the radial mixing profiles deviate from the more realistic predictions from 3D simulations. 3D convective O-burning simulations show that the radial convective velocity profile gradually decreases as it approaches the boundary of the O-shell rather than MLT predictions of a stiff boundary (Meakin and Arnett, 2007; Jones et al., 2017). The downturn can be seen both at the bottom of a shell (Herwig et al., 2006; Meakin and Arnett, 2007) or at the top (Jones et al., 2017). The decrease in radial velocity happens because mixing in these simulations are driven by convective plumes rather than the blobs of MLT.

In the middle of a convective region, plumes have strong radial velocities and travel towards the convective boundary, but as they reach the boundary, their radial velocity component diminishes while the other components of the velocity increase. This is different from MLT which predicts a stiff boundary where velocities drop to zero at the boundary. Using Equation 4 from Jones et al. (2017), the downturn to the radial velocity profile can be mimiced in 1D.

$$D_{\text{3D-insp.}} = \frac{1}{3} v_{\text{MLT}} \times \min\left(\ell, r - r_0\right) \tag{2.6}$$

where ℓ is the mixing length, r_0 is the Schwarzschild boundary at the bottom of the O-shell, and an additional factor of 1/3 is applied to match the MLT diffusion coefficient at the top of the shell.

MLT also underpredicts the strength of the convective velocities in the O-shell. Jones et al. (2017) found that convective velocities are stronger by a factor of ~ 30 when compared to the MLT predictions. Andrassy et al. (2020) in their 3D C-shell entrainment simulations that their convective velocities could be up to ~ 5 times stronger than Jones et al. (2017) depending on the luminosity from carbon and oxygen burning. It is possible the velocities could be higher as these simulations do not include feedback from a nuclear network and treatment of radiation pressure (Jones et al., 2017; Andrassy et al., 2020). Rizzuti et al. (2024a) find that velocities are boosted by a factor of ~ 10 due to the feedback from new reactions with the ingested material in their 3D O-C shell mergers. In our 1D models, we implement this by applying a boost factor of 3×, 10×, and 50× to the convective velocities as shown in Figure 2.4.

The entrainment of C-shell material could be different during the merger depending on the strength of burning (Jones et al., 2017; Andrassy et al., 2020). To investigate the impact of this, a range of rates are considered: 4×10^{-5} $\mathrm{M_\odot s^{-1}}$, 4×10^{-4} $\mathrm{M_\odot s^{-1}}$, and 4×10^{-3} $\mathrm{M_\odot s^{-1}}$, as well as a scenario with no entrainment. The mass of the C-shell is 0.8 $\mathrm{M_\odot}$ in the model considered, so for 110 s of merging the maximum possible entrainment rate would be 7×10^{-3} $\mathrm{M_\odot s^{-1}}$ similar to Ritter et al. (2018a).

3D simulations also show the convective profile can be quenched as material is ingested because of entropy entrainment (Miller Bertolami et al., 2006; Herwig et al., 2011, 2014). During H-ingestion into a He-shell, the energy feedback from the ingested material could cause a split in the convective profile with very small amount of entrainment (Herwig et al., 2011). Herwig et al. (2014) likewise found this effect could cause a drop in the radial velocity profile and reduce the entrainment of species and labelled the event Global Oscillation of Shell Hydrogen-Ingestion (GOSH). Similar effects during C-shell entrainment are possible,

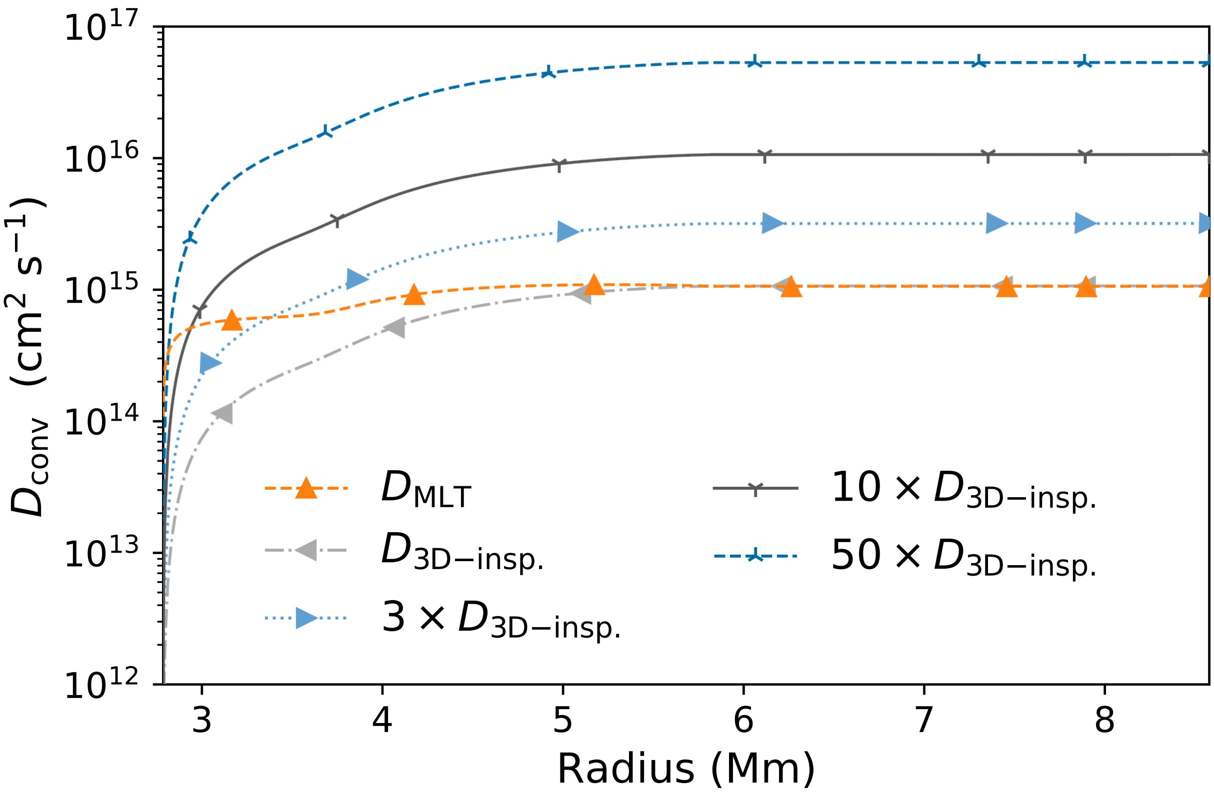

Figure 2.4: The diffusion coefficient profiles for the MLT and 3D-inspired gradual downturn scenarios. The dashed orange line is the D_{MLT} profile, and the dashed light grey line, dotted light blue line, solid grey line, and dashed dark blue line are the downturn profiles with boost factors of 1, 3, 10, and 50 respectively.

as Andrassy et al. (2020) found that strong oscillatory modes like GOSHs were present in their 3D simulations. Energy feedback events like this could explain supernova observations (Smith and Arnett, 2014). There is no clear prescription for how to implement this effect into 1D models, but it is clear that there would be decreased mixing because of a split. To investigate a possible convective quenching, we consider a GOSH-like event and a partial merger, where a dip (but not full split) occurs in the MLT diffusion profile:

$$D_{\text{quench}} = D_{\text{MLT}} - (D_{\text{MLT}} - c) \times \exp\left[-\frac{(r-a)^2}{w^2}\right] \tag{2.7}$$

where c is the extent, w is the width, and a is the center of the dip using a simple Gaussian. The GOSH-like convective splitting occurs at the location where $D_\alpha = 1$ for a significant burning event (Herwig et al., 2011). We centre this event at $a = 4.95$ Mm at a location of probable energetic feedback could occur. The O-shell could partially merge with the C-shell due to feedback effects, so we consider a partial merger at $a = 7.5$ Mm where the unmerged MLT profile has a dip as seen in Figure 2.3. A width of $w = 0.25$ Mm is used for both scenarios, which is approximately the distance between the top of the O-shell and the convective bump above it in Figure 2.3. For both scenarios we consider a weaker dip of $c = 10^{14}$ cm^{2}s^{-1} and a stronger dip of $c = 10^{13}$ cm^{2}s^{-1} to investigate the impact on nucleosynthesis. The profiles for the GOSH-like and partial merger scenarios are shown in Figure 2.5.

2.3.3 Determining impact of varying nuclear reactions

Many unstable n-deficient isotopes from Se to Po have not been measured experimentally and are determined by theoretical models, and the uncertainties of unmeasured reactions reactions for unstable isotopes are much greater than for stable isotopes. To determine the impact of nuclear physics for the γ process, we vary our adopted (γ, α), (γ, p), and (γ, n) photo-disintegration rates used by the NuGrid database (Pignatari et al., 2016b) for the unstable n-deficient isotopes by a random factor uniformly selected between 0.1 to 10 by a Monte Carlo method for 1000 cases. This applies the same approach used for (n, γ) rates for the i process developed by Denissenkov et al. (2018) and Denissenkov et al. (2021). This was done for the MLT and downturn mixing scenarios with an ingestion rate of 4×10^{-3} M$_\odot$s^{-1}. This approach also allows for the identification of reaction rates that are relevant for the production of an isotope using correlations. The Pearson coefficient describes correlations between $X/X_{\text{no variation}}$ and the variation factors where X is the final mass-averaged and

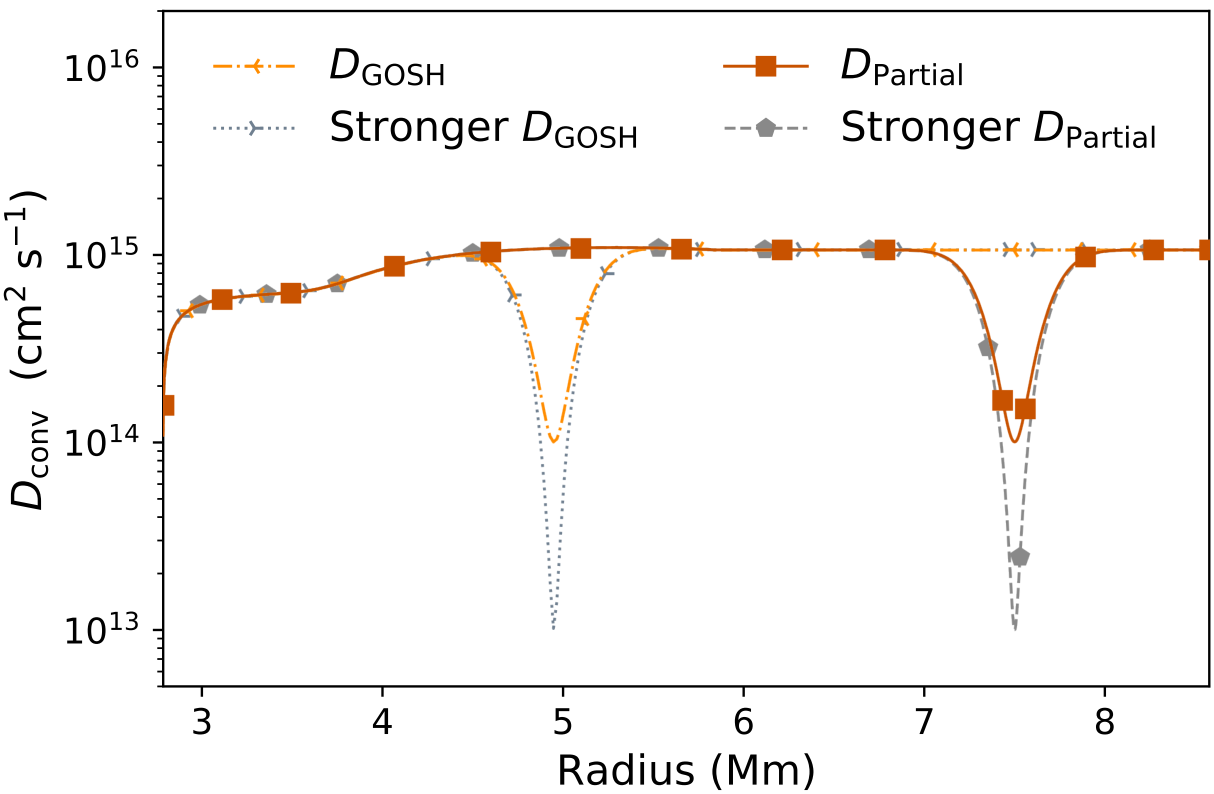

Figure 2.5: The quenched mixing scenario convective profiles. The dashed light orange line and dotted light grey line are the GOSH-like profiles with a dip centred at $r = 4.95$ Mm. The solid red line and dashed dark grey line are the partial merger profiles with a dip centred $r = 7.5$ Mm. The MLT profile is omitted for clarity.

Table 2.1: $\langle \mathrm{OP} \rangle$ for each mixing scenario and the average spread ($\mathrm{OP_{max}} - \mathrm{OP_{min}}$) for the Monte Carlo simulations for the p nuclei. All Monte Carlo simulations are calculated with an ingestion rate of 4×10^{-3} $\mathrm{M_\odot s^{-1}}$.

Scenario	No Ingestion	4×10^{-5} $\mathrm{M_\odot s^{-1}}$	4×10^{-4} $\mathrm{M_\odot s^{-1}}$	4×10^{-3} $\mathrm{M_\odot s^{-1}}$	MC Spread
MLT	−0.11	1.06	1.92	2.24	0.56
1× Downturn	0.05	1.12	1.98	2.58	0.59
3× Downturn	−0.23	1.28	2.18	2.83	0.69
10× Downturn	−1.23	1.18	2.10	2.89	0.76
50× Downturn	−5.47	0.88	1.81	2.72	0.79
GOSH-like	–	–	–	2.06	–
Stronger GOSH-like	–	–	–	1.78	–
Partial Merger	–	–	–	2.13	–
Stronger Partial Merger	–	–	–	1.91	–

decayed mass fraction of a Monte Carlo case and $X_{\text{no variation}}$ is the final decayed mass fraction for the default case where all varation factors are equal to 1. All correlations $|r_\mathrm{P}| \geq 0.15$ are reported in this study. In addition to the Pearson coefficient, a logarithmic slope ζ is also reported to determine the importance of a reaction on the final mass fraction of an isotope. The importance of considering ζ as well as additional caveats for the correlated rates are discussed in Appendix 2.6.1.

2.4 Results

An overview of our results in terms of overproduction factors and average spreads of overproduction factors are presented in Table 2.1.

2.4.1 Convective-reactive production of the p nuclei

Convective-reactive nucleosynthesis is characterized by a region where the timescales for advection and nuclear reactions are similar. In the γ process, heavier species are produced at cooler temperatures of 1.5−2 GK and are destroyed at higher temperatures, whereas lighter species are produced in temperatures up to 3.5 GK (Rauscher et al., 2013). Whether an isotope undergoes (γ, n) and (n, γ) reactions or contributes to the production of lighter p nuclei by (γ, p) and (γ, α) reactions depends on the temperature at that position. The convective-reactive environment of the O-shell allows for the production of both light and heavy p nuclei because the shell is not well-mixed, and therefore species are produced at their ideal temperatures. Figure 2.6 shows how different mass ranges of p nuclei can be produced and peak at different positions in the O-shell.

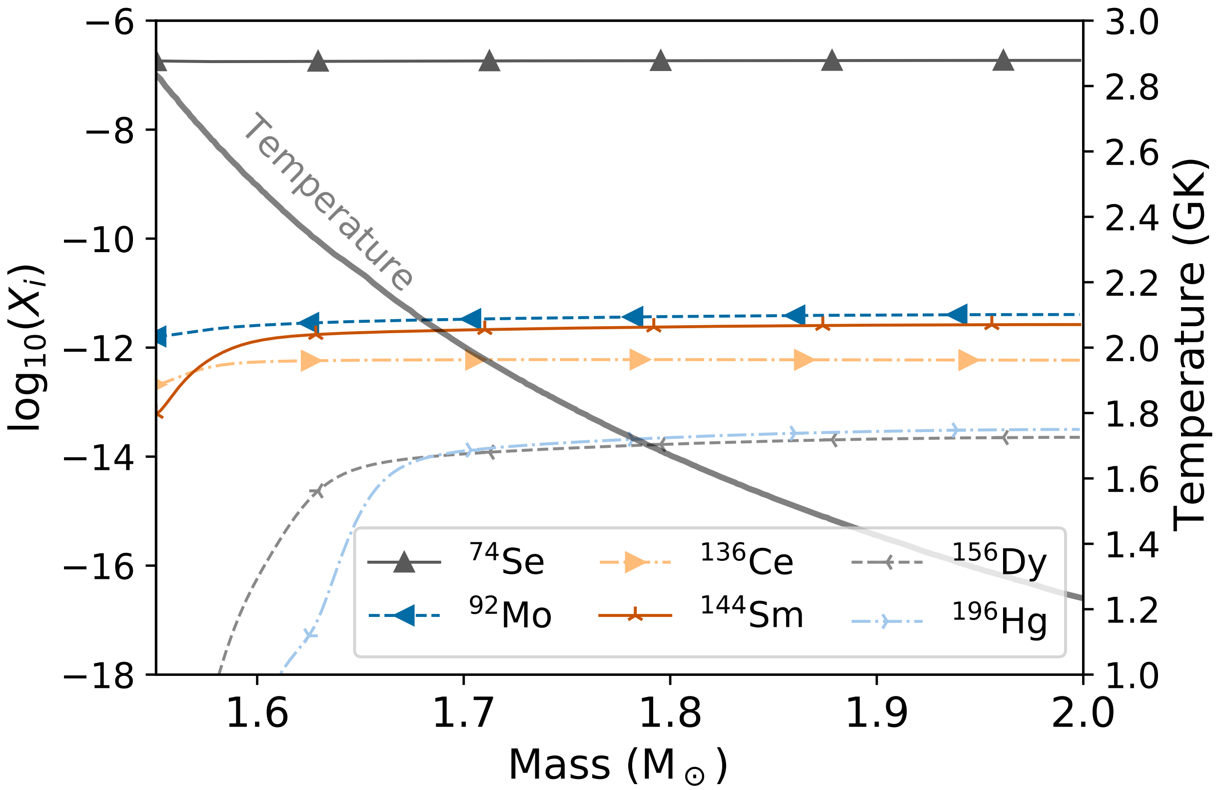

Figure 2.6: Mass fractions without radiogenic contributions at t = 110 sec for the MLT mixing scenario without C-shell ingestion.

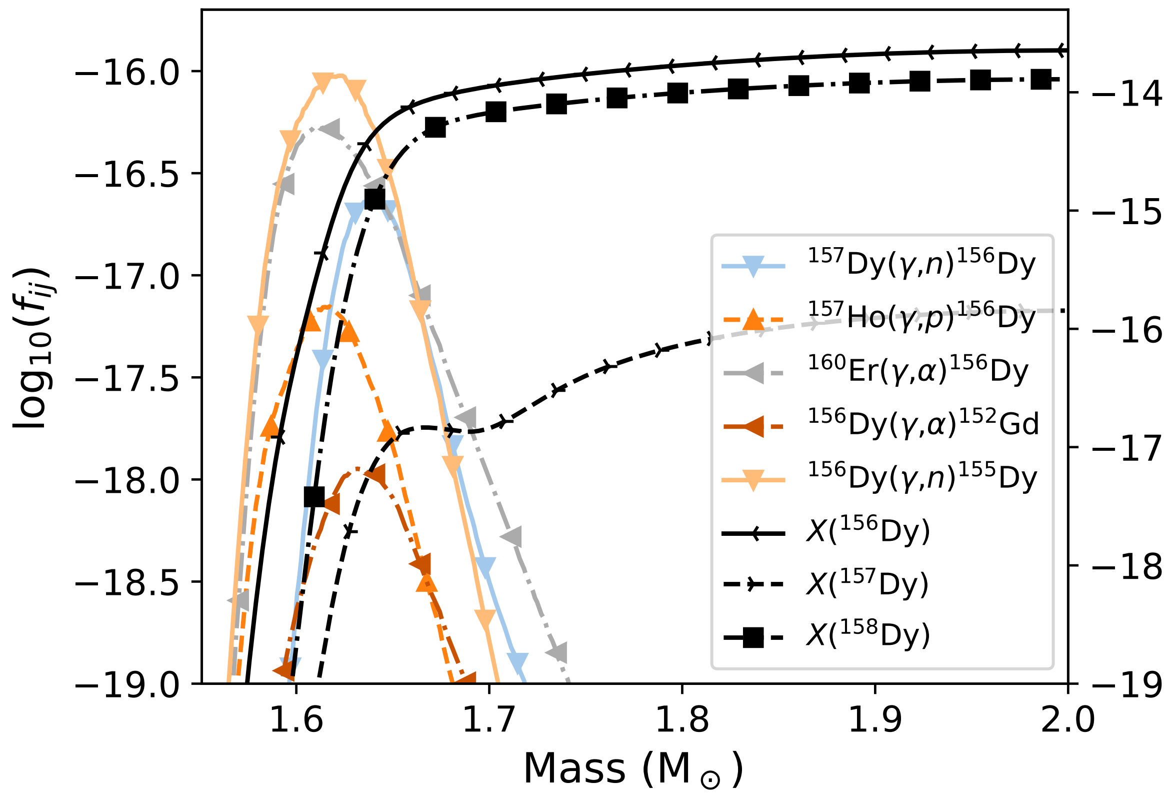

Figure 2.7: Reaction fluxes f_{ij} for ^{156}Dy and mass fractions X_i for $^{156-158}$Dy without radiogenic contributions for the MLT scenario with no ingestion after $t = 110$ s. The direction of reactions $i \rightarrow j$ are written as left to right in the legend.

Locations of peak destruction can be identified by a sudden drop in mass fraction as seen for ^{156}Dy, ^{196}Hg, and to a lesser extent ^{144}Sm. This is the location where $D_\alpha = 1$, which is different for each isotope (Herwig et al., 2011). This is contrary to the normal assumption of a well-mixed convective environment where $D_\alpha \ll 1$ or radiative burning where little to no mixing occurs, and is a key reason why the γ process in the O-shell produces the whole range of p nuclei. Figure 2.7 shows the dominant reactions that ^{156}Dy is produced and destroyed by in the simulation without C-shell ingestion. These reactions are shown as reaction fluxes f_{ij} which are defined as the net flux of a reaction between species i and j:

$$f_{ij} = \frac{X_i X_j}{A_i A_j} \rho N_A \left(\langle \sigma v \rangle_{ij} - \langle \sigma v \rangle_{ji} \right) \tag{2.8}$$

where X is the mass fraction, A is the atomic mass, ρ is the density, N_A is Avogadro's number $\langle \sigma v \rangle_{ij}$ is the reaction rate between species i and j.

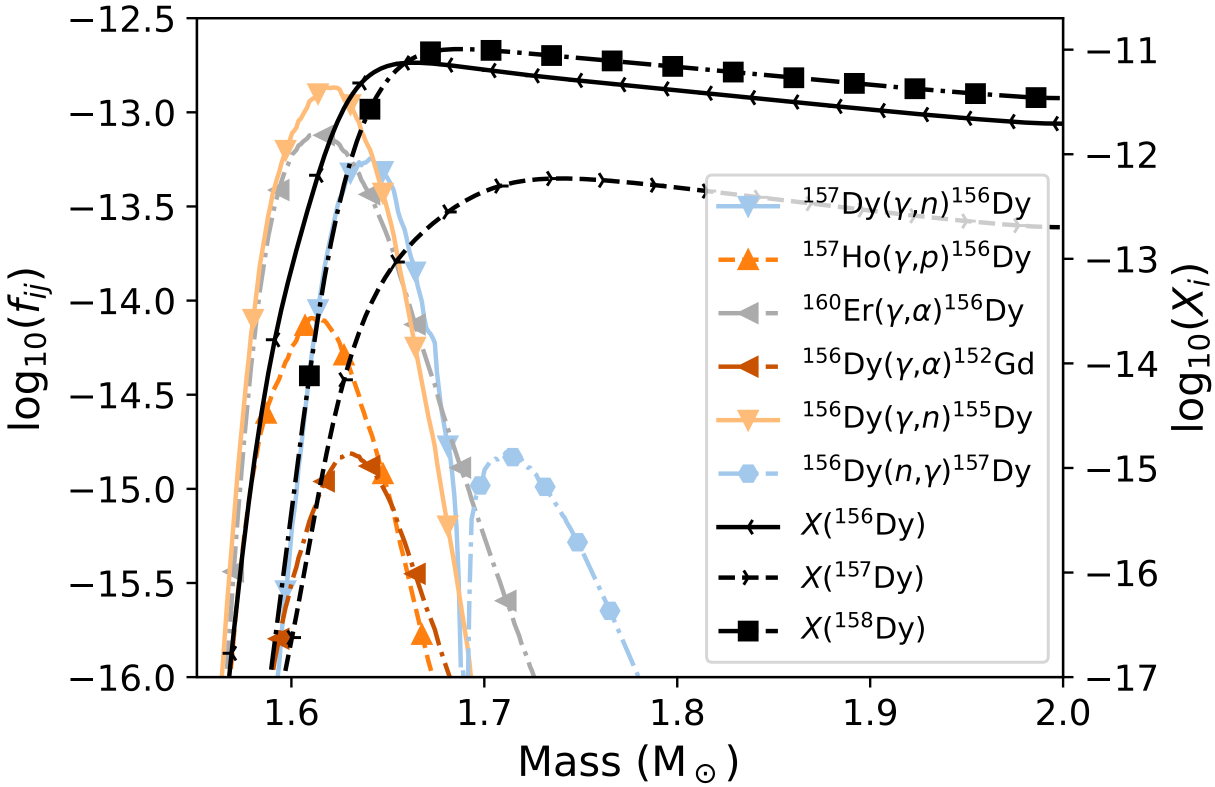

Figure 2.8: Reaction fluxes f_{ij} for ^{156}Dy and mass fractions X_i for $^{156-158}$Dy without radiogenic contributions for the MLT scenario with an ingestion rate of 4×10^{-3} M$_\odot$s^{-1} after $t = 110$ s. The direction of reactions $i \to j$ are written as left to right in the legend.

The reactions that ^{156}Dy undergoes in the O-shell depend on the location in the shell, however Figure 2.7 shows that the dominant destruction channel ^{156}Dy$(\gamma, n)^{155}$Dy net destroys ^{156}Dy, as Figure 2.11 will show. It is also evident that the mass fraction is not well-mixed as the gradient of the mass fraction is steep at the location of peak destruction.

Entraining C-shell material is important for this process, as heavier species can be gradually depleted by (γ, α) and (γ, p) reactions. Figure 2.8 shows the same as Figure 2.7, but with a C-shell ingestion rate of 4×10^{-3} M$_\odot$s^{-1}, the maximum considered in this study. Since the initial amount of ^{156}Dy in the ingested C-shell is negligible as shown in Figure 2.2, the merger provides other species that are used to produce ^{156}Dy.

There are several differences between Figures 2.7 and 2.8. First, f_{ij} and X_i are larger by several orders of magnitude and ^{156}Dy has a net production in the shell, as Figure 2.11 will show in Section 2.4.3. Second, the mass fraction of ^{156}Dy has a tilt up where it is net produced and then drops off sharply at the location of peak destruction instead of the decline

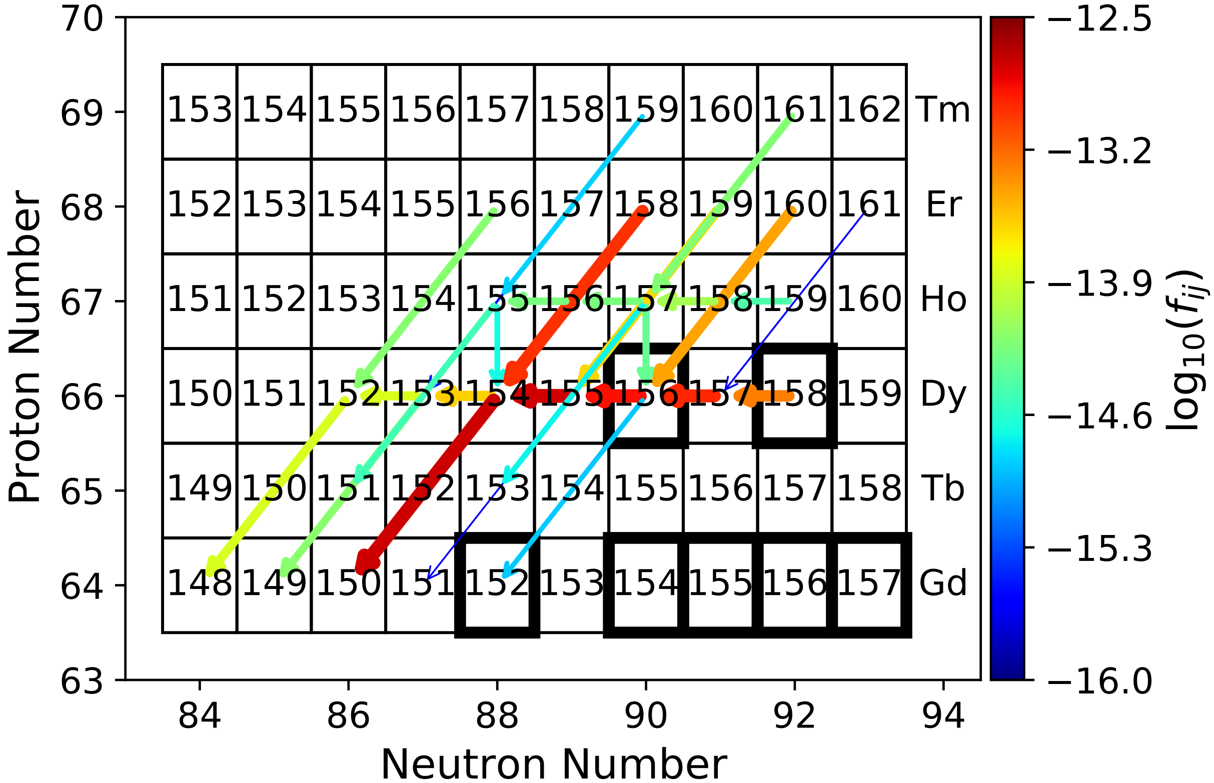

Figure 2.9: Chart of reactions between isotopes at $m = 1.64$ M$_\odot$ [$T = 2.28$ GK] for the same conditions as Figure 2.8. Both arrow colour and size indicate f_{ij}, and arrows point in the direction of the reaction.

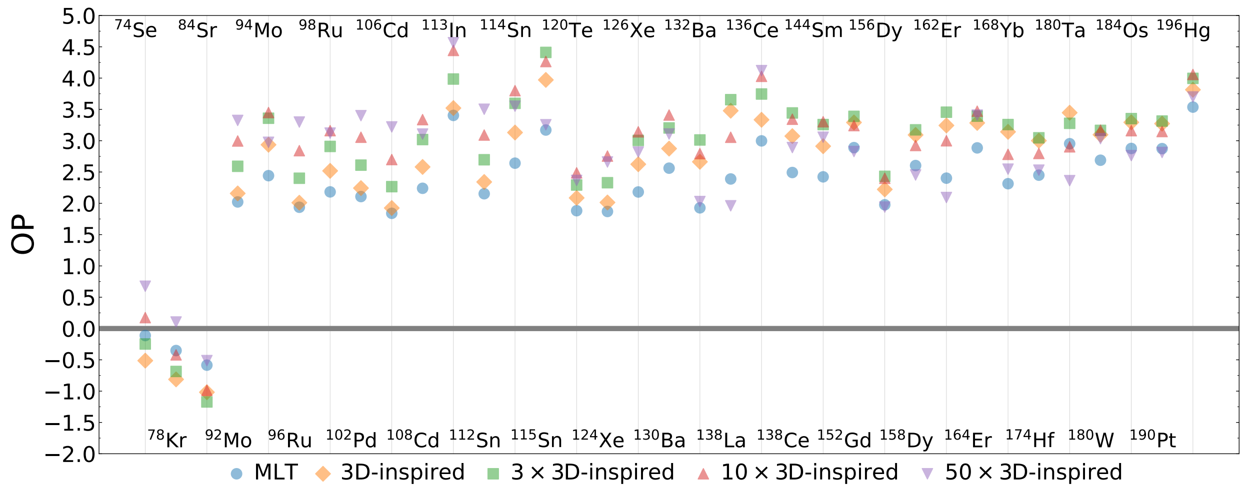

Figure 2.10: The overproduction compared to initial of the p nuclei for the MLT and 3D-inspired mixing scenarios. The average spread in production $\mathrm{OP}_{\mathrm{max}} - \mathrm{OP}_{\mathrm{min}} = 0.96$ dex. $\mathrm{OP} = 0$ is the initial amount.

seen in Figure 2.7.

^{156}Dy is produced because the ingestion of C-shell material allows for ^{158}Dy to be replenished as it advects into the shell. With a continual supply of ^{158}Dy, the chain $^{158}\mathrm{Dy}(\gamma, n)^{157}\mathrm{Dy}(\gamma, n)^{156}\mathrm{Dy}$ can occur and significantly contribute to the production of ^{156}Dy equal to $^{160}\mathrm{Er}(\gamma, \alpha)^{156}\mathrm{Dy}$.

Figure 2.8 also shows another feature of this convective-reactive environment: ^{156}Dy and ^{157}Dy co-produce each other. ^{156}Dy is advected from its location of peak production at $\sim$1.63 $\mathrm{M}_{\odot}$ both deeper into the shell where it is fully destroyed, and toward the top where it undergoes $^{156}\mathrm{Dy}(n, \gamma)^{157}\mathrm{Dy}$, which mildly contributes to the production of ^{157}Dy and peaks at $\sim$1.73 $\mathrm{M}_{\odot}$.

This demonstrates that in a convective-reactive environment isotopes are not well-mixed, and how the reaction rates vary with position. They also show how isotopes can produce each other at different locations in the shell. Finally, they show why the ingestion of C-shell material is necessary for significant production of the p nuclei in the O-shell.

2.4.2 Impact from a downturn and boosting mixing speeds

Here we present the impact of a 3D-inspired gradual downturn at the lower boundary and boosting mixing speeds as explained in Section 2.3.2 with the diffusion coefficients in Figure 2.4. These cases are calculated with an ingestion speed of 4×10^{-3} $\mathrm{M}_{\odot}\mathrm{s}^{-1}$ and the results are shown in Figure 2.10.

The MLT simulation has an $\langle \mathrm{OP} \rangle$ of 2.24 dex, and the downturn scenarios have $\langle \mathrm{OP} \rangle$ of 2.58 dex, 2.83 dex, 2.89 dex, and 2.72 dex for the $1\times$, $3\times$, $10\times$, and $50\times$3D-inspired mixing scenarios respectively. The average spread in production for each isotope $\mathrm{OP}_{\mathrm{max}} - \mathrm{OP}_{\mathrm{min}} =$ 0.96 dex, which shows that mixing speeds are important for the production of the p nuclei. This O-shell during the merger significantly produces all p nuclei except ^{74}Se, ^{78}Kr, and ^{84}Sr.

The 3D-inspired $1\times$ scenario favours the production of the heavier p nuclei compared to the MLT scenario because τ_{mix} decreases as the temperature. Because of this, more reactions occur at the cooler temperatures where the (n, γ) and (γ, n) reactions are more favoured than the (γ, α) and (γ, p) reactions. All isotopes are comparably produced to MLT or more produced except ^{74}Se, ^{78}Kr, and ^{84}Sr who require the hottest temperatures for their production.

As mixing speeds increase, the production increases in a non-linear and non-monotonic way. For the $50\times$ case, the average production of all p nuclei is lower than the $3\times$ and $10\times$ scenarios. This is because the mixing speeds are high enough that material is advected to the bottom of the O-shell fast enough despite the downturn, and correspondingly the lighter p nuclei are generally more favoured including ^{74}Se, ^{78}Kr, and ^{84}Sr. Production for individual isotopes also can be non-linear and non-monotonic. For example, ^{115}Sn, ^{132}Ba, and ^{138}La all increase for the $1\times$ and $3\times$ scenarios, but then their production is not as strong for the $10\times$ and $50\times$ scenarios.

Another result is that isotopic pairs of the same element are not affected the same way by the downturn compared to MLT, nor by the increase in mixing speed and we find that the ratio between these isotopes is dependent on the mixing scenario. Since these isotopes are connected by (γ, n) and (n, γ) reactions, if the location of $D_\alpha = 1$ for a reaction changes because of a change to the mixing speed or the presence of the decreasing diffusion profile, then the production of these isotopes will change. ^{94}Mo is produced more for all mixing scenarios except for the $50\times$ scenario, where ^{92}Mo has a larger OP, and likewise ^{115}Sn exhibits this behaviour when compared to ^{112}Sn and ^{114}Sn. We find that it is possible that the ratio can tend to unity as mixing speeds increase as seen for ^{96,98}Ru, ^{106,108}Cd, and ^{112,114}Sn. For ^{136,138}Ce the lighter isotope is always favoured as mixing speed increases and for ^{156,158}Dy the opposite is true. This demonstrates the importance of the decreasing diffusion profile and increasing mixing speed is for the production of the p nuclei.

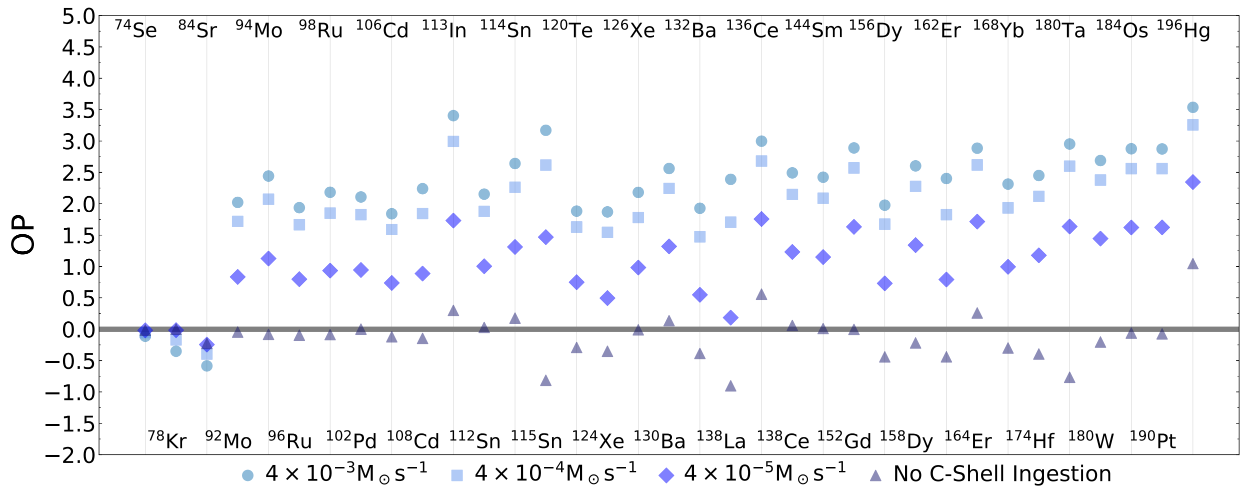

Figure 2.11: The overproduction compared to initial of the p nuclei for the MLT mixing scenario for no ingestion, 4×10^{-5} $\mathrm{M_\odot s^{-1}}$, 4×10^{-4} $\mathrm{M_\odot s^{-1}}$, and 4×10^{-3} $\mathrm{M_\odot s^{-1}}$. The average spread in production $\mathrm{OP_{max}} - \mathrm{OP_{min}} = 1.22$ dex excluding the no ingestion case. $\mathrm{OP} = 0$ is the initial amount.

2.4.3 Impact from varying the ingestion rate

Here we present the impact of entraining C-shell material with 4×10^{-5} $\mathrm{M_\odot s^{-1}}$, 4×10^{-4} $\mathrm{M_\odot s^{-1}}$, and 4×10^{-3} $\mathrm{M_\odot s^{-1}}$ as well as no entrainment as explained in Section 2.3.2. Figure 2.11 shows the results for the MLT mixing scenario and Figures 2.16–2.19 in Appendix 2.6.2 show the results for all downturn scenarios.

The results show that the production of the p nuclei is monotonically increased by the ingestion of C-shell material for all isotopes except ^{74}Se, ^{78}Kr, and ^{84}Sr who exhibit the opposite behaviour for all mixing scenarios. The only exception is that ^{74}Se production increases for the 10× and 50×3D-inspired scenarios with ingestion rate. This is because with higher ingestion rates, the more stable isotopes enter the shell as demonstrated in Section 2.4.1. The lightest three have the opposite behaviour because without ingestion they are destroyed less by (n,γ) reactions. The difference in OP between two ingestion rates is largely uniformly for ^{92}Mo–^{196}Hg. The average spread $\mathrm{OP_{max}} - \mathrm{OP_{min}}$ between the different ingestion rates for the MLT and 3D-inspired scenarios is 1.22, 1.58, 1.64, 1.78, and 1.84 dex. This shows that the non-linear and non-monotonic behaviour in Section 2.4.2 is because of the changing location of peak burning in the convective-reactive environment and not just more material being present.

Without ingestion of C-shell material, the MLT, 1×, and 3× scenarios see little to no production, but for the 19× and 50× scenarios there is a significant underproduction of

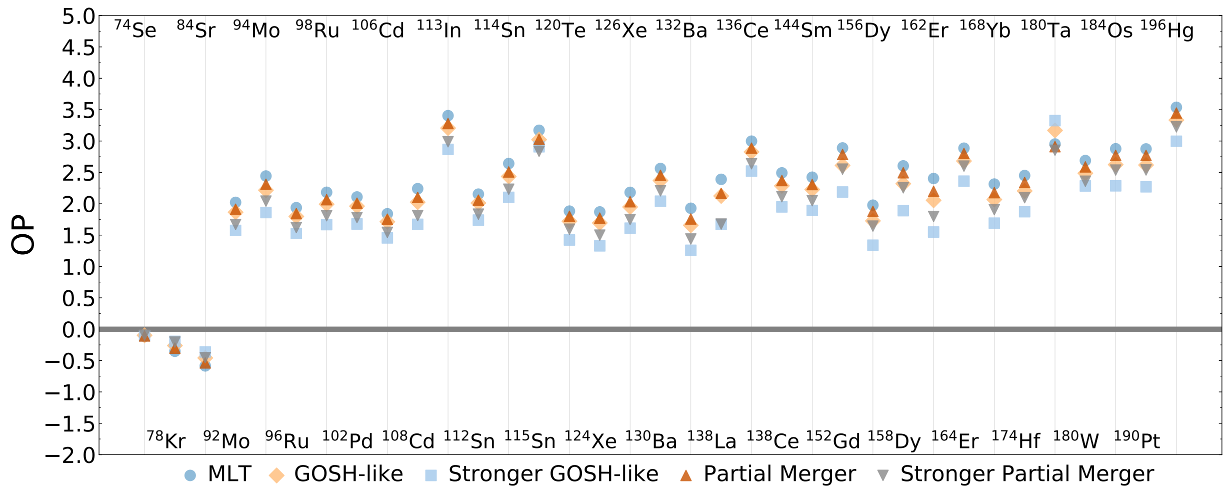

Figure 2.12: The overproduction compared to initial of the p nuclei for the MLT, GOSH-like, and partial merger scenarios with ingestion rate 4×10^{-3} $M_{\odot}s^{-1}$. The average spread in production $OP_{max} - OP_{min} = 0.51$ dex. OP = 0 is the initial amount.

the heavier p nuclei. ^{180}Ta in particular experiences a significant underproduction of $OP = -83$ dex for the 50×3D-inspired scenario. This is because in the fastest cases the species are advected to the bottom of the shell where the temperatures are purely destructive and there is no replenishment from the C-shell. This shows that entrainment of C-shell material is necessary for significant contributions from pre-explosive γ process.

The sign of the ratio of isotopic pairs is largely preserved across the ingestion rates for the MLT, 1×, 3×, and 50× scenarios, but the magnitude of the ratio does change. The few exceptions are ^{162,164}Er in the 1× and 3× scenarios who become roughly equal for the fastest ingestion rate, ^{115}Sn which becomes more abundant than ^{112,114}Sn in the 3× scenario. In the 10× scenario production of isotopic pairs is largely equal for the slower ingestion rates, but at the fastest ingestion rate the ratio can become unequal. This shows that the the ingestion rate matters for the magnitude of the ratio between isotopic pairs, but that the sign of the ratio depends on which diffusion profile is used.

2.4.4 Impact from dips from convective quenching

Here we present the impact of dips from convective quenching using the profiles shown in Figure 2.5 from Section 2.3.2 which represent GOSH-like feedback and a partial merger. The results are shown in Figure 2.12.

The MLT simulation has ⟨OP⟩ of 2.24 dex, the GOSH-like scenarios have ⟨OP⟩ of 2.06 dex and 1.78 dex, and the partial merger scenarios have ⟨OP⟩ of 2.13 dex and 1.91 dex.

We find that the GOSH-like scenarios suppresses the production more than the partial merger scenarios of equal dip depths, and that the deeper dips are suppress production more than the shallow with an average spread $\mathrm{OP}_{\mathrm{max}} - \mathrm{OP}_{\mathrm{min}} = 0.51$.

The dip functions as a barrier for the convective-reactive flow, and can section off parts of the O-shell. The partial merger scenario both limits the ingested C-shell material and slows down the material from deeper in the O-shell from advecting up to the top of the shell where very few reactions occur. The GOSH-like dip slows down the material from reaching their preferential temperatures, and also prevents material at the deepest part of the O-shell from mixing up to the top of the shell which keeps it at hotter temperatures where it can be destroyed. This is also why the deeper dips suppress production more, as the velocities are slower by an additional factor of 10 at the deepest point of the dip.

The suppression of production is largely uniform for all isotopes except for ^{74}Se, ^{78}Kr, ^{84}Sr, and ^{180}Ta. Additionally, it appears that isotopes $A \geq 138$ are more strongly affected by the stronger GOSH-like dip. The isotopes ^{74}Se, ^{78}Kr, and ^{84}Sr have a minor increase from these dips because the location of their production is at the bottom of the O-shell where the dips are not present. The boost in production of ^{180}Ta is because the peak production and destruction locations happen to be centered at the exact same location as the GOSH dip, which lowers its destruction and slightly boosts its production. Isotopic ratios are not significantly affected by the dips, although the magnitude of the ratios can increase slightly. This demonstrates the importance of both the location and magnitude of the convective dips in the O-shell for convective-reactive γ process.

2.4.5 Nuclear physics impact and mixing dependencies

Here we present the impact of varying our adopted nuclear physics rates for the MLT and 3D-inspired mixing scenarios for an ingestion rate of 4×10^{-3} $\mathrm{M_{\odot} s^{-1}}$. The result for the MLT mixing scenario is shown in Figure 2.13 and Table 2.4, and the 3D-inspired mixing scenarios are shown in Figures 2.20–2.23 and Tables 2.4–2.8 which can be found in Appendix 2.6.3.

The MLT mixing scenario has an average spread in production of $\mathrm{OP}_{\mathrm{max}} - \mathrm{OP}_{\mathrm{min}} =$ 0.56 dex, and the 3D-inspired scenarios have an average spread of 0.59 dex, 0.69 dex, 0.76 dex, and 0.79 dex for the 1×, 3×, 10×, and 50× scenarios respectively. We find that the spread in production increases with mixing speed because the material is able to reach hotter temperatures and the number of possible nucleosynthetic pathways changes.

In the MLT mixing scenario about a third of the isotopes are not affected in any significant way, but in the 3D-inspired scenarios only 5 are not affected although the specific isotopes

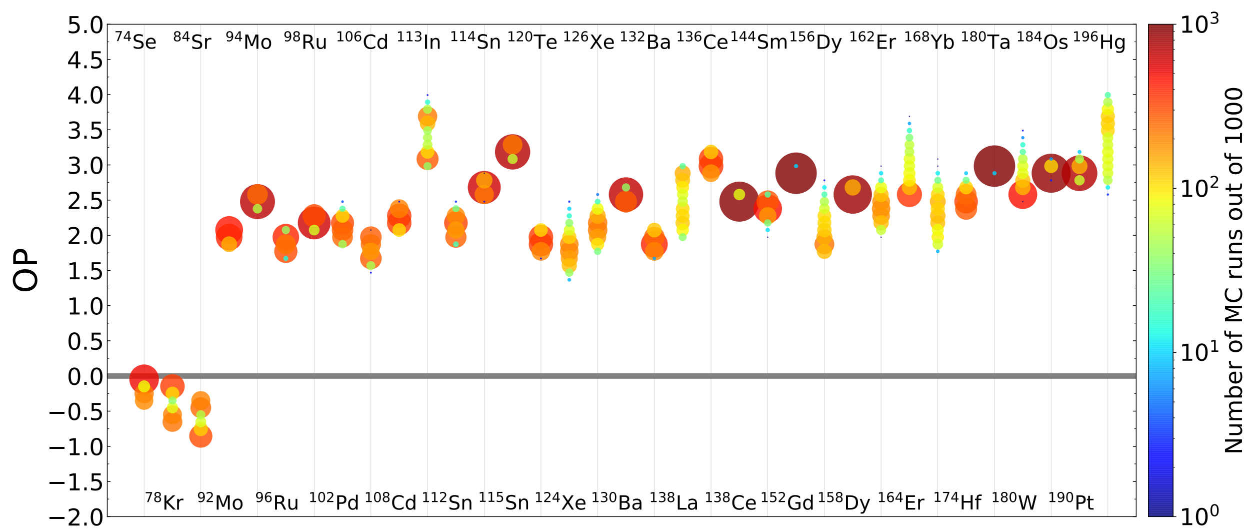

Figure 2.13: Histogram showing the spread due to varying (γ, p), (γ, n), (γ, α) and corresponding capture rates for unstable n-deficient isotopes from Se–Po for the MLT mixing scenario. Colour and size both correspond to the logarthimic binning of Monte Carlo runs. The average spread $\mathrm{OP}_{\max} - \mathrm{OP}_{\min} = 0.56$ dex. OP $= 0$ is the initial amount.

vary. The isotopes that appear the least affected across mixing scenario are ^{138}Ce, ^{152}Gd, ^{158}Dy, and ^{180}Ta.

The spread of an individual isotope is dependent on the mixing scenario. Species like ^{106}Cd, ^{156}Dy, and ^{180}W have a different spread in production for each mixing scenario. While the change can be monotonic for species like ^{180}W, ^{184}Os, and ^{190}Pt, for ^{106}Cd and ^{112}Sn, and ^{130}Ba they decrease for the 50×3D-inspired scenario.

This mixing scenario dependence for the spread is also seen for the distribution of OPas no isotope is double peaked across all mixing scenarios. As examples, ^{74}Se, ^{113}In, and ^{138}La in some mixing scenarios clearly are double peaked, indicating that there are distinctive branches in the nucleosynthetic pathways, but do not have it in others. Additionally, the magnitude of which peak is favoured is also dependent on the mixing scenario as seen for ^{78}Kr and ^{84}Sr.

Whether a particular reaction rate is correlated an isotope's final mass fraction along with the strength of the correlation is dependent on the mixing scenario. Table 2.2 lists the rates unique to single mixing scenario.

It is clearly important to consider the mixing conditions if an experiment is to be proposed to measure a reaction rate. As Table 2.2 shows for the 50×3D-inspired scenario, there are many reactions even for a single isotope that can be correlated uniquely in that mixing scenario. This clearly shows that a decreasing radial velocity profile and the exact magnitude of the mixing speeds are critical for understanding the nuclear reactions in this convective-reactive environment.

There are also correlated reactions that all mixing scenarios share as shown in Table 2.3. Additionally, all downturn cases share correlations not found in the MLT scenario: $X(^{115}\mathrm{Sn})$ with $^{110}\mathrm{Sn}(\gamma,\alpha)$ and $X(^{138}\mathrm{Ce})$ with $^{138}\mathrm{Nd}(\gamma,p)$. However, the shared correlations are not of equal strength across all mixing scenarios. As an example the final mass fraction of ^{156}Dy is correlated with $^{160}\mathrm{Er}(\gamma,\alpha)$, but in the 50×3D-inspired the correlation is much weaker. This underscores the possible difficulties in using 1D astrophysical sites to identify important reactions for nuclear physics experiments.

The spread in production for varying our adopted nuclear physics rates for each of the mixing scenarios is comparable to the spread seen from varying the mixing conditions. The results in Section 2.4.4 have a spread of 0.51 dex, Section 2.4.2 0.96 dex, and the maximum spread in Section 2.4.3 is 1.84 dex. The average spread in production from varying the nuclear physics rates ranges from 0.56 dex to 0.79 dex.

Rauscher et al. (2016) determined uncertainties for the explosive γ-process for a 15 $\mathrm{M}_\odot$ and two 25 $\mathrm{M}_\odot$ models with solar metallicity using a Monte Carlo method. The spread

Table 2.2: Reactions correlated with the production/destruction of an isotope unique to an individual mixing scenario.

Isotope	**Unique Correlated Reaction Rates**
MLT Mixing Case	
^{138}Ce	^{139}Pr(γ, p)
^{168}Yb	^{176}W(γ, α)
^{174}Hf	^{178}W(γ, n)
^{184}Os	^{186}Pt(γ, n), ^{188}Pt(γ, n)
3D-inspired Mixing Scenario	
^{113}In	^{114}In(γ, n)
^{152}Gd	^{150}Gd(γ, α), ^{196}Pb(γ, n)
^{180}Ta	^{179}Ta(γ, α)
3×3D-inspired Mixing Scenario	
^{180}W	^{181}Os(γ, n)
10×3D-inspired Mixing Scenario	
^{84}Sr	^{84}Rb(γ, n)
^{120}Te	^{119}Te(γ, n)
^{126}Xe	^{122}Xe(γ, n)
^{130}Ba	^{126}Ba(γ, p), ^{128}Ba(γ, α), ^{128}Ba(γ, p)
^{132}Ba	^{128}Ba(γ, α)
^{168}Yb	^{169}Hf(γ, n)
^{174}Hf	^{176}W(γ, α)
^{184}Os	^{185}Pt(γ, α)
50×3D-inspired Mixing Scenario	
^{92}Mo	^{100}Pd(γ, α), ^{100}Pd(γ, p), ^{110}Sn(γ, n), ^{110}Sn(γ, p)
^{96}Ru	^{97}Ru(γ, α), ^{110}Sn(γ, α), ^{110}Sn(γ, n), ^{110}Sn(γ, p)
^{102}Pd	^{104}Cd(γ, α), ^{104}Cd(γ, p)
^{106}Cd	^{104}Cd(γ, p), ^{110}Sn(γ, p)
^{108}Cd	^{110}Sn(γ, α)
^{112}Sn	^{110}Sn(γ, p)
^{115}Sn	^{122}Xe(γ, n)
^{120}Te	^{120}Xe(γ, α)
^{126}Xe	^{127}Ba(γ, n)
^{130}Ba	^{132}Ce(γ, α), ^{132}Ce(γ, n), ^{132}Ce(γ, p), ^{134}Ce(γ, α), ^{134}Ce(γ, n)
^{132}Ba	^{132}Ce(γ, α), ^{132}Ce(γ, n), ^{132}Ce(γ, p), ^{133}Ce(γ, n), ^{134}Ce(γ, α)
^{138}Ce	^{139}Nd(γ, n)
^{156}Dy	^{156}Er(γ, α), ^{158}Er(γ, α), ^{158}Er(γ, n)
^{162}Er	^{168}Hf(γ, n), ^{162}Yb(γ, α), ^{164}Yb(γ, α)
^{184}Os	^{184}Pt(γ, n)

Table 2.3: Reactions correlated with the production/destruction of an isotope shared across all mixing scenarios.

Isotope	**Shared Correlated Reaction Rates**
^{74}Se	^{75}Se(γ, n)
^{78}Kr	^{79}Kr(γ, n)
^{84}Sr	^{85}Sr(γ, n)
^{92}Mo	^{93}Mo(γ, n)
^{94}Mo	^{93}Mo(γ, n)
^{96}Ru	^{97}Ru(γ, n)
^{98}Ru	^{100}Pd(γ, α), ^{100}Pd(γ, p)
^{102}Pd	^{100}Pd(γ, α), ^{100}Pd(γ, p), ^{103}Pd(γ, n)
^{106}Cd	^{107}Cd(γ, n), ^{110}Sn(γ, α)
^{108}Cd	^{107}Cd(γ, n)
^{113}In	^{113}Sn(γ, n)
^{112}Sn	^{113}Sn(γ, n)
^{114}Sn	^{110}Sn(γ, α), ^{113}Sn(γ, n), ^{122}Xe(γ, α)
^{115}Sn	^{113}Sn(γ, n)
^{120}Te	^{122}Xe(γ, α), ^{122}Xe(γ, p)
^{124}Xe	^{122}Xe(γ, α), ^{122}Xe(γ, p)
^{138}La	^{137}La(γ, n)
^{136}Ce	^{138}Nd(γ, n), ^{138}Nd(γ, p), ^{140}Nd(γ, α)
^{144}Sm	^{196}Pb(γ, n), ^{142}Sm(γ, n), ^{142}Sm(γ, p), ^{143}Sm(γ, n)
^{152}Gd	^{152}Dy(γ, α)
^{156}Dy	^{160}Er(γ, α)
^{164}Er	^{164}Yb(γ, α), ^{164}Yb(γ, n)
^{174}Hf	^{174}W(γ, α)
^{180}Ta	^{179}Ta(γ, n)
^{180}W	^{180}Os(γ, α), ^{180}Os(γ, n), ^{196}Pb(γ, n)
^{184}Os	^{196}Pb(γ, n), ^{184}Pt(γ, α)
^{190}Pt	^{190}Hg(γ, α), ^{190}Hg(γ, n), ^{196}Pb(γ, n)
^{196}Hg	^{196}Pb(γ, n), ^{202}Pb(γ, n)

in their 90% probability interval for the 15 $M_\odot$ model was 0.61 dex, and for the 25 $M_\odot$ models were 0.63 dex and 0.99 dex. This is comparable to the maximum spread found in this work, although we are considering pre-explosive γ process. Rauscher et al. (2016) also provide tables of correlated rates, but only a handful of these rates appear in our Tables 2.4–2.8 including those they find with $r_\mathrm{P} \geq |0.65|$. This is because the convective-reactive environment we consider allows for different reaction pathways to be favoured depending on the mixing conditions.

2.5 Discussions and Conclusion

In this paper, we have shown that understanding the mixing details in the O-burning shell during an O-C shell merger is critical for accurately modelling the nucleosynthesis of the *p* nuclei. This work raises the importance of 3D hydrodynamic simulations for understanding the nucleosynthesis in O-C shell mergers (Bazan and Arnett, 1994; Yadav et al., 2020; Rizzuti et al., 2024a). We have demonstrated the convective-reactive nature of the nucleosynthesis in the O-shell, where the timescales for advection and reaction are comparable and our results show that:

- A gradual downturn motivated by 3D simulations increases production of the *p* nuclei, but increasing mixing speeds impacts the production in a non-linear and non-monotonic way.
- The ratio of isotopic pairs are sensitive to the mixing scenario.
- Increasing the entrainment rate of C-shell material increases the production of the *p* nuclei.
- Without entrainment, the *p* nuclei are negligibly produced or net destroyed.
- A dip in the convective profile can suppress the production of the *p* nuclei.
- The location and magnitude of the convective dip impacts the nucleosynthesis of the *p* nuclei.
- Varying the adopted reactions rates have a comparable spread in production to changing mixing conditions.
- The spread due to varying the reaction rates is dependent on the mixing scenario.

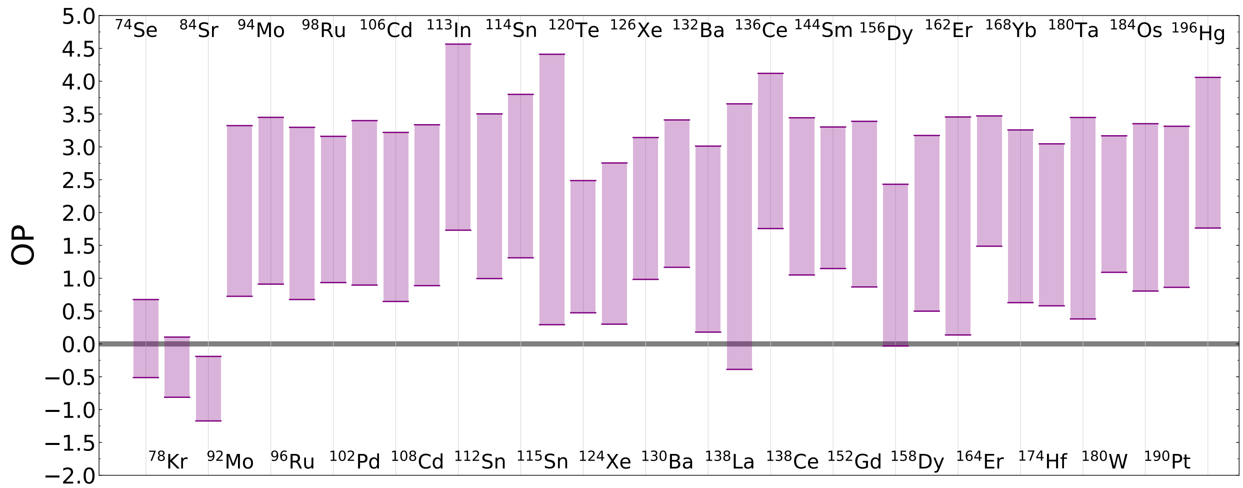

Figure 2.14: Bars representing the maximum and minimum OP across all mixing scenarios, excluding those without C-shell ingestion. The average spread $\mathrm{OP_{max}} - \mathrm{OP_{min}} = 2.45$ dex. $\mathrm{OP} = 0$ is the initial amount.

- Whether a reaction rate is correlated with an isotope is dependent on the mixing scenario, and some are unique to a single mixing scenario.

Figure 2.14 shows the maximum spread for the p nuclei across all mixing scenarios considered in this paper, excluding the case of no merger, have an average spread of 2.45 dex. This shows the significant impact of the macrophysical uncertainties in 1D stellar models on nucleosynthesis, and this highlights the importance of understanding hydrodynamic models better as the shell merger dominates production of the p nuclei (Roberti et al., 2023, 2024).

Although we find that the mixing conditions significantly impact the results of Section 2.4.5, not all scenarios are equally likely to represent the conditions in a merger. The MLT mixing scenario, 1×, and 50× case may not be representative of the conditions for realistic O-C shell mergers. 3D hydrodynamic simulations show the O-shell has a downturn to radial convective velocities and mixing speeds roughly 3−10 times larger than what MLT predicts (Jones et al., 2017; Andrassy et al., 2020; Rizzuti et al., 2024a). This suggests that the 3× and 10× downturn mixing scenarios are likely more representative of the conditions in a merger. The exploration done in this paper shows the importance of understanding the mixing conditions, but not all scenarios are equally likely to represent the conditions in a merger.

The results in this work are important for the interpretation of presolar grains. Fok et al. (2024) argue that the nucleosynthesis in O-C shell mergers matter when interpreting the $^{29}\mathrm{Si}/^{28}\mathrm{Si}$ ratio seen in grains. As shown in this work, the ratio of isotopic pairs is sensitive to the mixing conditions in the O-shell. This means that comparing the grains data to the results of this work can be used as a diagnostic tool to constrain the mixing details of O-C shell mergers and connect measured isotopic ratios to 3D hydrodynamics.

There are limitations to the results provided in this work and further extensions that could be done. We have focused on the impact of mixing conditions and varying reaction rates, but do not investigate how different stellar conditions are relevant to the nucleosynthesis or if changing the stable seeds from the C-shell impact these results. The distribution of stable seeds and the metallicity of the star are of significant importance for the nucleosynthesis of the p nuclei (Travaglio et al., 2015; Battino et al., 2020). It is possible that earlier in stellar evolution that weak s process in the C-shell could modify the stable seeds (Pignatari et al., 2010), which could be relevant for the lighter p nuclei. The NuGrid 15 $M_\odot$ $Z = 0.02$ model does have the largest overproduction factor compared to the other NuGrid models with shell mergers (Ritter et al., 2018a). However, the other models have $Z = 0.01$, and since there is a metallicity dependency of the γ process Travaglio et al. (2015), this is not surprising and it is likely that the spread in production due to the mixing conditions would be similar.

Other stellar models could have different O-shell sizes and temperature profiles due to different mixing prescriptions, initial mass, and rotations, which would directly affect locations of peak burning and how the mixing conditions impact the nucleosynthesis. However, if the shell is convective-reactive, a spread in production comparing mixing scenarios would still be expected.

Finally, explosive γ process can contribute 10% to 40% of the final yields of the p nuclei even with a shell merger (Roberti et al., 2023). However, O-C shell mergers are critical to the nucleosynthesis of a massive star prior to the CCSN regardless of explosive energy (Roberti et al., 2024). Even if the results in Figure 2.14 do not represent the whole of the nucleosynthesis of the p nuclei, they are still critical for understanding the nucleosynthesis in the O-shell prior to the CCSN.

The results of this work have implications beyond the p nuclei. The light odd-Z isotopes ^{31}P, ^{35}Cl, ^{39}K, and ^{45}Sc are also produced during O-C shell mergers and are likewise impacted by varying the mixing conditions (Ritter et al., 2018a; Roberti et al., 2025). Observations have found P-enhanced stars (Masseron et al., 2020; Brauner et al., 2023, 2024) which could be explained by a previous massive star having a O-C shell merger. Additionally, ^{40}K has been shown to be relevant to the heating of planets early in their formation (Frank et al., 2014; O'Neill et al., 2020) The shell merger also has a distinct impact on galactic chemical evolution (GCE) models (Ritter et al., 2018a), and massive star models show this feature regardless of metallicity and stellar evolution model (Roberti et al., 2025). Further work is needed to understand the impact of the macrophysical uncertainties in 1D stellar models on the nucleosynthesis of these light odd-Z isotopes.

2.6 Appendix

2.6.1 Correlations of nuclear reaction rates

The Pearson correlation coefficient, r_{P}, is insufficient to quantify the importnace of a correlated rate as demonstrated by Figure 2.15. A strong correlation does not correspond directly to a signifcant change to the final mass fraction of a species. In addition to the correlation coefficient, a slope ζ is provided that better describes this behaviour. ζ is the slope of a linear regression between $\log_{10}(X/X_{\text{no variation}})$ and $\log_{10}(\text{variation factor})$ where X is the final decayed mass fraction for Monte Carlo rates and $X_{\text{no variation}})$ is the final decayed mass fraction for the default case where all variation factors are equal to 1.

It is clear from Figure 2.15 that similarly correlations do not directly mean that the reaction rate is relevant for the production of a species, but the slope ζ also does not directly point to this. The bottom right plot showing the impact on the final mass fraction of ^{196}Hg from varying the input rate for ^{196}Pb(γ, n) has both a strong slope and correlation, but the bottom left plot has a strong correlation and a weak slope. Reaction rates that are both highly correlated and have a strong slope will have direct impact on the production of species.

Another caveat is that this method of varying the reaction rates in a Monte Carlo way does not distinguish between the photo-disintegration and corresponding capture rate. All correlated rates are reported according to their photo-disintegration rates, but as shown by the upper left plot of Figure 2.15 for ^{74}Se and ^{75}Se$(\gamma, n)^{74}$Se this results sometimes in saying that a reaction that would be considered a production term is strongly anti-correlated with production. This is because when we apply the variation factor, we do so for both ^{75}Se(γ, n) and ^{74}Se(n, γ). As shown in Section 2.4.1, the reactions in this shell are not balanced and both could be relevant, although as Figure 2.8 shows for heavier species the (γ, n) rate is typically the dominant reaction.

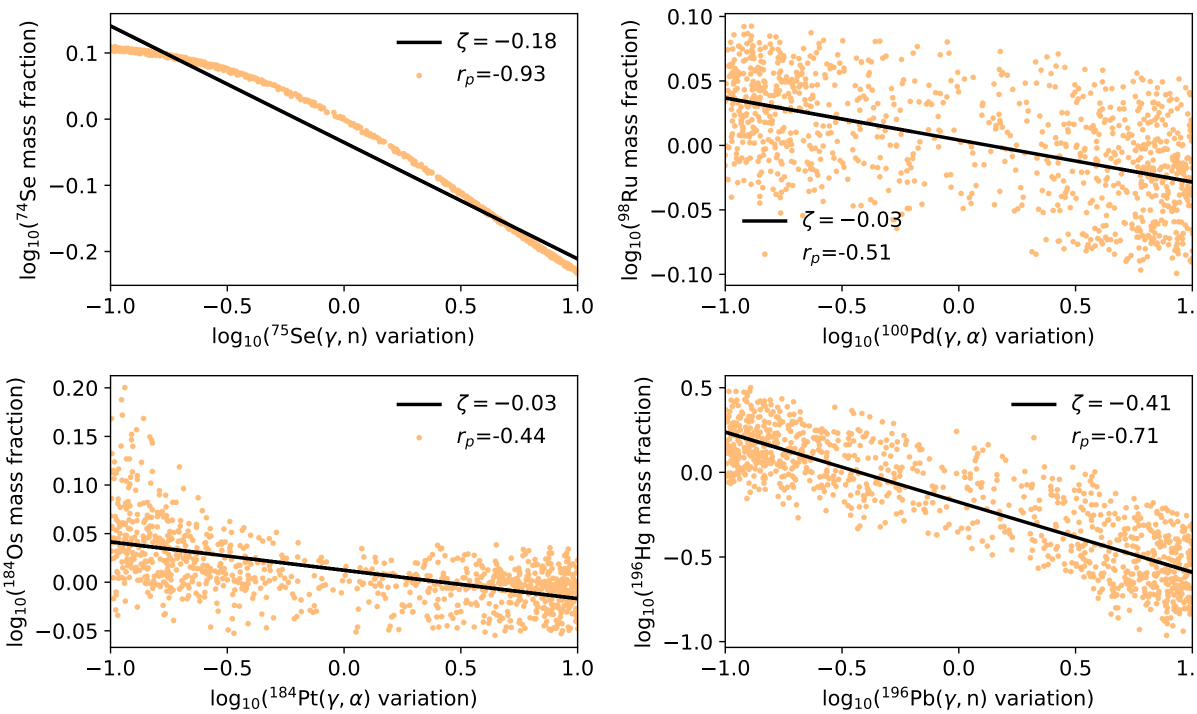

Figure 2.15: Examples of strongly correlated mass fractions with reaction rates for four species for the MLT mixing scenario. The orange dots are the values and the black line is the best linear fit to OP_{var} and $\log_{10}$(variation factor). The top left plot and bottom right plot show strong correlations for ^{74}Se and ^{196}Hg that directly impact the mass fraction. The top right plot shows a correlation for ^{98}Ru where there is considerable spread in the mass fraction. The bottom left plot shows a correlation for ^{184}Os where the mass fraction is strongly affected only for multiplication factors less than 1.

2.6.2 Results of varying ingestion rate

Here we provide the results for varying the ingestion rates for the downturn mixing scenarios.

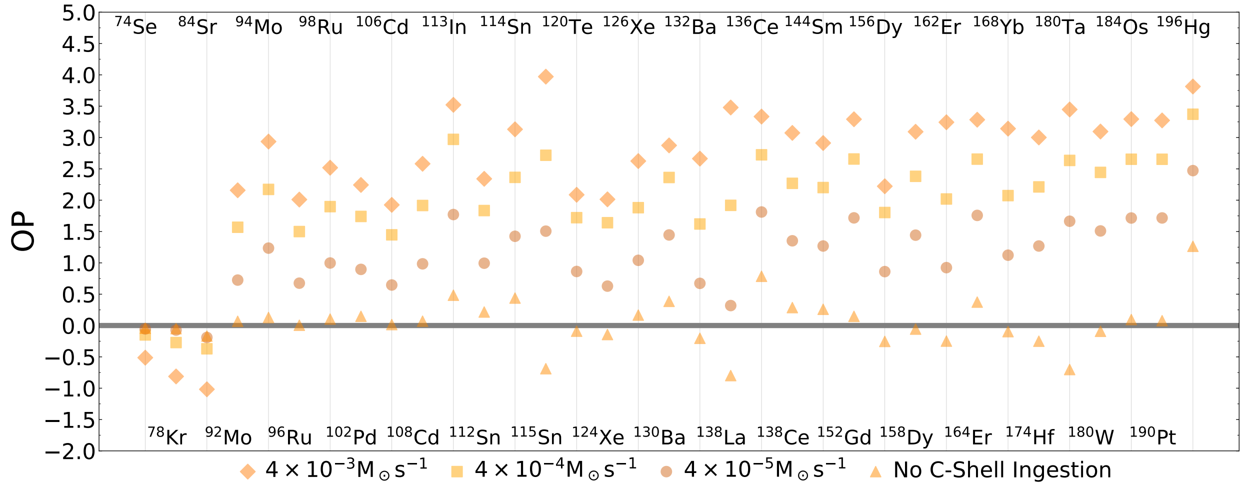

Figure 2.16: The overproduction compared to initial of the p nuclei for the 3D-inspired mixing scenario for no ingestion, 4×10^{-5} $M_\odot s^{-1}$, 4×10^{-4} $M_\odot s^{-1}$, and 4×10^{-3} $M_\odot s^{-1}$. The average spread in production $OP_{max} - OP_{min} = 1.58$ dex. $OP = 0$ is the initial amount.

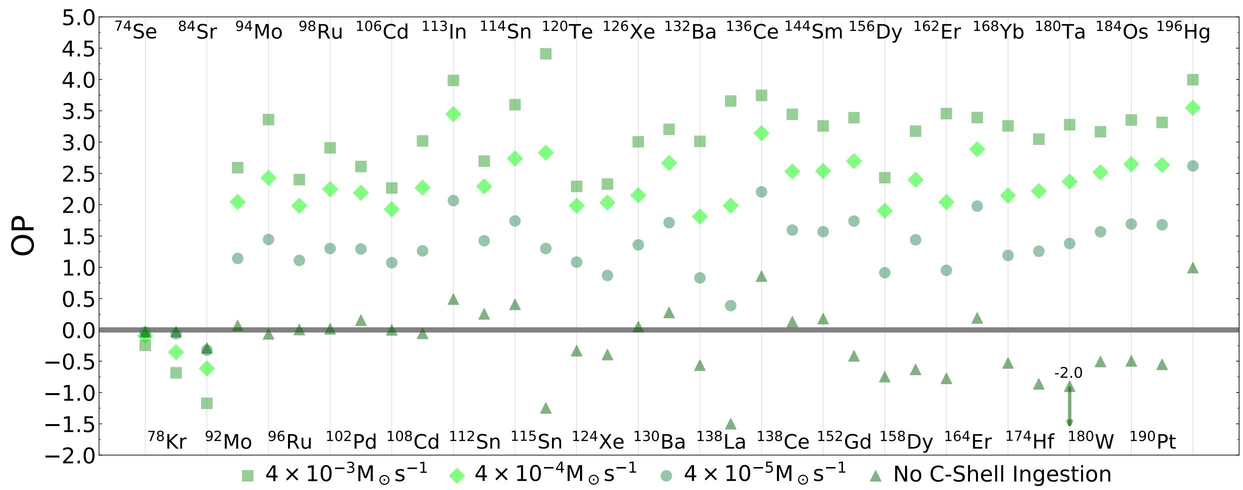

Figure 2.17: The overproduction compared to initial of the p nuclei for the 3×3D-inspired mixing scenario for no ingestion, 4×10^{-5} $M_\odot s^{-1}$, 4×10^{-4} $M_\odot s^{-1}$, and 4×10^{-3} $M_\odot s^{-1}$. The average spread in production $OP_{max} - OP_{min} = 1.64$ dex excluding the no ingestion case. $OP = 0$ is the initial amount.

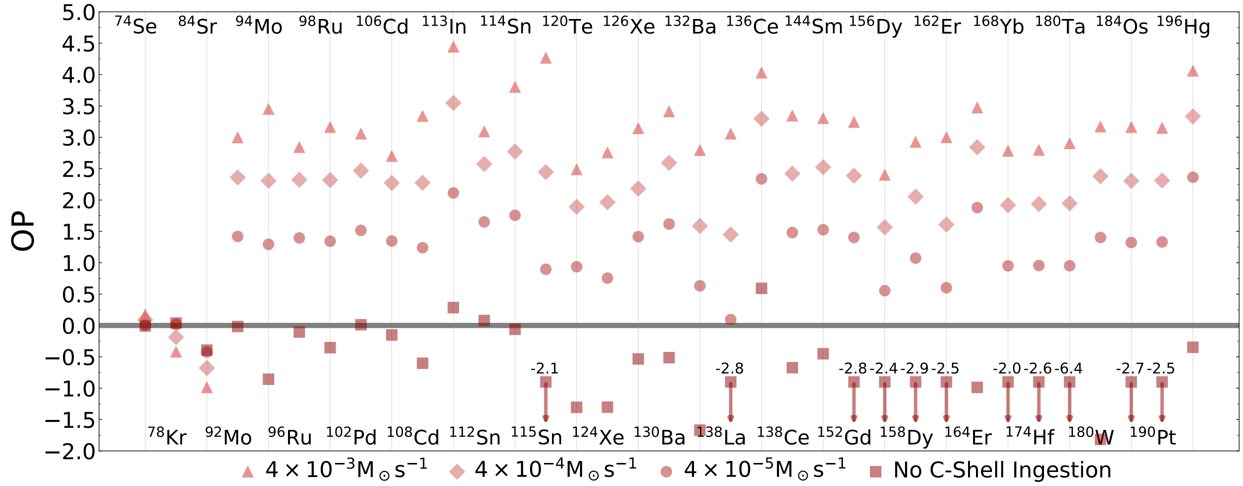

Figure 2.18: The overproduction compared to initial of the p nuclei for the 10×3D-inspired mixing scenario for no ingestion, 4×10^{-5} $\mathrm{M_{\odot}s^{-1}}$, 4×10^{-4} $\mathrm{M_{\odot}s^{-1}}$, and 4×10^{-3} $\mathrm{M_{\odot}s^{-1}}$. Arrows denote OP out of bounds and the true OP is written above. The average spread in production $\mathrm{OP_{max}} - \mathrm{OP_{min}} = 1.78$ dex excluding the no ingestion case. OP = 0 is the initial amount.

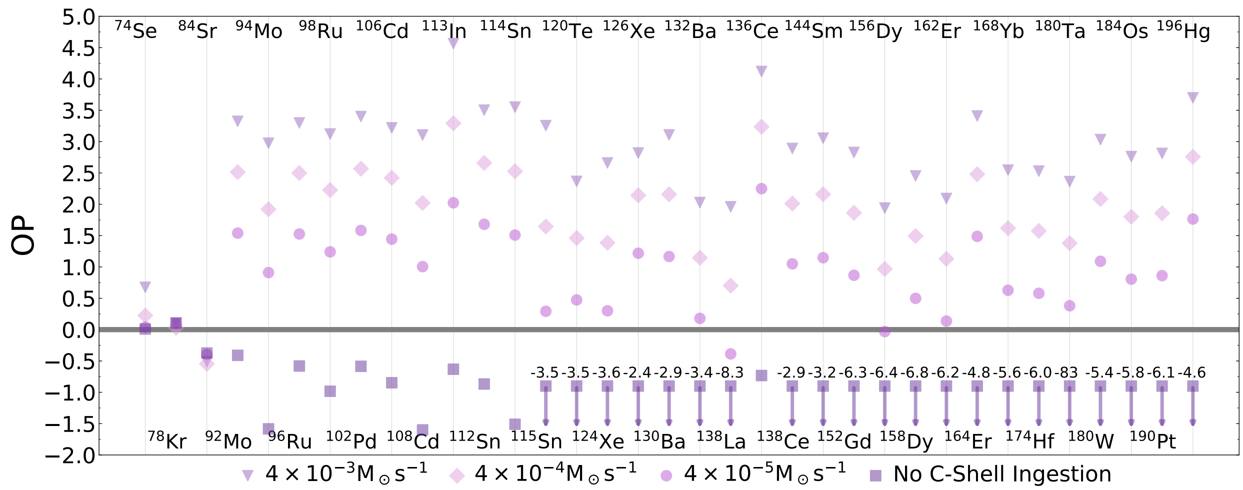

Figure 2.19: The overproduction compared to initial of the p nuclei for the 50×3D-inspired mixing scenario for no ingestion, 4×10^{-5} $\mathrm{M_{\odot}s^{-1}}$, 4×10^{-4} $\mathrm{M_{\odot}s^{-1}}$, and 4×10^{-3} $\mathrm{M_{\odot}s^{-1}}$. Arrows denote OP out of bounds and the true OP is written above. The average spread in production $\mathrm{OP_{max}} - \mathrm{OP_{min}} = 1.84$ dex excluding the no ingestion case. OP = 0 is the initial amount.

2.6.3 Results of varying the input nuclear physics

Here we provide the results for varying the input nuclear reactions for the downturn mixing scenarios and the reaction rate correlation tables for the MLT and downturn scenarios.

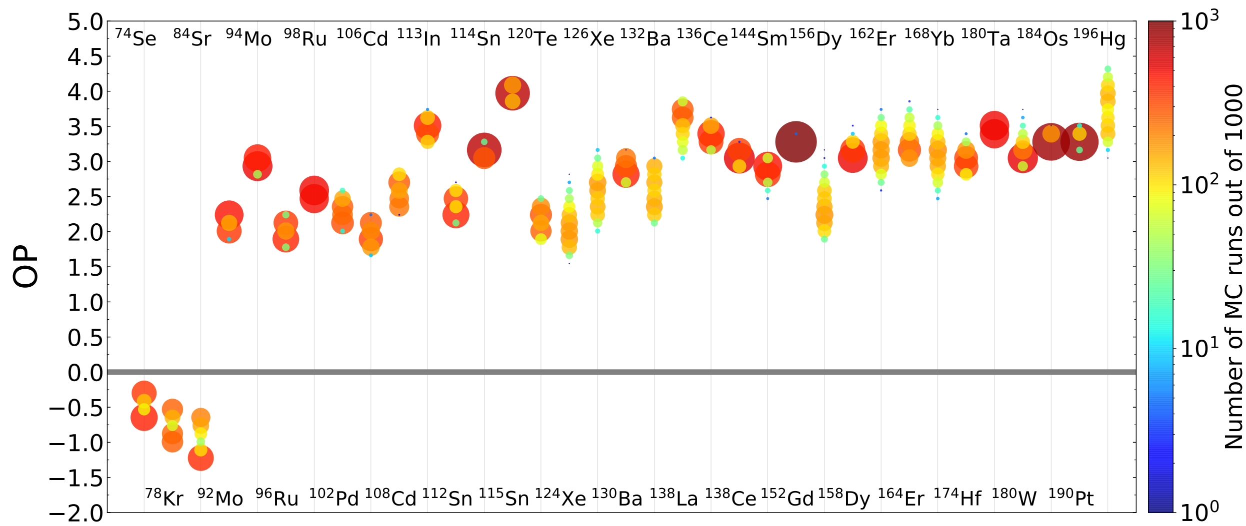

Figure 2.20: Histogram showing the spread due to varying (γ, p), (γ, n), (γ, α) and corresponding capture rates for unstable n-deficient isotopes from Se–Po for the 3D-inspired mixing scenario. Colour and size both correspond to the logarthimic binning of Monte Carlo runs. The average spread $\mathrm{OP_{max}} - \mathrm{OP_{min}} = 0.59$ dex. OP = 0 is the initial amount.

Table 2.4: Correlations and ζ slopes between mass fraction and reaction rates for the MLT mixing scenario.

Isotope	**Reaction**	$\mathbf{r_P}$	ζ	**Isotope**	**Reaction**	$\mathbf{r_P}$	ζ
^{74}Se	^{75}Se(γ, n)	−0.93	−0.18	^{144}Sm	^{142}Sm(γ, n)	−0.19	−0.02
^{78}Kr	^{79}Kr(γ, n)	−0.88	−0.28		^{142}Sm(γ, p)	−0.17	−0.02
^{84}Sr	^{85}Sr(γ, n)	−0.88	−0.28		^{143}Sm(γ, n)	−0.25	−0.03
^{92}Mo	^{93}Mo(γ, n)	−0.94	−0.07		^{146}Sm(γ, n)	0.20	0.03
	^{110}Sn(γ, α)	0.16	0.01		^{150}Gd(γ, n)	0.17	0.02
^{94}Mo	^{93}Mo(γ, n)	0.97	0.06		^{150}Gd(γ, α)	−0.15	−0.02
^{96}Ru	^{97}Ru(γ, n)	−0.88	−0.12		^{196}Pb(γ, n)	0.47	0.06
	^{100}Pd(γ, α)	0.20	0.03		^{202}Pb(γ, n)	0.21	0.03
^{98}Ru	^{97}Ru(γ, n)	0.36	0.02	^{152}Gd	^{152}Dy(γ, α)	−0.40	−0.01
	^{100}Pd(γ, p)	0.62	0.04		^{154}Dy(γ, α)	−0.15	−0.00
	^{100}Pd(γ, α)	−0.51	−0.03		^{160}Er(γ, α)	0.39	0.00

Continued on next page

Table 2.4 – continued from previous page

Isotope	Reaction	$\mathbf{r_P}$	ζ	Isotope	Reaction	$\mathbf{r_P}$	ζ
^{102}Pd	^{100}Pd(γ,p)	−0.29	−0.05	^{156}Dy	^{159}Er(γ,n)	−0.18	−0.06
	^{100}Pd(γ,α)	−0.30	−0.05		^{160}Er(γ,α)	0.74	0.26
	^{103}Pd(γ,n)	−0.66	−0.12		^{202}Pb(γ,n)	0.18	0.05
^{106}Cd	^{107}Cd(γ,n)	−0.86	−0.16	^{158}Dy	^{158}Er(γ,α)	−0.23	−0.01
	^{110}Sn(γ,α)	0.23	0.05		^{160}Er(γ,α)	0.56	0.01
^{108}Cd	^{107}Cd(γ,n)	0.72	0.10		^{196}Pb(γ,n)	0.16	0.00
	^{109}Cd(γ,n)	−0.46	−0.06		^{202}Pb(γ,n)	0.18	0.00
	^{110}Sn(γ,p)	0.16	0.02	^{162}Er	^{159}Er(γ,n)	−0.18	−0.06
^{113}In	^{113}Sn(γ,n)	0.91	0.37		^{160}Er(γ,n)	−0.18	−0.06
^{112}Sn	^{110}Sn(γ,α)	−0.17	−0.03		^{160}Er(γ,α)	−0.26	−0.07
	^{113}Sn(γ,n)	−0.83	−0.15		^{161}Er(γ,n)	0.21	0.06
^{114}Sn	^{110}Sn(γ,α)	−0.15	−0.01		^{166}Yb(γ,α)	0.53	0.14
	^{113}Sn(γ,n)	0.74	0.06		^{196}Pb(γ,n)	0.25	0.07
	^{122}Xe(γ,n)	−0.18	−0.01		^{202}Pb(γ,n)	0.17	0.05
	^{122}Xe(γ,p)	0.20	0.02	^{164}Er	^{164}Yb(γ,n)	−0.24	−0.09
	^{122}Xe(γ,α)	0.41	0.04		^{164}Yb(γ,α)	−0.58	−0.32
^{115}Sn	^{113}Sn(γ,n)	0.80	0.05		^{196}Pb(γ,n)	0.17	0.05
	^{122}Xe(γ,p)	0.17	0.01	^{168}Yb	^{168}Hf(γ,α)	−0.28	−0.14
	^{122}Xe(γ,α)	0.35	0.02		^{172}Hf(γ,α)	0.60	0.24
^{120}Te	^{121}Te(γ,n)	−0.71	−0.09		^{176}W(γ,α)	0.20	0.07
	^{122}Xe(γ,p)	0.45	0.05		^{196}Pb(γ,n)	0.21	0.07
	^{122}Xe(γ,α)	−0.32	−0.04	^{174}Hf	^{174}W(γ,n)	−0.21	−0.03
^{124}Xe	^{122}Xe(γ,n)	−0.17	−0.04		^{174}W(γ,α)	−0.40	−0.08
	^{122}Xe(γ,p)	−0.24	−0.06		^{178}W(γ,n)	−0.15	−0.03
	^{122}Xe(γ,α)	−0.45	−0.15		^{178}W(γ,α)	0.42	0.07
	^{123}Xe(γ,n)	0.16	0.03	^{180}Ta	^{179}Ta(γ,n)	−0.91	−0.02
	^{125}Xe(γ,n)	−0.46	−0.16	^{180}W	^{180}Os(γ,n)	−0.28	−0.07
^{126}Xe	^{122}Xe(γ,α)	−0.34	−0.09		^{180}Os(γ,α)	−0.52	−0.21
	^{125}Xe(γ,n)	0.49	0.13		^{196}Pb(γ,n)	0.21	0.05
	^{127}Xe(γ,n)	−0.25	−0.07	^{184}Os	^{185}Os(γ,n)	0.33	0.02
	^{126}Ba(γ,p)	−0.18	−0.04		^{184}Pt(γ,α)	−0.44	−0.03
	^{126}Ba(γ,α)	−0.32	−0.09		^{186}Pt(γ,n)	0.17	0.01
^{130}Ba	^{126}Ba(γ,α)	−0.17	−0.01		^{186}Pt(γ,α)	−0.16	−0.01

Continued on next page

Table 2.4 – continued from previous page

Isotope	Reaction	$\mathbf{r_P}$	ζ	Isotope	Reaction	$\mathbf{r_P}$	ζ
	^{128}Ba(γ,n)	−0.19	−0.01		^{188}Pt(γ,n)	−0.17	−0.01
	^{129}Ba(γ,n)	0.23	0.01		^{196}Pb(γ,n)	0.16	0.01
	^{131}Ba(γ,n)	−0.79	−0.05	^{190}Pt	^{190}Hg(γ,n)	−0.28	−0.03
^{132}Ba	^{131}Ba(γ,n)	0.66	0.09		^{190}Hg(γ,α)	−0.51	−0.06
	^{133}Ba(γ,n)	−0.61	−0.08		^{196}Pb(γ,n)	0.20	0.02
^{138}La	^{137}La(γ,n)	−0.71	−0.37	^{196}Hg	^{196}Pb(γ,n)	−0.71	−0.41
^{136}Ce	^{138}Nd(γ,n)	−0.38	−0.05		^{197}Pb(γ,n)	−0.16	−0.06
	^{138}Nd(γ,p)	0.65	0.08		^{200}Pb(γ,n)	0.19	0.10
	^{138}Nd(γ,α)	−0.16	−0.02		^{202}Pb(γ,n)	0.30	0.14
	^{140}Nd(γ,α)	0.34	0.04				
^{138}Ce	^{137}Ce(γ,n)	0.51	0.02				
	^{139}Ce(γ,n)	−0.43	−0.01				
	^{139}Pr(γ,p)	0.29	0.01				
	^{138}Nd(γ,n)	−0.30	−0.01				
	^{138}Nd(γ,α)	−0.22	−0.01				

Table 2.5: Correlations and ζ slopes between mass fraction and reaction rates for the 3D-inspired mixing scenario.

Isotope	Reaction	$\mathbf{r_P}$	ζ	Isotope	Reaction	$\mathbf{r_P}$	ζ
^{74}Se	^{75}Se(γ,n)	−0.83	−0.22	^{152}Gd	^{150}Gd(γ,α)	−0.18	−0.00
^{78}Kr	^{79}Kr(γ,n)	−0.79	−0.24		^{151}Gd(γ,n)	−0.20	−0.00
^{84}Sr	^{85}Sr(γ,n)	−0.80	−0.34		^{152}Dy(γ,α)	−0.31	−0.01
^{92}Mo	^{93}Mo(γ,n)	−0.91	−0.14		^{154}Dy(γ,α)	−0.17	−0.00
^{94}Mo	^{93}Mo(γ,n)	0.93	0.10		^{160}Er(γ,α)	0.39	0.01
^{96}Ru	^{97}Ru(γ,n)	−0.82	−0.14		^{196}Pb(γ,n)	0.16	0.00
	^{100}Pd(γ,α)	0.21	0.04	^{156}Dy	^{157}Dy(γ,n)	−0.28	−0.12
^{98}Ru	^{97}Ru(γ,n)	0.50	0.02		^{159}Er(γ,n)	−0.17	−0.05
	^{100}Pd(γ,p)	0.58	0.03		^{160}Er(γ,α)	0.68	0.26
	^{100}Pd(γ,α)	−0.31	−0.01		^{196}Pb(γ,n)	0.16	0.05
	^{110}Sn(γ,α)	0.22	0.01	^{158}Dy	^{157}Dy(γ,n)	0.24	0.03
^{102}Pd	^{100}Pd(γ,p)	−0.18	−0.04		^{159}Dy(γ,n)	−0.21	−0.03
	^{100}Pd(γ,α)	−0.21	−0.04		^{159}Er(γ,n)	−0.17	−0.02

Continued on next page

Table 2.5 – continued from previous page

Isotope	Reaction	r_P	ζ	Isotope	Reaction	r_P	ζ
	^{103}Pd(γ,n)	−0.72	−0.16		^{160}Er(γ,α)	0.66	0.07
^{106}Cd	^{107}Cd(γ,n)	−0.75	−0.16		^{196}Pb(γ,n)	0.16	0.02
	^{110}Sn(γ,α)	0.31	0.07		^{202}Pb(γ,n)	0.20	0.02
^{108}Cd	^{107}Cd(γ,n)	0.20	0.05	^{162}Er	^{159}Er(γ,n)	−0.25	−0.10
	^{109}Cd(γ,n)	−0.83	−0.20		^{159}Er(γ,α)	−0.16	−0.05
^{113}In	^{114}In(γ,n)	−0.32	−0.05		^{160}Er(γ,n)	−0.23	−0.09
	^{110}Sn(γ,p)	−0.16	−0.02		^{160}Er(γ,α)	−0.44	−0.15
	^{110}Sn(γ,α)	−0.27	−0.04		^{161}Er(γ,n)	0.16	0.06
	^{113}Sn(γ,n)	0.60	0.10		^{166}Yb(γ,α)	0.33	0.11
	^{169}Lu(γ,n)	0.15	0.02		^{196}Pb(γ,n)	0.27	0.10
^{112}Sn	^{113}Sn(γ,n)	−0.80	−0.18		^{202}Pb(γ,n)	0.16	0.06
^{114}Sn	^{110}Sn(γ,α)	−0.17	−0.01	^{164}Er	^{164}Yb(γ,n)	−0.26	−0.07
	^{113}Sn(γ,n)	0.65	0.05		^{164}Yb(γ,α)	−0.56	−0.18
	^{122}Xe(γ,p)	0.21	0.02		^{196}Pb(γ,n)	0.21	0.06
	^{122}Xe(γ,α)	0.44	0.03		^{202}Pb(γ,n)	0.18	0.05
^{115}Sn	^{110}Sn(γ,α)	−0.16	−0.01	^{168}Yb	^{164}Yb(γ,α)	−0.19	−0.09
	^{113}Sn(γ,n)	0.74	0.06		^{166}Yb(γ,n)	−0.22	−0.08
	^{122}Xe(γ,p)	0.16	0.01		^{166}Yb(γ,α)	−0.38	−0.13
	^{122}Xe(γ,α)	0.33	0.03		^{167}Yb(γ,n)	0.20	0.07
^{120}Te	^{121}Te(γ,n)	−0.76	−0.19		^{172}Hf(γ,α)	0.43	0.16
	^{122}Xe(γ,p)	0.26	0.05		^{196}Pb(γ,n)	0.28	0.09
	^{122}Xe(γ,α)	−0.23	−0.05		^{202}Pb(γ,n)	0.19	0.06
^{124}Xe	^{122}Xe(γ,p)	−0.19	−0.05	^{174}Hf	^{172}Hf(γ,α)	−0.40	−0.07
	^{122}Xe(γ,α)	−0.39	−0.15		^{174}W(γ,α)	−0.19	−0.04
	^{125}Xe(γ,n)	−0.54	−0.23		^{178}W(γ,α)	0.51	0.09
^{126}Xe	^{122}Xe(γ,α)	−0.35	−0.14		^{182}Os(γ,α)	0.18	0.03
	^{125}Xe(γ,n)	0.22	0.09		^{196}Pb(γ,n)	0.19	0.03
	^{127}Xe(γ,n)	−0.63	−0.27	^{180}Ta	^{179}Ta(γ,n)	−0.91	−0.07
^{130}Ba	^{131}Ba(γ,n)	−0.82	−0.13		^{179}Ta(γ,α)	−0.15	−0.01
^{132}Ba	^{131}Ba(γ,n)	0.36	0.14	^{180}W	^{180}Os(γ,n)	−0.27	−0.05
	^{133}Ba(γ,n)	−0.76	−0.30		^{180}Os(γ,α)	−0.51	−0.13
^{138}La	^{133}La(γ,p)	−0.20	−0.07		^{196}Pb(γ,n)	0.23	0.04
	^{135}La(γ,n)	−0.35	−0.12	^{184}Os	^{182}Os(γ,α)	−0.38	−0.02

Continued on next page

Table 2.5 – continued from previous page

Isotope	Reaction	$\mathbf{r_P}$	ζ	Isotope	Reaction	$\mathbf{r_P}$	ζ
	^{136}La(γ,n)	0.26	0.08		^{184}Pt(γ,α)	−0.33	−0.02
	^{137}La(γ,n)	−0.34	−0.14		^{186}Pt(γ,α)	−0.15	−0.01
^{136}Ce	^{137}Ce(γ,n)	−0.49	−0.07		^{188}Pt(γ,α)	0.23	0.01
	^{138}Nd(γ,n)	−0.27	−0.03		^{196}Pb(γ,n)	0.19	0.01
	^{138}Nd(γ,p)	0.48	0.06	^{190}Pt	^{190}Hg(γ,n)	−0.30	−0.02
	^{140}Nd(γ,α)	0.38	0.04		^{190}Hg(γ,α)	−0.50	−0.05
^{138}Ce	^{137}Ce(γ,n)	0.65	0.07		^{196}Pb(γ,n)	0.20	0.02
	^{139}Ce(γ,n)	−0.52	−0.05	^{196}Hg	^{196}Pb(γ,n)	−0.68	−0.35
	^{138}Nd(γ,p)	0.16	0.01		^{197}Pb(γ,n)	−0.26	−0.10
^{144}Sm	^{142}Sm(γ,n)	−0.15	−0.02		^{200}Pb(γ,n)	0.17	0.07
	^{142}Sm(γ,p)	−0.16	−0.02		^{202}Pb(γ,n)	0.24	0.10
	^{143}Sm(γ,n)	−0.19	−0.03				
	^{146}Sm(γ,n)	0.19	0.03				
	^{146}Sm(γ,α)	−0.22	−0.03				
	^{150}Gd(γ,n)	0.21	0.03				
	^{150}Gd(γ,α)	−0.20	−0.03				
	^{196}Pb(γ,n)	0.46	0.06				
	^{202}Pb(γ,n)	0.20	0.03				

Table 2.6: Correlations and ζ slopes between mass fraction and reaction rates for the 3×3D-inspired mixing scenario.

Isotope	Reaction	$\mathbf{r_P}$	ζ	Isotope	Reaction	$\mathbf{r_P}$	ζ
^{74}Se	^{75}Se(γ,n)	−0.81	−0.24	^{144}Sm	^{142}Sm(γ,n)	−0.23	−0.02
^{78}Kr	^{79}Kr(γ,n)	−0.76	−0.20		^{142}Sm(γ,p)	−0.23	−0.03
^{84}Sr	^{85}Sr(γ,n)	−0.75	−0.26		^{143}Sm(γ,n)	−0.30	−0.04
^{92}Mo	^{93}Mo(γ,n)	−0.91	−0.23		^{146}Sm(γ,α)	−0.17	−0.02
^{94}Mo	^{93}Mo(γ,n)	0.92	0.17		^{150}Gd(γ,n)	0.17	0.02
^{96}Ru	^{97}Ru(γ,n)	−0.84	−0.25		^{150}Gd(γ,α)	−0.16	−0.02
^{98}Ru	^{97}Ru(γ,n)	0.65	0.05		^{196}Pb(γ,n)	0.44	0.05
	^{100}Pd(γ,p)	0.47	0.03		^{202}Pb(γ,n)	0.17	0.02
	^{100}Pd(γ,α)	−0.23	−0.02	^{152}Gd	^{151}Gd(γ,n)	−0.17	−0.00
	^{110}Sn(γ,α)	0.21	0.01		^{152}Dy(γ,α)	−0.39	−0.01

Continued on next page

Table 2.6 – continued from previous page

Isotope	Reaction	$\mathbf{r_P}$	ζ	Isotope	Reaction	$\mathbf{r_P}$	ζ
^{102}Pd	^{100}Pd(γ,p)	−0.20	−0.06		^{154}Dy(γ,α)	−0.19	−0.00
	^{100}Pd(γ,α)	−0.22	−0.06		^{160}Er(γ,α)	0.33	0.01
	^{103}Pd(γ,n)	−0.71	−0.26	^{156}Dy	^{157}Dy(γ,n)	−0.21	−0.08
^{106}Cd	^{107}Cd(γ,n)	−0.78	−0.27		^{159}Er(γ,n)	−0.18	−0.06
	^{110}Sn(γ,α)	0.21	0.08		^{160}Er(γ,α)	0.69	0.25
^{108}Cd	^{107}Cd(γ,n)	0.24	0.08		^{196}Pb(γ,n)	0.15	0.05
	^{109}Cd(γ,n)	−0.80	−0.24		^{202}Pb(γ,n)	0.15	0.05
^{113}In	^{110}Sn(γ,p)	−0.22	−0.04	^{158}Dy	^{157}Dy(γ,n)	0.20	0.02
	^{110}Sn(γ,α)	−0.33	−0.06		^{159}Er(γ,n)	−0.19	−0.02
	^{113}Sn(γ,n)	0.62	0.15		^{160}Er(γ,α)	0.67	0.07
	^{169}Lu(γ,n)	0.16	0.03		^{196}Pb(γ,n)	0.16	0.02
^{112}Sn	^{113}Sn(γ,n)	−0.78	−0.31		^{202}Pb(γ,n)	0.20	0.02
^{114}Sn	^{110}Sn(γ,p)	−0.17	−0.02	^{162}Er	^{159}Er(γ,n)	−0.29	−0.12
	^{110}Sn(γ,α)	−0.26	−0.03		^{159}Er(γ,α)	−0.17	−0.05
	^{113}Sn(γ,n)	0.70	0.08		^{160}Er(γ,n)	−0.22	−0.09
	^{122}Xe(γ,p)	0.16	0.02		^{160}Er(γ,α)	−0.40	−0.13
	^{122}Xe(γ,α)	0.27	0.03		^{161}Er(γ,n)	0.18	0.08
	^{169}Lu(γ,n)	0.15	0.01		^{166}Yb(γ,α)	0.32	0.11
^{115}Sn	^{110}Sn(γ,p)	−0.17	−0.02		^{196}Pb(γ,n)	0.27	0.10
	^{110}Sn(γ,α)	−0.25	−0.03		^{202}Pb(γ,n)	0.15	0.06
	^{113}Sn(γ,n)	0.73	0.08	^{164}Er	^{164}Yb(γ,n)	−0.24	−0.08
	^{122}Xe(γ,α)	0.24	0.03		^{164}Yb(γ,α)	−0.56	−0.24
	^{169}Lu(γ,n)	0.15	0.01		^{165}Yb(γ,n)	−0.17	−0.06
^{120}Te	^{121}Te(γ,n)	−0.78	−0.24		^{196}Pb(γ,n)	0.19	0.06
	^{122}Xe(γ,p)	0.24	0.06		^{202}Pb(γ,n)	0.15	0.06
	^{122}Xe(γ,α)	−0.22	−0.06	^{168}Yb	^{164}Yb(γ,α)	−0.22	−0.10
^{124}Xe	^{122}Xe(γ,p)	−0.20	−0.07		^{166}Yb(γ,n)	−0.18	−0.07
	^{122}Xe(γ,α)	−0.36	−0.18		^{166}Yb(γ,α)	−0.34	−0.12
	^{125}Xe(γ,n)	−0.54	−0.33		^{167}Yb(γ,n)	0.19	0.07
^{126}Xe	^{122}Xe(γ,p)	−0.19	−0.07		^{172}Hf(γ,α)	0.43	0.16
	^{122}Xe(γ,α)	−0.39	−0.18		^{196}Pb(γ,n)	0.29	0.10
	^{125}Xe(γ,n)	0.22	0.11		^{202}Pb(γ,n)	0.19	0.06
	^{127}Xe(γ,n)	−0.57	−0.29	^{174}Hf	^{172}Hf(γ,α)	−0.31	−0.06

Continued on next page

Table 2.6 – continued from previous page

Isotope	Reaction	$\mathbf{r_P}$	ζ	Isotope	Reaction	$\mathbf{r_P}$	ζ
^{130}Ba	^{126}Ba(γ,α)	−0.16	−0.02		^{174}W(γ,n)	−0.19	−0.03
	^{131}Ba(γ,n)	−0.81	−0.15		^{174}W(γ,α)	−0.30	−0.07
^{132}Ba	^{131}Ba(γ,n)	0.41	0.15		^{178}W(γ,α)	0.46	0.08
	^{133}Ba(γ,n)	−0.74	−0.27		^{182}Os(γ,α)	0.17	0.03
^{138}La	^{133}La(γ,p)	−0.19	−0.08		^{196}Pb(γ,n)	0.20	0.04
	^{135}La(γ,n)	−0.31	−0.12	^{180}Ta	^{179}Ta(γ,n)	−0.89	−0.07
	^{136}La(γ,n)	0.25	0.08	^{180}W	^{180}Os(γ,n)	−0.26	−0.08
	^{137}La(γ,n)	−0.35	−0.16		^{180}Os(γ,α)	−0.50	−0.21
^{136}Ce	^{137}Ce(γ,n)	−0.47	−0.07		^{181}Os(γ,n)	−0.16	−0.04
	^{138}Nd(γ,n)	−0.31	−0.04		^{196}Pb(γ,n)	0.22	0.06
	^{138}Nd(γ,p)	0.52	0.07	^{184}Os	^{182}Os(γ,α)	−0.19	−0.02
	^{140}Nd(γ,α)	0.32	0.04		^{185}Os(γ,n)	0.17	0.01
^{138}Ce	^{137}Ce(γ,n)	0.66	0.08		^{184}Pt(γ,α)	−0.45	−0.04
	^{139}Ce(γ,n)	−0.37	−0.04		^{186}Pt(γ,α)	−0.16	−0.01
	^{138}Nd(γ,n)	−0.16	−0.02		^{188}Pt(γ,α)	0.16	0.01
	^{138}Nd(γ,p)	0.28	0.03		^{196}Pb(γ,n)	0.20	0.01
	^{140}Nd(γ,α)	0.18	0.02	^{190}Pt	^{190}Hg(γ,n)	−0.30	−0.04
					^{190}Hg(γ,α)	−0.51	−0.07
					^{196}Pb(γ,n)	0.19	0.02
				^{196}Hg	^{196}Pb(γ,n)	−0.68	−0.42
					^{197}Pb(γ,n)	−0.28	−0.13
					^{202}Pb(γ,n)	0.20	0.10

Table 2.7: Correlations and ζ slopes between mass fraction and reaction rates for the 10×3D-inspired mixing scenario.

Isotope	Reaction	$\mathbf{r_P}$	ζ	Isotope	Reaction	$\mathbf{r_P}$	ζ
^{74}Se	^{75}Se(γ,n)	−0.87	−0.30	^{138}Ce	^{137}Ce(γ,n)	0.58	0.05
^{78}Kr	^{79}Kr(γ,n)	−0.81	−0.35		^{139}Ce(γ,n)	−0.19	−0.01
^{84}Sr	^{84}Rb(γ,n)	0.16	0.06		^{138}Nd(γ,n)	−0.31	−0.02
	^{85}Sr(γ,n)	−0.81	−0.33		^{138}Nd(γ,p)	0.37	0.03
^{92}Mo	^{93}Mo(γ,n)	−0.93	−0.17		^{138}Nd(γ,α)	−0.17	−0.01
^{94}Mo	^{93}Mo(γ,n)	0.94	0.21		^{140}Nd(γ,α)	0.20	0.01

Continued on next page

Table 2.7 – continued from previous page

Isotope	Reaction	r_P	ζ	Isotope	Reaction	r_P	ζ
^{96}Ru	^{97}Ru(γ,n)	−0.88	−0.25	^{144}Sm	^{142}Sm(γ,n)	−0.31	−0.04
^{98}Ru	^{97}Ru(γ,n)	0.68	0.07		^{142}Sm(γ,p)	−0.29	−0.03
	^{100}Pd(γ,p)	0.45	0.05		^{143}Sm(γ,n)	−0.41	−0.05
	^{100}Pd(γ,α)	−0.29	−0.03		^{196}Pb(γ,n)	0.35	0.04
	^{110}Sn(γ,α)	0.18	0.02	^{152}Gd	^{152}Dy(γ,α)	−0.50	−0.02
^{102}Pd	^{100}Pd(γ,p)	−0.29	−0.08		^{154}Dy(γ,α)	−0.16	−0.00
	^{100}Pd(γ,α)	−0.30	−0.08		^{158}Er(γ,n)	0.15	0.00
	^{103}Pd(γ,n)	−0.65	−0.21		^{160}Er(γ,α)	0.19	0.00
^{106}Cd	^{107}Cd(γ,n)	−0.83	−0.30	^{156}Dy	^{159}Er(γ,n)	−0.19	−0.06
	^{110}Sn(γ,α)	0.19	0.07		^{160}Er(γ,α)	0.70	0.26
^{108}Cd	^{107}Cd(γ,n)	0.63	0.15		^{202}Pb(γ,n)	0.16	0.05
	^{109}Cd(γ,n)	−0.50	−0.10	^{158}Dy	^{159}Dy(γ,n)	0.19	0.01
^{113}In	^{110}Sn(γ,p)	−0.17	−0.05		^{158}Er(γ,n)	−0.16	−0.01
	^{110}Sn(γ,α)	−0.24	−0.08		^{158}Er(γ,α)	−0.28	−0.01
	^{113}Sn(γ,n)	0.82	0.34		^{159}Er(γ,n)	−0.18	−0.01
^{112}Sn	^{110}Sn(γ,α)	−0.17	−0.05		^{160}Er(γ,α)	0.56	0.02
	^{113}Sn(γ,n)	−0.78	−0.33		^{196}Pb(γ,n)	0.17	0.01
^{114}Sn	^{110}Sn(γ,p)	−0.18	−0.02		^{202}Pb(γ,n)	0.17	0.01
	^{110}Sn(γ,α)	−0.27	−0.04	^{162}Er	^{159}Er(γ,n)	−0.28	−0.10
	^{113}Sn(γ,n)	0.76	0.12		^{159}Er(γ,α)	−0.16	−0.05
	^{122}Xe(γ,α)	0.15	0.03		^{160}Er(γ,n)	−0.19	−0.07
	^{169}Lu(γ,n)	0.16	0.02		^{160}Er(γ,α)	−0.29	−0.09
^{115}Sn	^{110}Sn(γ,p)	−0.18	−0.02		^{161}Er(γ,n)	0.18	0.06
	^{110}Sn(γ,α)	−0.27	−0.04		^{166}Yb(γ,α)	0.40	0.12
	^{113}Sn(γ,n)	0.77	0.12		^{196}Pb(γ,n)	0.26	0.09
	^{169}Lu(γ,n)	0.16	0.02		^{202}Pb(γ,n)	0.15	0.05
^{120}Te	^{119}Te(γ,n)	−0.20	−0.04	^{164}Er	^{164}Yb(γ,n)	−0.23	−0.10
	^{121}Te(γ,n)	−0.66	−0.13		^{164}Yb(γ,α)	−0.59	−0.43
	^{122}Xe(γ,p)	0.42	0.07		^{165}Yb(γ,n)	−0.17	−0.09
	^{122}Xe(γ,α)	−0.35	−0.06		^{196}Pb(γ,n)	0.16	0.07
^{124}Xe	^{122}Xe(γ,n)	−0.22	−0.08	^{168}Yb	^{168}Hf(γ,n)	−0.16	−0.05
	^{122}Xe(γ,p)	−0.27	−0.10		^{168}Hf(γ,α)	−0.40	−0.21
	^{122}Xe(γ,α)	−0.44	−0.23		^{169}Hf(γ,n)	−0.17	−0.05

Continued on next page

Table 2.7 – continued from previous page

Isotope	Reaction	r_P	ζ	Isotope	Reaction	r_P	ζ
	^{125}Xe(γ,n)	−0.40	−0.22		^{172}Hf(γ,α)	0.41	0.19
^{126}Xe	^{122}Xe(γ,n)	−0.21	−0.06		^{196}Pb(γ,n)	0.22	0.08
	^{122}Xe(γ,p)	−0.26	−0.08	^{174}Hf	^{174}W(γ,n)	−0.23	−0.06
	^{122}Xe(γ,α)	−0.46	−0.19		^{174}W(γ,α)	−0.52	−0.19
	^{125}Xe(γ,n)	0.38	0.16		^{176}W(γ,α)	−0.15	−0.05
	^{127}Xe(γ,n)	−0.26	−0.11	^{180}Ta	^{179}Ta(γ,n)	−0.88	−0.03
^{130}Ba	^{126}Ba(γ,p)	−0.18	−0.02	^{180}W	^{180}Os(γ,n)	−0.30	−0.12
	^{126}Ba(γ,α)	−0.31	−0.03		^{180}Os(γ,α)	−0.54	−0.39
	^{128}Ba(γ,n)	−0.26	−0.03		^{196}Pb(γ,n)	0.21	0.08
	^{128}Ba(γ,p)	−0.16	−0.01	^{184}Os	^{184}Pt(γ,α)	−0.49	−0.10
	^{128}Ba(γ,α)	−0.27	−0.02		^{185}Pt(γ,α)	−0.16	−0.02
	^{129}Ba(γ,n)	0.21	0.02		^{196}Pb(γ,n)	0.16	0.02
	^{131}Ba(γ,n)	−0.51	−0.05	^{190}Pt	^{190}Hg(γ,n)	−0.27	−0.06
^{132}Ba	^{128}Ba(γ,α)	−0.18	−0.02		^{190}Hg(γ,α)	−0.52	−0.15
	^{131}Ba(γ,n)	0.63	0.12		^{196}Pb(γ,n)	0.18	0.04
	^{133}Ba(γ,n)	−0.56	−0.10	^{196}Hg	^{196}Pb(γ,n)	−0.75	−0.53
^{138}La	^{137}La(γ,n)	−0.65	−0.35		^{202}Pb(γ,n)	0.20	0.12
^{136}Ce	^{138}Nd(γ,n)	−0.39	−0.06				
	^{138}Nd(γ,p)	0.65	0.10				
	^{138}Nd(γ,α)	−0.16	−0.03				
	^{140}Nd(γ,α)	0.31	0.05				

Table 2.8: Correlations and ζ slopes between mass fraction and reaction rates for the 50×3D-inspired mixing scenario.

Isotope	Reaction	r_P	ζ	Isotope	Reaction	r_P	ζ
^{74}Se	^{75}Se(γ,n)	−0.95	−0.10	^{130}Ba	^{126}Ba(γ,α)	−0.19	−0.01
^{78}Kr	^{79}Kr(γ,n)	−0.91	−0.28		^{132}Ce(γ,n)	−0.23	−0.01
^{84}Sr	^{85}Sr(γ,n)	−0.92	−0.26		^{132}Ce(γ,p)	0.32	0.01
^{92}Mo	^{93}Mo(γ,n)	−0.57	−0.03		^{132}Ce(γ,α)	−0.20	−0.01
	^{100}Pd(γ,p)	0.19	0.01		^{134}Ce(γ,n)	−0.20	−0.01
	^{100}Pd(γ,α)	0.20	0.01		^{134}Ce(γ,α)	0.41	0.02
	^{110}Sn(γ,n)	0.16	0.01	^{132}Ba	^{132}Ce(γ,n)	−0.28	−0.04

Continued on next page

Table 2.8 – continued from previous page

Isotope	Reaction	r_P	ζ	Isotope	Reaction	r_P	ζ
	^{110}Sn(γ,p)	0.23	0.01		^{132}Ce(γ,p)	−0.25	−0.03
	^{110}Sn(γ,α)	0.37	0.02		^{132}Ce(γ,α)	−0.31	−0.04
^{94}Mo	^{93}Mo(γ,n)	0.89	0.08		^{133}Ce(γ,n)	−0.19	−0.03
^{96}Ru	^{97}Ru(γ,n)	−0.62	−0.05		^{134}Ce(γ,α)	−0.18	−0.02
	^{97}Ru(γ,α)	−0.18	−0.01	^{138}La	^{137}La(γ,n)	−0.75	−0.45
	^{100}Pd(γ,α)	0.33	0.03	^{136}Ce	^{138}Nd(γ,n)	−0.42	−0.09
	^{110}Sn(γ,n)	0.16	0.01		^{138}Nd(γ,p)	0.66	0.14
	^{110}Sn(γ,p)	0.16	0.01		^{140}Nd(γ,α)	0.25	0.06
	^{110}Sn(γ,α)	0.23	0.02	^{138}Ce	^{138}Nd(γ,n)	−0.33	−0.04
^{98}Ru	^{100}Pd(γ,p)	0.70	0.13		^{138}Nd(γ,p)	−0.31	−0.04
	^{100}Pd(γ,α)	−0.53	−0.10		^{138}Nd(γ,α)	−0.24	−0.02
^{102}Pd	^{100}Pd(γ,p)	−0.39	−0.06		^{139}Nd(γ,n)	−0.31	−0.04
	^{100}Pd(γ,α)	−0.38	−0.05	^{144}Sm	^{142}Sm(γ,n)	−0.36	−0.06
	^{103}Pd(γ,n)	−0.23	−0.04		^{142}Sm(γ,p)	−0.29	−0.04
	^{104}Cd(γ,p)	0.17	0.03		^{143}Sm(γ,n)	−0.50	−0.09
	^{104}Cd(γ,α)	−0.27	−0.04		^{196}Pb(γ,n)	0.19	0.03
^{106}Cd	^{104}Cd(γ,p)	−0.23	−0.03	^{152}Gd	^{152}Dy(γ,α)	−0.50	−0.05
	^{107}Cd(γ,n)	−0.62	−0.07		^{158}Er(γ,n)	0.18	0.01
	^{110}Sn(γ,p)	0.20	0.02	^{156}Dy	^{156}Er(γ,α)	−0.44	−0.27
	^{110}Sn(γ,α)	0.40	0.05		^{158}Er(γ,n)	0.19	0.09
^{108}Cd	^{107}Cd(γ,n)	0.61	0.11		^{158}Er(γ,α)	−0.18	−0.07
	^{110}Sn(γ,p)	0.49	0.08		^{160}Er(γ,α)	0.25	0.15
	^{110}Sn(γ,α)	−0.33	−0.06	^{158}Dy	^{158}Er(γ,n)	−0.28	−0.02
^{113}In	^{113}Sn(γ,n)	0.94	0.44		^{158}Er(γ,α)	−0.53	−0.05
^{112}Sn	^{110}Sn(γ,p)	−0.25	−0.03	^{162}Er	^{162}Yb(γ,α)	−0.49	−0.30
	^{110}Sn(γ,α)	−0.34	−0.04		^{164}Yb(γ,α)	−0.16	−0.07
	^{113}Sn(γ,n)	−0.65	−0.08		^{168}Hf(γ,n)	0.17	0.07
^{114}Sn	^{110}Sn(γ,α)	−0.21	−0.02	^{164}Er	^{164}Yb(γ,n)	−0.31	−0.17
	^{113}Sn(γ,n)	0.78	0.08		^{164}Yb(γ,α)	−0.56	−0.51
	^{122}Xe(γ,n)	−0.28	−0.03		^{202}Pb(γ,n)	0.16	0.09
	^{122}Xe(γ,p)	0.16	0.02	^{168}Yb	^{168}Hf(γ,n)	−0.28	−0.12
	^{122}Xe(γ,α)	0.23	0.03		^{168}Hf(γ,α)	−0.55	−0.57
	^{169}Lu(γ,n)	0.15	0.01	^{174}Hf	^{174}W(γ,n)	−0.30	−0.13

Continued on next page

Table 2.8 – continued from previous page

Isotope	Reaction	$\mathbf{r_P}$	ζ	Isotope	Reaction	$\mathbf{r_P}$	ζ
^{115}Sn	^{110}Sn(γ,α)	-0.21	-0.02		^{174}W(γ,α)	-0.53	-0.37
	^{113}Sn(γ,n)	0.80	0.08	^{180}Ta	^{179}Ta(γ,n)	-0.87	-0.00
	^{122}Xe(γ,n)	-0.27	-0.03	^{180}W	^{180}Os(γ,n)	-0.38	-0.21
	^{122}Xe(γ,p)	0.15	0.02		^{180}Os(γ,α)	-0.52	-0.46
	^{122}Xe(γ,α)	0.22	0.03		^{196}Pb(γ,n)	0.19	0.11
	^{169}Lu(γ,n)	0.15	0.01	^{184}Os	^{184}Pt(γ,n)	-0.16	-0.05
^{120}Te	^{120}Xe(γ,α)	-0.22	-0.05		^{184}Pt(γ,α)	-0.55	-0.29
	^{122}Xe(γ,p)	0.58	0.11		^{196}Pb(γ,n)	0.16	0.04
	^{122}Xe(γ,α)	-0.49	-0.10	^{190}Pt	^{190}Hg(γ,n)	-0.30	-0.13
^{124}Xe	^{122}Xe(γ,n)	-0.28	-0.10		^{190}Hg(γ,α)	-0.49	-0.30
	^{122}Xe(γ,p)	-0.31	-0.11		^{196}Pb(γ,n)	0.20	0.08
	^{122}Xe(γ,α)	-0.43	-0.21	^{196}Hg	^{196}Pb(γ,n)	-0.76	-0.60
	^{123}Xe(γ,n)	0.18	0.06		^{202}Pb(γ,n)	0.18	0.11
^{126}Xe	^{126}Ba(γ,p)	-0.35	-0.17				
	^{126}Ba(γ,α)	-0.48	-0.32				
	^{127}Ba(γ,n)	-0.22	-0.12				

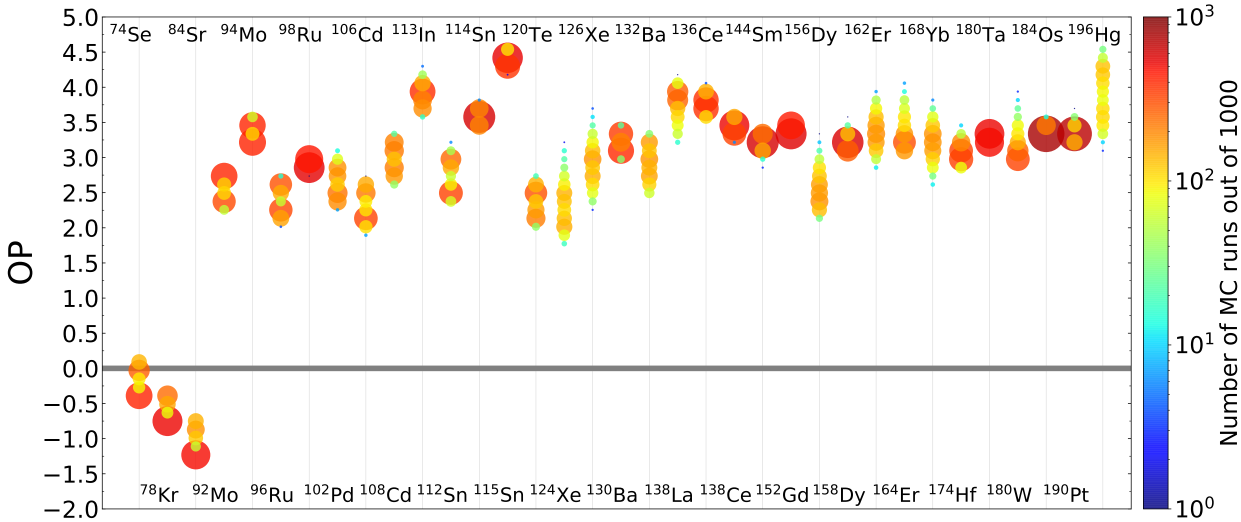

Figure 2.21: Histogram showing the spread due to varying (γ, p), (γ, n), (γ, α) and corresponding capture rates for unstable n-deficient isotopes from Se–Po for the 3×3D-inspired mixing scenario. Colour and size both correspond to the logarthimic binning of Monte Carlo runs. The average spread $\mathrm{OP}_{\max} - \mathrm{OP}_{\min} = 0.69$ dex. $\mathrm{OP} = 0$ is the initial amount.

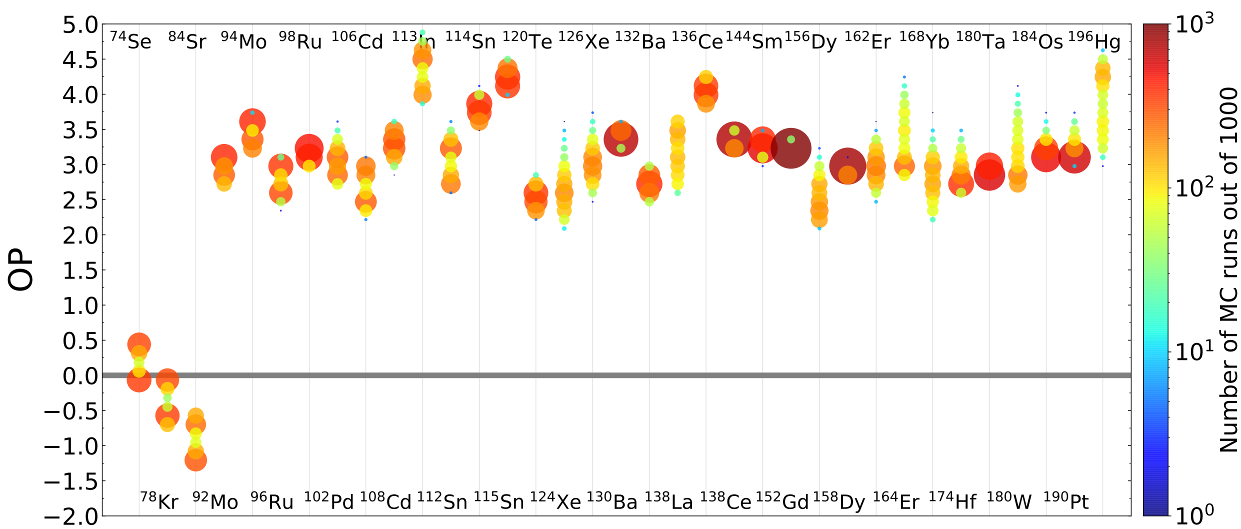

Figure 2.22: Histogram showing the spread due to varying (γ, p), (γ, n), (γ, α) and corresponding capture rates for unstable n-deficient isotopes from Se–Po for the 10×3D-inspired mixing scenario. Colour and size both correspond to the logarthimic binning of Monte Carlo runs. The average spread $\mathrm{OP}_{\max} - \mathrm{OP}_{\min} = 0.76$ dex. $\mathrm{OP} = 0$ is the initial amount.

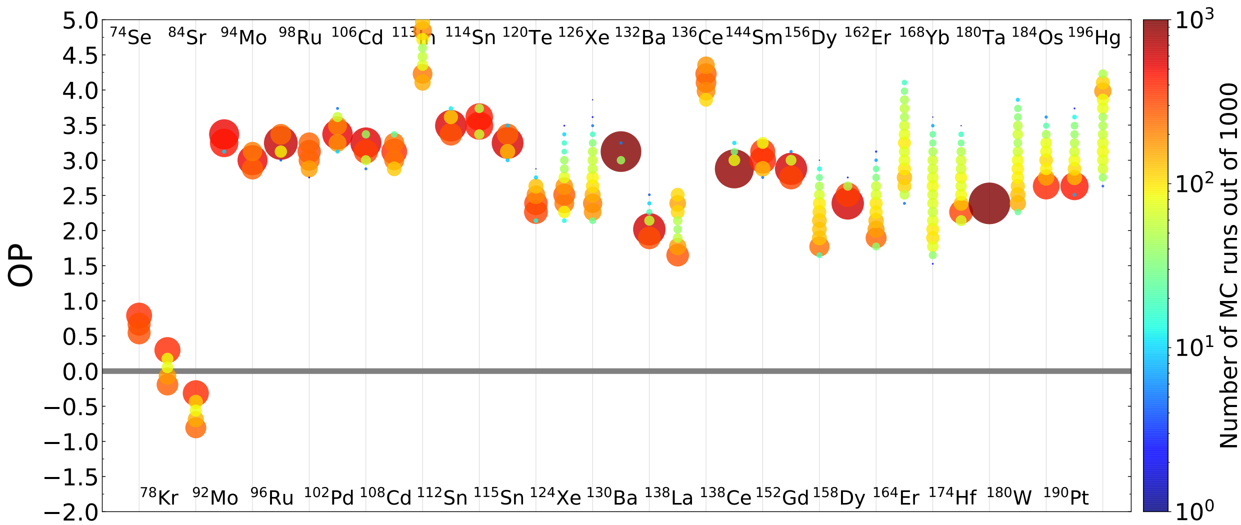

Figure 2.23: Histogram showing the spread due to varying (γ,p), (γ,n), (γ,α) and corresponding capture rates for unstable n-deficient isotopes from Se–Po for the 50×3D-inspired mixing scenario. Colour and size both correspond to the logarthimic binning of Monte Carlo runs. The average spread $\mathrm{OP_{max}} - \mathrm{OP_{min}} = 0.79$ dex. OP = 0 is the initial amount.

Chapter 3

Investigations Beyond the p-Nuclei

The work presented in Section 3.1 presents the preliminary results of exploratory research done on the light odd-Z elements produced in O-C shell mergers solely by myself. The work presented in Section 3.2 presents the preliminary results of collaborative research done with Dr. Falk Herwig on the advective-reactive r-process around a black hole.

3.1 Light Odd-Z Isotope Production

The p-nuclei are not the sole unique nucleosynthetic feature of O-C shell mergers. The light odd-Z elements P, Cl, K, and Sc have all been noted as products as well Ritter et al. (2018a); Roberti et al. (2025). These elements are of interest because they are underproduced in galactic chemical evolution calculations (Ritter et al., 2018a), there are observations of P-enhanced stars (Masseron et al., 2020; Brauner et al., 2023, 2024), and K in particular is relevant for the habitability of exoplanets Frank et al. (2014); O'Neill et al. (2020).

Using the same models described in Chapter 2, I present the preliminary results of how the light odd-Z elements are produced in O-C shell mergers.

As Figure 3.1 shows, the light odd-Z elements are all clearly effected by the presence of a convective downturn. Similar to the p-nuclei, the impact is non-monotonic and non-linear, as ^{31}P, ^{35}Cl, ^{40}K, and ^{45}Sc all decrease in production at the fastest mixing speeds. K and Sc show the most significant impact from the convective downturn with a spread in production of $1.5 - 2$ orders of magnitude.

[JI: I have more plots and tables I can add here if we want to do that]

- Tables for each mixing case showing the relevant reactions that each species undergoes
- Plots showing how the convective profiles change the mass fraction profiles of K-39, K-40, K-41 including GOSH and partial merger (with cool dips)

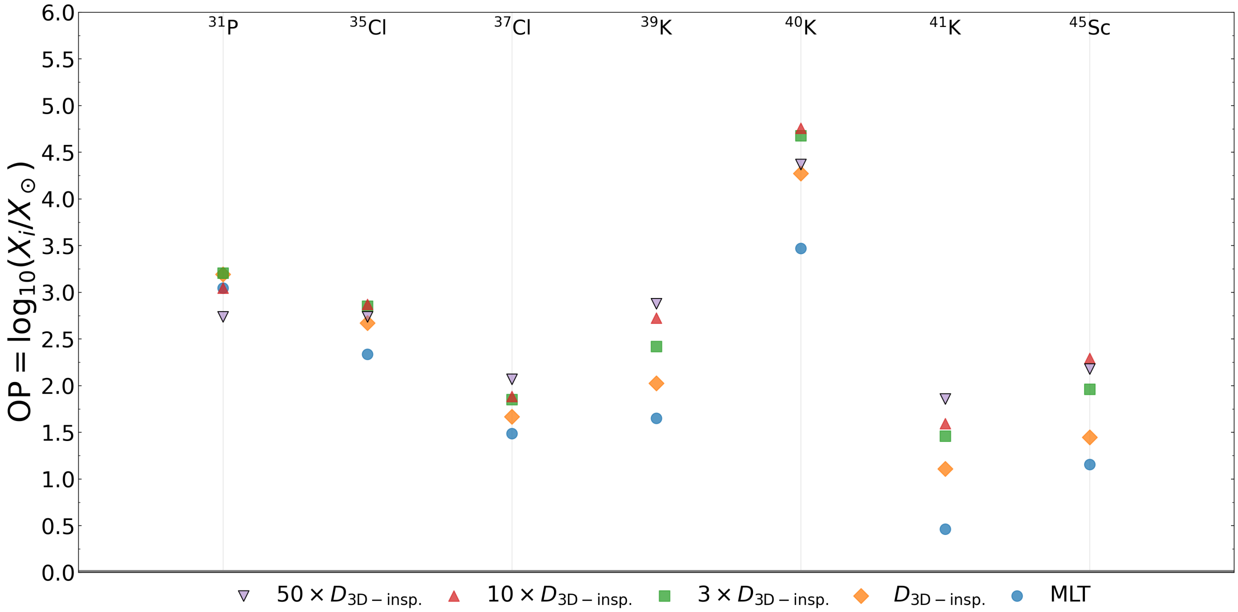

Figure 3.1: Impact on the production of the light odd-Z stable isotopes of P, Cl, K, and Sc comparing MLT and convective dowturn models.

- I can make the impact on production with GOSH and partial merger too
- I have plots showing the impact from ingestion
- XRISM collaboration Ne/O vs K/Ar plots that show a clear trend for both the mixing speeds and ingestion rate

3.2 Advective-Reactive r-process

[*JI: This section is very much a work in progress. I kind of forget what we did. :p*]

Advective-reactive nucleosynthesis is the generalization of the convective-reactive framework to include transport of material beyond convection. In this work, we take the `SkyNet` post-processed trajectories [JI: done by Rodrigo ? citation ?] that escape a black hole formed by neutron star mergers and determine whether the advective-reactive framework is relevant to r-process nucleosynthesis.

Over time, the distance between any two trajectories increases as they move away from the black hole. That means there is a time limit on how long the trajectories are able to exchange material. We defin a distance L_{mix} analogously to ℓ_{mix} and say that any two trajectories that are within this distance can exchange material.

$$L_{\text{mix}} = \sqrt{D t_{\text{nuc}}} \tag{3.1}$$

where $D = 3 \times 10^16 \text{cm}^2/\text{s}$ is the diffusion coefficient and t_{nuc} was given an analytic form based on an eyeballed linear fitting of $\tau = X/(dX/dt)$ vs t plots for some neutron heavy In isotopes. Because of this, we get L_{mix} as a function time. If for a time $t \geq t_{\text{mix}}$ the pairwise distance between two trajectories is less than $L_{\text{mix}}(t)$ then we say that they can exchange material. The results of this analysis are shown in Figure 3.2. It appears that 7.5-15% of trajectories could be able to exchange material for the optimistic cases.

Much more work is needed to determine whether this matters because if both trajectories are in NSE they would have roughly the same pattern anyways. We were in the early stages of seeing if we could post-process the trajectories with `ppn` rather than `SkyNet` and see if maybe we could figure out if the trajectories are exchanging and at least one is not in NSE. Also we wanted to ask if the trajectories that were eaten by the black hole could've exchanged material with the ones that escaped.

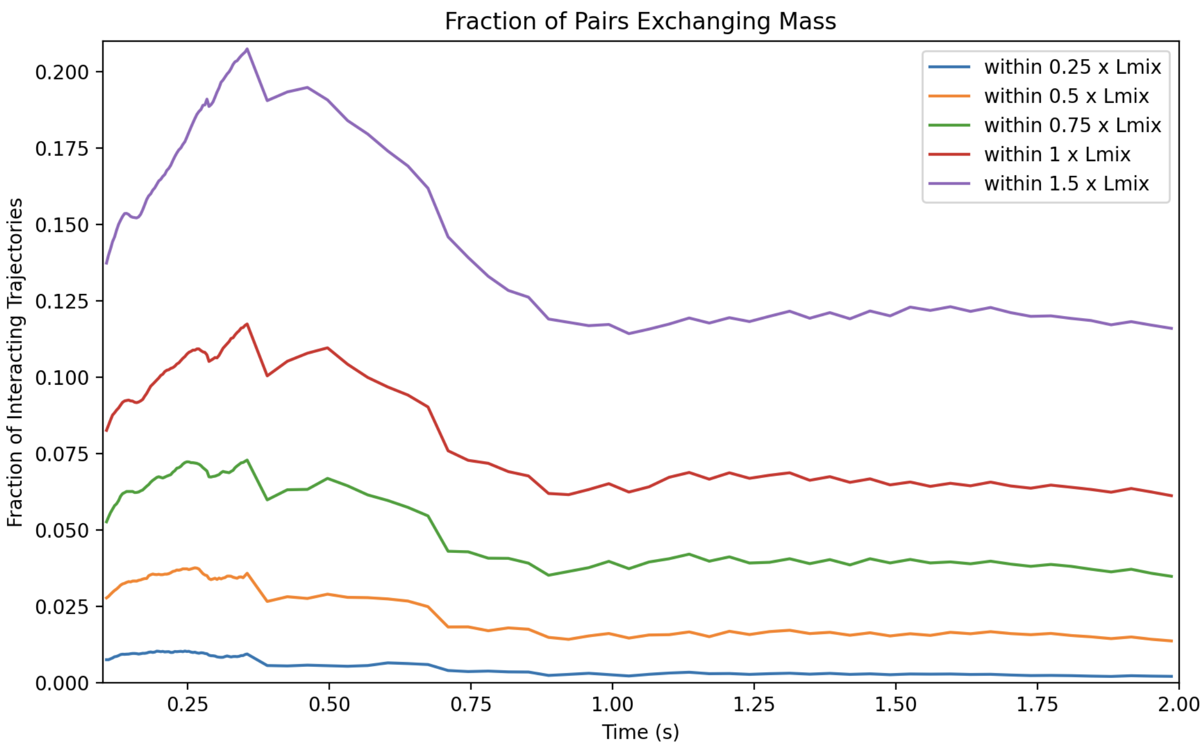

Figure 3.2: Fraction of trajectories with distance for enough time to exchange mass. Each coloured line represents a size for L_{mix}.

Chapter 4

Summary and Conclusion

4.1 Mixing, Nuclear Physics, and the p-Nuclei

In this work, I have explored how changing the mixing assumptions made in 1-D stellar evolution models can have a significant impact on the final nucleosynthesis of the p-nuclei in O-C shell mergers. I have shown that the convective downturn, which is a feature of 3-D hydrodynamic simulations, can have a significant impact on the nucleosynthesis of the p-nuclei. The convective downturn leads to a non-monotonic and non-linear impact on the production of the p-nuclei, with isotopes of the same element being effected differently, which has direct implications for grains analysis. I have also shown the impact of the C-shell entrainment rate is also significant. While most species are uniformly boosted by the entrainment, the large variation depending on the true entrainment rate means that there exists large uncertainties there. Potential convective quenching in the profile either GOSH-like at peak Ne-burning or at the top of the shell due to hydrodynamic feedback only partially merging the O- and C-shells decrease the production of the p-nuclei also was shown. Finally, I have shown that while the impact from varying the input nuclear physics has a comparable impact to any individual mixing scenario, different scenearios result in different nuclear physics impacts. Addtionally, whether a species and reaction rate are correlated is largely dependent on the mixing scenario with few shared correlated rates. This demonstrates the importance of understanding the mixing conditions in the O-C shell merger environment to accurately model the nucleosynthesis of the p-nuclei.

4.2 Light Odd-Z Isotope Production and Macrophysics

Although the work is preliminary, current results show that the light odd-Z elements are also significantly impacted by the 1-D mixing asssumptions showing all the same features as the p-nuclei. This shows the importance of understanding the mixing conditions in O-C shell mergers for O-C merger nucleosynthesis generically, and not just the more obscure p-nuclei. The convective-reactive environment is a complex scenario for nucleosynthesis, and this emphasizes the importance of understanding and connecting 3-D hydrodynamic simulations to nucleosynthesis.

Bibliography

Andrassy, R., Herwig, F., Woodward, P., and Ritter, C. (2020). 3D hydrodynamic simulations of C ingestion into a convective O shell. *Monthly Notices of the Royal Astronomical Society*, 491:972–992.

Arcones, A. and Montes, F. (2011). Production of Light-element Primary Process Nuclei in Neutrino-driven Winds. *The Astrophysical Journal*, 731:5.

Arnett, W. D. and Meakin, C. (2011). Toward Realistic Progenitors of Core-collapse Supernovae. *The Astrophysical Journal*, 733:78.

Arnett, W. D., Meakin, C., Hirschi, R., Cristini, A., Georgy, C., Campbell, S., Scott, L. J. A., Kaiser, E. A., Viallet, M., and Mocák, M. (2019). 3D Simulations and MLT. I. Renzini's Critique. *The Astrophysical Journal*, 882:18.

Arnould, M. (1976). Possibility of synthesis of proton-rich nuclei in highly evolved stars. II. *Astronomy and Astrophysics*, 46:117–125.

Arnould, M. and Goriely, S. (2003). The p-process of stellar nucleosynthesis: Astrophysics and nuclear physics status. *Physics Reports*, 384(1):1–84.

Battino, U., Pignatari, M., Travaglio, C., Lederer-Woods, C., Denissenkov, P., Herwig, F., Thielemann, F., and Rauscher, T. (2020). Heavy elements nucleosynthesis on accreting white dwarfs: Building seeds for the p-process. *Monthly Notices of the Royal Astronomical Society*, 497:4981–4998.

Battino, U., Tattersall, A., Lederer-Woods, C., Herwig, F., Denissenkov, P., Hirschi, R., Trappitsch, R., den Hartogh, J. W., Pignatari, M., and NuGrid Collaboration (2019). NuGrid stellar data set - III. Updated low-mass AGB models and s-process nucleosynthesis with metallicities Z= 0.01, Z = 0.02, and Z = 0.03. *Monthly Notices of the Royal Astronomical Society*, 489:1082–1098.

Bazan, G. and Arnett, D. (1994). Convection, Nucleosynthesis, and Core Collapse. *The Astrophysical Journal*, 433:L41.

Bisterzo, S., Gallino, R., Straniero, O., Cristallo, S., and Käppeler, F. (2011). The s-process in low-metallicity stars - II. Interpretation of high-resolution spectroscopic observations with asymptotic giant branch models. *Monthly Notices of the Royal Astronomical Society*, 418:284–319.

Böhm-Vitense, E. (1958). Über die Wasserstoffkonvektionszone in Sternen verschiedener Effektivtemperaturen und Leuchtkräfte. Mit 5 Textabbildungen. *Zeitschrift fur Astrophysik*, 46:108.

Brauner, M., Masseron, T., García-Hernández, D. A., Pignatari, M., Womack, K. A., Lugaro, M., and Hayes, C. R. (2023). Unveiling the chemical fingerprint of phosphorus-rich stars: I. In the infrared region of APOGEE-2. *Astronomy & Astrophysics*, 673:A123.

Brauner, M., Pignatari, M., Masseron, T., García-Hernández, D. A., and Lugaro, M. (2024). Unveiling the chemical fingerprint of phosphorus-rich stars II. Heavy-element abundances from UVES/VLT spectra.

Burbidge, E. M., Burbidge, G. R., Fowler, W. A., and Hoyle, F. (1957). Synthesis of the Elements in Stars. *Reviews of Modern Physics*, 29(4):547–650.

Cameron, A. G. W. and Fowler, W. A. (1971). Lithium and the s-PROCESS in Red-Giant Stars. *The Astrophysical Journal*, 164:111.

Chapman, S. (1961). Scale Times and Scale Lengths of Variables: With Geomagnetic and Ionospheric Illustrations. *Proceedings of the Physical Society*, 77:424–432.

Collins, C., Müller, B., and Heger, A. (2018). Properties of convective oxygen and silicon burning shells in supernova progenitors. *Monthly Notices of the Royal Astronomical Society*, 473:1695–1704.

Cox, J. P. and Giuli, R. T. (1968). *Principles of Stellar Structure*. Gordon and Breach Science Publishers.

Denissenkov, P., Perdikakis, G., Herwig, F., Schatz, H., Ritter, C., Pignatari, M., Jones, S., Nikas, S., and Spyrou, A. (2018). The impact of (n, γ) reaction rate uncertainties of unstable isotopes near N = 50 on the i-process nucleosynthesis in He-shell flash white dwarfs. *Journal of Physics G Nuclear Physics*, 45:055203.

Denissenkov, P. A., Herwig, F., Perdikakis, G., and Schatz, H. (2021). The impact of (n,γ) reaction rate uncertainties of unstable isotopes on the i-process nucleosynthesis of the elements from Ba to W. *Monthly Notices of the Royal Astronomical Society*, 503:3913–3925.

Dillmann, I., Rauscher, T., Heil, M., Käppeler, F., Rapp, W., and Thielemann, F. K. (2008). P-Process simulations with a modified reaction library. *Journal of Physics G Nuclear Physics*, 35:014029.

Dimotakis, P. E. (2005). Turbulent Mixing. *Annual Review of Fluid Mechanics*, 37:329–356.

Fok, H. K., Pignatari, M., Côté, B., and Trappitsch, R. (2024). Silicon Isotopic Composition of Mainstream Presolar SiC Grains Revisited: The Impact of Nuclear Reaction Rate Uncertainties. *The Astrophysical Journal*, 977:L24.

Frank, E. A., Meyer, B. S., and Mojzsis, S. J. (2014). A radiogenic heating evolution model for cosmochemically Earth-like exoplanets. *Icarus*, 243:274–286.

Freytag, B., Ludwig, H. G., and Steffen, M. (1996). Hydrodynamical models of stellar convection. The role of overshoot in DA white dwarfs, A-type stars, and the Sun. *Astronomy and Astrophysics*, 313:497–516.

Fröhlich, C., Martínez-Pinedo, G., Liebendörfer, M., Thielemann, F. K., Bravo, E., Hix, W. R., Langanke, K., and Zinner, N. T. (2006). Neutrino-Induced Nucleosynthesis of A>64 Nuclei: The Np Process. *Physical Review Letters*, 96:142502.

Goriely, S., Arnould, M., Borzov, I., and Rayet, M. (2001). The puzzle of the synthesis of the rare nuclide 138La. *Astronomy and Astrophysics*, 375:L35–L38.

Goriely, S., José, J., Hernanz, M., Rayet, M., and Arnould, M. (2002). He-detonation in sub-Chandrasekhar CO white dwarfs: A new insight into energetics and p-process nucleosynthesis. *Astronomy and Astrophysics*, 383:L27–L30.

Henyey, L. G., Forbes, J. E., and Gould, N. L. (1964). A New Method of Automatic Computation of Stellar Evolution. *The Astrophysical Journal*, 139:306.

Herwig, F. (2000). The evolution of AGB stars with convective overshoot.

Herwig, F. (2023). Astr 501: Stellar interiors. Lecture notes, University of Victoria.

Herwig, F., Freytag, B., Hueckstaedt, R. M., and Timmes, F. X. (2006). Hydrodynamic Simulations of He Shell Flash Convection. *The Astrophysical Journal*, 642(2):1057.

Herwig, F., Pignatari, M., Woodward, P. R., Porter, D. H., Rockefeller, G., Fryer, C. L., Bennett, M., and Hirschi, R. (2011). CONVECTIVE–REACTIVE PROTON–12C COMBUSTION IN SAKURAI'S OBJECT (V4334 SAGITTARII) AND IMPLICATIONS FOR THE EVOLUTION AND YIELDS FROM THE FIRST GENERATIONS OF STARS. *The Astrophysical Journal*, 727(2):89.

Herwig, F., Woodward, P. R., Lin, P.-H., Knox, M., and Fryer, C. (2014). GLOBAL NON-SPHERICAL OSCILLATIONS IN THREE-DIMENSIONAL 4π SIMULATIONS OF THE H-INGESTION FLASH. *The Astrophysical Journal Letters*, 792(1):L3.

Iliadis, C. (2015). *Nuclear Physics of Stars*. Wiley-VCH.

Jones, S., Andrassy, R., Sandalski, S., Davis, A., Woodward, P., and Herwig, F. (2017). Idealized hydrodynamic simulations of turbulent oxygen-burning shell convection in 4π geometry. *Monthly Notices of the Royal Astronomical Society*, 465:2991–3010.

Joyce, M. and Tayar, J. (2023). A Review of the Mixing Length Theory of Convection in 1D Stellar Modeling. *Galaxies*, 11(3):75.

Käppeler, F., Gallino, R., Bisterzo, S., and Aoki, W. (2011). The s process: Nuclear physics, stellar models, and observations. *Reviews of Modern Physics*, 83:157–194.

Kippenhahn, R., Weigert, A., and Weiss, A. (2013). *Stellar Structure and Evolution*. Springer.

Langer, N., El Eid, M. F., and Fricke, K. J. (1985). Evolution of massive stars with semi-convective diffusion. *Astronomy and Astrophysics*, 145:179–191.

Masseron, T., García-Hernández, D. A., Santoveña, R., Manchado, A., Zamora, O., Manteiga, M., and Dafonte, C. (2020). Phosphorus-rich stars with unusual abundances are challenging theoretical predictions. *Nature Communications*, 11:3759.

Meakin, C. A. and Arnett, D. (2006). Active Carbon and Oxygen Shell Burning Hydrodynamics. *The Astrophysical Journal*, 637:L53–L56.

Meakin, C. A. and Arnett, D. (2007). Turbulent Convection in Stellar Interiors. I. Hydrodynamic Simulation. *The Astrophysical Journal*, 667:448–475.

Miller Bertolami, M. M., Althaus, L. G., Serenelli, A. M., and Panei, J. A. (2006). New evolutionary calculations for the born again scenario. *Astronomy and Astrophysics*, 449:313–326.

Müller, B. (2016). The Status of Multi-Dimensional Core-Collapse Supernova Models. *Publications of the Astronomical Society of Australia*, 33:e048.

Müller, B. (2020). Hydrodynamics of core-collapse supernovae and their progenitors. *Living Reviews in Computational Astrophysics*, 6:3.

Mumpower, M. R., Kawano, T., Sprouse, T. M., Vassh, N., Holmbeck, E. M., Surman, R., and Möller, P. (2018). β-delayed Fission in r-process Nucleosynthesis. *The Astrophysical Journal*, 869:14.

O'Neill, C., Lowman, J., and Wasiliev, J. (2020). The effect of galactic chemical evolution on terrestrial exoplanet composition and tectonics. *Icarus*, 352:114025.

Paxton, B., Bildsten, L., Dotter, A., Herwig, F., Lesaffre, P., and Timmes, F. (2010). MODULES FOR EXPERIMENTS IN StelLAR ASTROPHYSICS (MESA). *The Astrophysical Journal Supplement Series*, 192(1):3.

Pignatari, M., Gallino, R., Heil, M., Wiescher, M., Käppeler, F., Herwig, F., and Bisterzo, S. (2010). THE WEAK s-PROCESS IN MASSIVE STARS AND ITS DEPENDENCE ON THE NEUTRON CAPTURE CROSS SECTIONS. *The Astrophysical Journal*, 710(2):1557.

Pignatari, M., Göbel, K., Reifarth, R., and Travaglio, C. (2016a). The production of proton-rich isotopes beyond iron: The γ-process in stars. *International Journal of Modern Physics E*, 25(04):1630003.

Pignatari, M., Herwig, F., Hirschi, R., Bennett, M., Rockefeller, G., Fryer, C., Timmes, F. X., Ritter, C., Heger, A., Jones, S., Battino, U., Dotter, A., Trappitsch, R., Diehl, S., Frischknecht, U., Hungerford, A., Magkotsios, G., Travaglio, C., and Young, P. (2016b). NuGrid Stellar Data Set. I.Stellar Yields from H to Bi for Stars with Metallicities Z = 0.02 and Z = 0.01. *The Astrophysical Journal Supplement Series*, 225:24.

Prandtl, L. (1925). 7. Bericht über Untersuchungen zur ausgebildeten Turbulenz. *ZAMM - Journal of Applied Mathematics and Mechanics / Zeitschrift für Angewandte Mathematik und Mechanik, vol. 5, issue 2, pp. 136-139*, 5(2):136.

Prialnik, D. (2009). *An Introduction to the Theory of Stellar Structure and Evolution*. Cambridge University Press.

Rapp, W., Görres, J., Wiescher, M., Schatz, H., and Käppeler, F. (2006). Sensitivity of p-Process Nucleosynthesis to Nuclear Reaction Rates in a 25 Msolar Supernova Model. *The Astrophysical Journal*, 653:474–489.

Rauscher, T., Dauphas, N., Dillmann, I., Fröhlich, C., Fülöp, Z., and Gyürky, G. (2013). Constraining the astrophysical origin of the p-nuclei through nuclear physics and meteoritic data. *Reports on Progress in Physics*, 76(6):066201.

Rauscher, T., Heger, A., Hoffman, R. D., and Woosley, S. E. (2002). Nucleosynthesis in Massive Stars with Improved Nuclear and Stellar Physics. *The Astrophysical Journal*, 576:323–348.

Rauscher, T., Nishimura, N., Hirschi, R., Cescutti, G., Murphy, A. St. J., and Heger, A. (2016). Uncertainties in the production of p nuclei in massive stars obtained from Monte Carlo variations. *Monthly Notices of the Royal Astronomical Society*, 463(4):4153–4166.

Rayet, M., Arnould, M., and Prantzos, N. (1990). The p-process revisited. *Astronomy and Astrophysics*, 227:271–281.

Renzini, A. (1987). Some embarrassments in current treatments of convective overshooting. *Astronomy and Astrophysics*, 188:49–54.

Ritter, C., Andrassy, R., Côté, B., Herwig, F., Woodward, P. R., Pignatari, M., and Jones, S. (2018a). Convective-reactive nucleosynthesis of K, Sc, Cl and p-process isotopes in O-C shell mergers. *Monthly Notices of the Royal Astronomical Society*, 474:L1–L6.

Ritter, C., Herwig, F., Jones, S., Pignatari, M., Fryer, C., and Hirschi, R. (2018b). NuGrid stellar data set - II. Stellar yields from H to Bi for stellar models with MZAMS = 1-25 M$\odot$ and Z = 0.0001-0.02. *Monthly Notices of the Royal Astronomical Society*, 480:538–571.

Rizzuti, F., Hirschi, R., Varma, V., Arnett, W. D., Georgy, C., Meakin, C., Mocák, M., Murphy, A. S., and Rauscher, T. (2024a). Shell mergers in the late stages of massive star evolution: New insight from 3D hydrodynamic simulations. *Monthly Notices of the Royal Astronomical Society*, 533:687–704.

Rizzuti, F., Hirschi, R., Varma, V., Arnett, W. D., Georgy, C., Meakin, C., Mocák, M., Murphy, A. S., and Rauscher, T. (2024b). Stellar Evolution and Convection in 3D Hydrodynamic Simulations of a Complete Burning Phase. *Galaxies*, 12:87.

Roberti, L., Pignatari, M., Brinkman, H. E., Jeena, S. K., Sieverding, A., Falla, A., Limongi, M., Chieffi, A., and Lugaro, M. (2025). The Occurrence and Impact of Carbon-Oxygen Shell Mergers in Massive Stars.

Roberti, L., Pignatari, M., Fryer, C., and Lugaro, M. (2024). The γ-process nucleosynthesis in core-collapse supernovae - II. Effect of the explosive recipe. *Astronomy & Astrophysics*, 686:L8.

Roberti, L., Pignatari, M., Psaltis, A., Sieverding, A., Mohr, P., Fülöp, Zs., and Lugaro, M. (2023). The γ-process nucleosynthesis in core-collapse supernovae: I. A novel analysis of γ-process yields in massive stars. *Astronomy & Astrophysics*, 677:A22.

Sackmann, I. J. and Boothroyd, A. I. (1992). The Creation of Superrich Lithium Giants. *The Astrophysical Journal*, 392:L71.

Schatz, H., Aprahamian, A., Barnard, V., Bildsten, L., Cumming, A., Ouellette, M., Rauscher, T., Thielemann, F. K., and Wiescher, M. (2001). End Point of the rp Process on Accreting Neutron Stars. *Physical Review Letters*, 86:3471–3474.

Schatz, H., Aprahamian, A., Görres, J., Wiescher, M., Rauscher, T., Rembges, J. F., Thielemann, F. K., Pfeiffer, B., Möller, P., Kratz, K. L., Herndl, H., Brown, B. A., and Rebel, H. (1998). Rp-process nucleosynthesis at extreme temperature and density conditions. *Physics Reports*, 294(4):167–263.

Sieverding, A., Martínez-Pinedo, G., Huther, L., Langanke, K., and Heger, A. (2018). The ν-Process in the Light of an Improved Understanding of Supernova Neutrino Spectra. *The Astrophysical Journal*, 865:143.

Smith, N. and Arnett, W. D. (2014). PREPARING FOR AN EXPLOSION: HYDRODYNAMIC INSTABILITIES AND TURBULENCE IN PRESUPERNOVAE. *The Astrophysical Journal*, 785(2):82.

Travaglio, C., Gallino, R., Rauscher, T., Röpke, F. K., and Hillebrandt, W. (2015). Testing the Role of SNe Ia for Galactic Chemical Evolution of p-nuclei with Two-dimensional Models and with s-process Seeds at Different Metallicities. *The Astrophysical Journal*, 799:54.

Travaglio, C., Röpke, F. K., Gallino, R., and Hillebrandt, W. (2011). Type Ia Supernovae as Sites of the p-process: Two-dimensional Models Coupled to Nucleosynthesis. *The Astrophysical Journal*, 739:93.

Woosley, S. E. and Heger, A. (2007). Nucleosynthesis and remnants in massive stars of solar metallicity. *Physics Reports*, 442:269–283.

Woosley, S. E., Heger, A., and Weaver, T. A. (2002). The evolution and explosion of massive stars. *Reviews of Modern Physics*, 74:1015–1071.

Woosley, S. E. and Hoffman, R. D. (1992). The alpha -Process and the r-Process. *The Astrophysical Journal*, 395:202.

Woosley, S. E. and Howard, W. M. (1978). The p-processes in supernovae. *The Astrophysical Journal Supplement Series*, 36:285–304.

Xiong, Z., Martínez-Pinedo, G., Just, O., and Sieverding, A. (2024). Production of p Nuclei from r-Process Seeds: The $\ensuremath{\nu}r$ Process. *Physical Review Letters*, 132(19):192701.

Yadav, N., Müller, B., Janka, H. T., Melson, T., and Heger, A. (2020). Large-scale Mixing in a Violent Oxygen–Neon Shell Merger Prior to a Core-collapse Supernova. *The Astrophysical Journal*, 890(2):94.